

HelixCode CLI-Agent Fusion — Programme Continuation Guide

Last updated: 2026-05-29 (close-out¹⁵¹ — HXC-035 FIXED via §11.4.102 systematic-debugging (the new rule's first real use). The /auth/register 400 (surfaced by HXC-029) root-caused INLINE (subagent dispatch was crashing on an Anthropic API-500 storm — A2/B2/D all crashed early, left clean state, no salvage lost): confirmed via direct psql INSERT → column "display_name" does not exist (the error swallowed by the i18n translator into the generic 400). ROOT CAUSE: createSchemaSQL CREATE TABLE users omitted display_name, and the ADD-column migration in InitializeSchema() runs only in the if schemaExists branch, so a FRESH DB never gets it. FIX: +1 line display_name VARCHAR(255) in helix_code/internal/database/database.go. VERIFIED: rebuilt+restarted server → register HTTP 201 (was 400) + login → session token; evidence docs/qa/HXC-035/. Unblocks HXC-029 authenticated paths. NOTE: an Anthropic API-500 instability crashed 5 subagents this window — main-stream inline work continued unaffected; resume parallel subagent waves (HXC-029 CLI banks, HXC-034 wave 2) when the API stabilizes. Open set 3: HXC-029 (CLI/Playwright/Android banks remain), HXC-031, HXC-034. PRIOR close-out¹⁵⁰ — PARALLEL STREAMS landed (operator "work several streams in parallel"). Stream A / HXC-029: 3 more API banks (entity-management 35 PASS / admin-operations 22 / security-validation 16) self-driving vs the LIVE server, each 3× deterministic + mutation-proof; helix_qa 3e7d339 pushed all remotes; 4/18 banks now verified. Stream B / HXC-034 wave 1: §11.4.102 cascaded into ALL 10 dependencies/HelixDevelopment/* submodules (debate_orchestrator/doc_processor/helix_llm/helix_memory/helix_specifier/llm_orchestrator/llm_provider/llms_verifier/models/vision_engine), committed + pushed + ls-remote-verified (2 diverged mirrors cleanly rebased, no force); 10 meta pointers bumped this commit. Stream C: github catch-up detached. HXC-035 FILED (High): the bank verification surfaced a real product defect — POST /api/v1/auth/register → 400 internal_auth_failed_create_user on the live DB (blocks all authenticated paths); systematic-debugging dispatched per §11.4.102. Open set 5: HXC-029 (14 banks left), HXC-031,**

HXC-034 (waves 2+: vasic-digital/* + top-level), HXC-035 (debugging), + note debate_orchestrator needs §11.4.98/99/101 backfill. PRIOR close-out¹⁴⁹ — CONSTITUTION §11.4.102 ADDED + HXC-029 first bank verified + github-throughput root-caused. (1) **§11.4.102** (operator mandate): mandatory superpowers:systematic-debugging activation on ANY issue (Iron Law — no fix without root-cause investigation; composes §11.4.6/§11.4.7) + using-superpowers always-loaded + plugin/dependency-availability — added to constitution Constitution.md/CLAUDE.md/AGENTS.md/QWEN.md (+exports), committed 656b43a, pushed+ls-remote-verified to the constitution upstream (over :443). Meta pointer bumped 8ec2b38→656b43a (this commit). Cascade into 68 owned submodules + CM-COVENANT-114-102-PROPAGATION gate filed as **HXC-034** (large follow-up). (2) **HXC-029** first bank: helixcode-full-qa-api.json (helix_qa f5c6182) self-driving vs the LIVE server (PG+Redis+helixcode :8080 stood up) — 3 deterministic runs 32 PASS/0 FAIL/2 honest-SKIP + mutation-proof; committed 26acc959, pushed. (3) **github-push root cause FOUND** via instrumented push (systematic-debugging per the new §11.4.102): upload throughput to github collapsed to **~173 B/s** (gitlab fast; reachability+MTU+auth+port all fine) → large binary-export packs time out = the broken-pipe; environmental (ISP/route), handled via gitlab-primary + detached-retry. Open set 3: HXC-029 (17 banks remain), HXC-031, HXC-034. PRIOR close-out¹⁴⁸ — HXC-030 §11.4.99 OPERATOR-DOC SWEEP COMPLETE + CLOSED. 8 doc batches (subagent-driven per §11.4.70 — worktree then serial-non-worktree once worktree stale-base re-work surfaced) footered **38/38 (100%)** operator-facing instruction/guide/manual/setup/troubleshooting/tutorial docs, each via LIVE WebFetch of official sources (go.dev/postgresql.org/redis/ollama/opentelemetry/platform.claude.com) — never training data. Real stale fixes: Go 1.24→1.26.3, golang:1.21→1.26 Dockerfiles, go1.21.5→1.26.3, postgres 14→15, Anthropic/OpenAI doc-redirect URLs; honest §11.4.99(B) negative findings where sources 403'd; CONST-036 model IDs flagged-not-guessed. New scripts/gates/sources_verified_gate.sh wired **G13 ENFORCING** (was advisory; flipped at 100%) — future operator doc without a footer FAILs the sweep; scope correctly excludes qa_evidence/helix_qa/architecture/coverage/materials (evidence/internal, not instructions). git push: gitlab reliable; github intermittently broken-pipes (host↔github) but a DETACHED persistent-retry loop converges it every time (caught up at attempt 7) — all commits on BOTH remotes. Open set now 2: HXC-029 (18 auto-executable bank rewrites, needs infra), HXC-031 (Codex/Cline ports). PRIOR close-out¹⁴⁷ — HXC-033 RESOLVED + CLOSED: codegraph 0.9.7 own-org index restored. ROOT CAUSE (operator's data-compat hypothesis CONFIRMED): 0.9.7 requires explicit codegraph init before index; the pre-0.9.7 DB was incompatible. Fix (operator-directed wipe+reindex): wiped gitignored DB (codegraph.db/-wal/-shm; 1.7 GB stale WAL) + codegraph init (tracked config.json preserved) + codegraph index .. Result Files 75,663 / Nodes 1,272,492 (≈ HXC-017 baseline;

edges finalize async). **§11.4.79 probe PASS via codegraph query:** NewBundleTranslator→dependencies/HelixDevelopment/llm_orchestrator + vasic-digital (10 own-org hits); third-party LLama_CPP excluded. Corrected two of my own mis-diagnoses per §11.4.6: the “crash” was a faulty pgrep pattern (process was healthy, just slow); “own-org unreachable” was the wrong CLI verb (search→query in 0.9.7) + a stale MCP-server DB. Bonus: de-bloated docs/codegraph/Status.md 3.66 MB→8 KB (ANSI-spinner garbage from codegraph_sync.sh) + generated its HTML/PDF siblings. Non-blocking follow-ups: restart codegraph MCP server (tools/codegraph) to serve the fresh DB; fix codegraph_sync.sh ANSI dump (§11.4.26 constitution-submodule). **Parallel streams this window (operator “multiple background streams”):** **HXC-029 bank-classification (f99ae5b5 — 18/19 auto-executable, 0 genuinely-manual, NO live §11.4.98 violation) + HXC-030 §11.4.99 doc batch 2 (f23585b7 — PROVIDER_FEATURES/SECURITY/DEPLOYMENT WebFetch-verified) landed concurrently.** github push lagging on broken-pipe (gitlab current; will reconcile). Open set 3: HXC-029/030/031. PRIOR close-out¹⁴⁶ — codegraph 0.9.7 → HXC-033 FILED (Operator-blocked). Operator installed codegraph 0.9.7; §11.4.80 post-update sync FAILED: the install reset the gitignored index DB and every rebuild attempt died — full codegraph index . KILLED mid-run twice (no diagnostic; 54,207 then --force 4,630 files vs HXC-017’s 76,044), codegraph sync . exit 1 (8,461). MCP codegraph_search BundleTranslator resolves ONLY helix_code, NOT the own-org llm_orchestrator symbol → own-org submodules dropped from the index = §11.4.79 regression. Host memory ample (not §12.6 OOM); tracked .codegraph/config.json intact (own-org includes + §11.4.10 credential excludes present — not a config regression). Root cause UNCONFIRMED per §11.4.6 (0.9.7 crashes with no actionable diagnostic). Self-resolution exhausted (quiet/force/sync/memory/flags/config all checked); per §11.4.101 surfaced to operator (downgrade / upstream-bug / accept-degraded — autonomous downgrade would violate §11.4.80’s latest-installed). Documented in docs/codegraph/Status.md; evidence in qa-results/codegraph_.log. *Open set now 4: HXC-029/030/031 + HXC-033(blocked). PRIOR close-out¹⁴⁵ — HXC-032 FIXED + CLOSED (operator “do everything now”). Resolved the LLMOrchestrator broken-merge build-breaker: all 26 committed conflict hunks across 5 Go files (translator.go/translator_test.go/multi_pool.go/cmd/orchestrator/main.go/challenges/runner/main.go) resolved to the HEAD (i18n-migrated) side — the consumer-consistent lineage (Pkg()/SetPkgTranslator()/TPlural). Made the package coherent: bundle.go BundleTranslator gained an honest TPlural (interpolates count; no CLDR plural-form selection claimed per §11.9); automation_test.go TestAutomation_UpstreamsScripts aligned to the lowercase upstreams/ rename (4350384) that caused the conflict. **Verification: submodule go build+go vet+go test ./... 10/10 packages PASS (was build failure), zero conflict markers, downstream helix_agent go build ./... exit 0 (was broken). Submodule***

committed d3956ad, pushed origin/master 1e198e3..d3956ad (FF, no force), verified via ls-remote; meta .gitmodules pointer bumped same meta commit. HXC-032 closed Fixed (→ Fixed.md) per CONST-057. Open set now 3: HXC-029 (§11.4.98 long-tail), HXC-030 (§11.4.99 full sweep), HXC-031 (Codex/Cline ports). DB 145 items validate OK + WAL-checkpointed; G11/G12/clarity PASS. PRIOR close-out¹⁴⁴ — §11.4.99 DOC RECONCILIATION + HXC-031 RECLASSIFIED + HXC-032 FILED (operator “do it all”). (1) HXC-030 §11.4.99 increment (commit 68443b8c): reconciled stale Go 1.24→1.26.3 (verified vs go.dev/dl: go1.26.3 latest stable; 1.24 past support) + PostgreSQL 14→15 across 8 operator-facing docs (DEPLOYMENT_GUIDE, troubleshooting/guide, user_manual/README, general/DEVELOPER_GUIDE + SLASH_COMMANDS, 3 tutorials), each with a
 ## Sources verified 2026-05-29 footer; README already done in HXC-028; full §11.4.99 30-doc sweep continues. (2) HXC-031 RECLASSIFIED: the “deferred build-breaking CONST-052 renames” were stale prose — investigation found NO owned-org renames remain (HelixDevelopment/ leaves already lowercase; only third-party HuggingFace_Hub/LLama_CPP/Ollama + 1 third-party Upstreams/ remain, all CONST-052-exempt; parent org-dirs are intentional carve-outs per HXC-001). Only the Codex/Cline reference-agent ports genuinely remain. (3) HXC-032 FILED (Bug, High): scripts/const052_verify_refs.sh surfaced committed git conflict markers in 5 LLMOrchestrator Go files (26 hunks: translator.go/translator_test.go/multi_pool.go/cmd/orchestrator/main.go/challenges/runner/main.go) breaking helix_agent build — a §11.4 build-layer bluff, ALREADY on origin/master; root cause = i18n-migration lineage (8032035) merged with CONST-052 rename 4350384 markers-unresolved; verified-correct side = HEAD (i18n-migrated — consumers need Pkg()/SetPkgTranslator()/TPlural); deferred to a dedicated build+test-verified pass per §11.4.101 rather than rush-push 26 hunks to the remote (submodule left at its coherent committed state). DB 144 items validate OK + WAL-checkpointed; G11/G12/clarity PASS; tracker exports regenerated. PRIOR close-out¹⁴³ — HXC-029 §11.4.98 CONFIRMED-VIOLATION FIXES (operator “do it all now, yes”, local-only). Fixed both hard findings from the HXC-029 audit: (1) helix_code/tests/regression/server_timeout_test.go TestServerStability was t.Skip("run manually") over an unimplemented body — a §11.4 / §11.4.98 PASS-bluff — now a REAL self-driving test (real net/http server with config timeout semantics, real HTTP I/O, asserts idle-survival; PASS -count=3 ~1.75s each); also repaired the sibling TestServerTimeoutConfiguration CWD-dependent require.NoError(config.Load) hard-fail → skip-with-reason; whole tests/regression package now GREEN (was RED at HEAD, pre-existing). (2) helix_code/tests/e2e/scripts/clean.sh interactive read -p → --force/CLEAN_FORCE=1/TTY-gated self-driving (keeps results non-interactively, no hang; bash -n clean). HXC-029 → In progress; remaining = 19 HelixQA manual-review-required banks + ~24 integration files long-tail. DB resynced (143 items, validate

OK, WAL-checkpointed); summaries+exports regenerated; G11/G12/clarity gates PASS. PRIOR close-out¹⁴² — DOC-SYNC + FORWARD-WORK FILING (operator “do it all”, local-only, no push). **Items 1-3 (safe)**: (1) closed the long-standing §11.4.91 gap — the previously hand-maintained summaries had a “TODO: create” generator; built scripts/generate_issues_summary.sh + scripts/generate_fixed_summary.sh (mechanical, parse Issues.md/Fixed.md headings + Status/Type/Discovered, --check mode) and regenerated the stale summaries (were “1 open / round-464”; now **3 open / 34 closed**, Fixed_Summary Bug=25/Feature=69/Task=12/total=106); added scripts/gates/summary_sync_gate.sh wired as **G12** in verify-all-constitution-rules.sh (CM-{ISSUES,FIXED}-SUMMARY-SYNC freshness; paired §1.1 mutation proven PASS→stale→FAIL→restore→PASS) — this is the HXC-013-follow-up auto-sync seam; existing G11 (CM-WORKABLE-ITEMS-MD-DB-IN-SYNC) + §11.4.91 clarity gate both re-confirmed PASS (clarity 36/0). (2) resolved the helix_qa working-tree drift — 8 third-party tools/opensource/* nested-submodule gitlinks restored to recorded SHAs (stale index.lock cleared per §11.4.82(H)); meta tree clean. (3) confirmed Issues.md/Fixed.md are byte-identical to the DB (no tracker drift — earlier “drift” was a stale *summary*, now fixed). **Items 4-6 (filed per operator)**: HXC-029 (§11.4.98 forward — full-automation compliance sweep of every live/integration/e2e/Challenge test), HXC-030 (§11.4.99 forward — latest-source doc cross-reference sweep across all operator docs) **dispatched as background subagents** (§11.4.70/89) to produce first-deliverable audit ledgers under docs/qa/HXC-029|030/; HXC-031 (deferred build-breaking CONST-052 renames + Codex/Cline ports) filed **EXECUTION HELD** per §11.4.101 (high-blast-radius, awaiting per-batch operator go-ahead). DB resynced 140→143 items, validate OK, byte-identical round-trip, WAL-checkpointed (§11.4.95). PRIOR close-out¹⁴¹ — ALL TRACKER ITEMS CLOSED; open set was EMPTY (137 items: 72 Implemented + 41 Fixed + 24 Completed, DB-validated). This close-out lands **HXC-013** (SQLite single-source-of-truth, §11.4.93/95) + **HXC-025** (constitution §11.4.98/99/101 cascade). **HXC-013**: the constitution submodule’s non-functional workable-items scaffold is now a FUNCTIONAL Go binary (built+pushed UPSTREAM per §11.4.74 at constitution e460a5d) — sync md-to-db/db-to-md/validate/diff, byte-identical round-trip (0 diff on the real 137-item tracker via body_md blobs), go test passing; HelixCode now ships the TRACKED docs/workable_items.db (§11.4.95, WAL-checkpointed, .gitignore !docs/workable_items.db) + constitution pin → e460a5d. Operator-decisions resolved with the scope’s recommended safe-defaults per §11.4.101. Follow-up: auto-sync wiring + CM-WORKABLE-ITEMS gate. **HXC-025**: constitution fetched 15cd4bc→6017af9 (new universal anchors §11.4.98 full-automation-anti-bluff, §11.4.99 latest-source-doc-xref, §11.4.101 autonomous-decision; §11.4.100 ATMOSphere-demoted, not cascaded); cascaded all 3 into HelixCode’s 5 root govfiles + ALL 68 owned submodules (6 parallel subagent waves + pilot; each submodule

committed+pushed to its org remotes, append-only, non-ff union-merged, no force-push); verify-governance-cascade.sh extended to 27 anchors → 0 failures (root + 68 submodules); CONST-055 sweep G1-G10 → 0 failures. New constitution mandates §11.4.98 (live-test full-automation audit) + §11.4.99 (doc latest-source xref) imply forward work. **CUMULATIVE SESSION (close-outs 139-141): 34 real production bugs fixed across 35 packages; HXC-013/014/014b/015/018/019/024/025 all closed; ~45 commits across HelixCode + constitution + 68 submodules, all pushed. Test infra (podman PG×2/Redis/Ollama) left running for re-runs.** PRIOR close-out¹⁴⁰ — operator authorised the infra batch + HXC-018 + HXC-019, all DONE plus bonus HXC-015 + HXC-024. **HXC-018** (§11.4.90 obsolete-status + §11.4.91 summary-clarity gates + colorizer) → sweep G8/G9. **HXC-019** (§11.4.83 docs/qa evidence) promoted from advisory to ENFORCING release gate (-enforce -since baseline + [no-qa-evidence] opt-out) → sweep G7 + release-gate-test.sh; my own HXC-014 commits now carry docs/qa/HXC-014/transcript.md. **HXC-015** (§11.4.81 cross-platform parity) manifest + gate → sweep G10 + install.sh Windows-branch fix + §3.2.1 doc-fix (sandbox.go rlimit re-assessed as honestly fail-closed, F14.5-deferred). **Infra batch:** real podman PG+Redis+Ollama; redis(13)/database(10)/server(8 DDoS-class)/verifier(14)/ollama(9) integration-tagged stress+chaos all PASS -race; make stress-chaos-infra added. **HXC-024** (new): fixed pre-existing broken internal/llm -tags=integration build (deduped MockProvider, removed 10 dead provider tests grep-verified gone, kept KoboldAI). **+5 more real bugs this iteration** (verifier EventPublisher panic-crash; Ollama CONST-035 empty-generation bluff [/api/chat message.content not parsed] + isRunning race + 800-goroutine conn leak) → **session total 34 real production bugs fixed.** **HXC-014 CLOSED Completed** (35 pkgs covered; honest out-of-scope: cloud-LLM endpoints need API keys, stateless utilities have no resilience surface). Also fixed a HXC-014b regression (10 TestTr_RecoversFromNil unit tests asserted the racy self-heal write the sweep removed; masked because sweep verification ran only -run 'Stress|Chaos' not full unit suites — full go test ./internal/... re-run exit 0). ~9 commits this iteration, all pushed to 4 remotes; tracker+CONTINUATION+memory synced. **Sole remaining open item: HXC-013** (SQLite-backed workable-items tracker §11.4.93/95 — large cross-project effort: the constitution submodule's workable-items Go binary is a non-functional scaffold needing upstream construction per §11.4.26, high blast radius across all consumers; carries operator decisions w/ stated recommendations). PRIOR close-out¹³⁹ — HXC-014 §11.4.85 stress+chaos in-process phase + HXC-014b systemic i18n hardening. Subagent-driven (§11.4.70), 11 batches of 3 parallel disjoint-scope agents covered 31 in-process packages (worker/load_balancer/task/session/event/memory/context/rules/repomap/discovery/tools/workflow/hooks/agent/monitoring/commands/mcp/notification/performance/security/providers/llm-mgr/editor/template/cognee/focus/config/persistence/auth/deployment/

project) with stress (sustained p50/p95/p99 + ≥ 10 -goroutine concurrency + boundary) + chaos (panic/cancel/corrupt/pressure) suites, all under -race with captured evidence; make stress-chaos wired to all 31. **29 REAL production bugs surfaced + fixed** — a systemic concurrency defect class: (a) panic-in-goroutine with no recover() crashing the process [event/hooks/mcp $\times$ 2/notification/monitoring/commands/security/template/focus/persistence]; (b) non-reentrant RWMutex re-locked under read-lock = deadlock [worker/hooks/providers]; (c) map/var mutated+read with no/unused mutex = data race [agent/performance/memory/config(no mutex at all)/project/deployment]; plus an auth JWT DoS (forged token $\rightarrow$ panic). Each anti-bluff-proven: revert $\rightarrow$ DATA RACE/deadlock/SIGSEGV/panic $\rightarrow$ restore $\rightarrow$ PASS. **HXC-014b**: the copy-pasted unguarded translator.go i18n seam (data race + panic-crash) fixed across all 54 packages (translatorMu + recover), mutation-proven in internal/logging. 13 commits, all pushed to 4 remotes; tracker+exports synced. Remaining HXC-014: infra batch (server/redis/database — needs make test-infra-up docker stack; podman available) + low-concurrency long-tail. Prior open items HXC-013/015/018/019 untouched. PRIOR close-out¹³⁸ preserved below: round 464: HXC-010 closed Completed ($\rightarrow$ Fixed.md) — operator supplied OpenAI-compatible router credentials; both CodeGraph per-agent Challenges cg-challenge-05-kimi.sh + cg-challenge-07-qwen.sh re-run produce true tier-1 PASS — Kimi CLI and Qwen Code each genuinely invoke codegraph_search and receive real graph data from the scanned HelixCode code-graph; Kimi driven via an openai_legacy provider, Qwen via --auth-type openai, both against SiliconFlow; API keys injected via environment variables only, never written to any tracked file (CONST-042 leak-audit CLEAN); all 7 CodeGraph anti-bluff Challenges now true-end-to-end verified across Claude Code, OpenCode, Crush, Kimi CLI, Qwen Code; migrated to docs/Fixed.md. Open set is now 1 item — HXC-001 (Task, In progress)).**

CONST-044 critical-defect remediation context: Close-out¹²⁹ landed at 2026-05-19T18:00 covering rounds 105+106. Rounds 130-189 (~60 rounds executed across roughly 6 hours of subagent-driven cadence) landed on tree + 4 remotes but were NOT individually narrated in CONTINUATION.md. Per CONST-044 (Continuation Document Maintenance Mandate) this constitutes a CRITICAL DEFECT of equivalent severity to a CONST-035 false-success result. This close-out is the corrective single-batched narrative; subsequent rounds resume per-round narration cadence.

Verbatim 2026-05-19 operator mandate (preserved per CONST-049 §11.4.17): *“all existing tests and Challenges do work in anti-bluff manner - they MUST confirm that all tested codebase really works as expected! We had been in position that all tests do execute with success and all*

Challenges as well, but in reality the most of the features does not work and can't be used! This MUST NOT be the case and execution of tests and Challenges MUST guarantee the quality, the completion and full usability by end users of the product!"

Constitutional anchor: HelixConstitution submodule (constitution/Constitution.md + constitution/CLAUDE.md + constitution/AGENTS.md) is the **canonical root** per CONST-059. This catch-up narrative extends the consumer-side docs/CONTINUATION.md and does NOT modify any canonical-root file.

Round-by-round catch-up table (rounds 130-189)

Round	Scope	Commit SHA(s)	LOC / tests
130	security i18n kickoff (Phase 4 round 23) — 27→2 violations, 26 PrivEscCheck Desc/Details + 1 Summary template	fd81a84 + meta 6119741	+342 LOC; mutat falsifiability
131	helix_code/cmd/cli/main.go × 10 migration (Phase 4 round 24) — Option B (cmd-local i18n pkg)	3a01303 + baseline 7f78077	+10 IDs; baseline refreshed
132	AutoTemp i18n kickoff (Phase 4 round 25)	(submodule TBD) + meta 20344f5	pointer-only; repo truncated
133	Auth i18n kickoff (Phase 4 round 26)	(submodule TBD) + meta 4e78c99	pointer-only; repo pending
134	helix_code/cmd/server/main.go × 10 migration (Phase 4 round 27) — HTTP server entry, Option B	69189d0	+10 IDs; mutation
135	PliniusCommon i18n kickoff (Phase 4 round 28) — infrastructure-only, 36 bundle keys, 64×256 concurrent-safe	fbbe695 + meta ae6699b	+250 LOC
136	helix_code/applications/desktop (Phase 4 round 29) — Fyne GUI, content absorbed alongside android	b5a9487 (content absorbed; CONST-043 preserved)	content in tree
137	helix_code/applications/terminal_ui × up to 10 (Phase 4 round 30) — tview/tcell TUI, 10 sidebar items+title+status	4eba31b	+296 LOC; baseli
138	helix_code/applications/ios infrastructure-only (Phase 4 round	27d121b (mislabelled)	6 tests PASS

Round	Scope	Commit SHA(s)	LOC / tests
	31) — Swift native (CONST-052 Apple exemption)	round-139 due to race)	
139	helix_code/applications/android infrastructure-only (Phase 4 round 32) — Kotlin/Java native (CONST-052 lang exemption)	b5a9487 (re-commit after parallel 27d121b race)	Go bridge surface
140	helix_code/applications/aurora_os × up to 10 (Phase 4 round 33) — ALL 6 PLATFORM APPS NOW COVERED	75f35f6	Aurora platform; pending
141	helix_code/cmd/config_test × 12 (Phase 4 round 34) — snake_case correction; 4 pre-existing CONST-046 eliminated	83993ac	+504 LOC; 11 tests+mutation; h-4
142	helix_code/cmd/security_test × 10 (Phase 4 round 35) — 17→8 violations	57d34c8	+423 LOC; tests+mutation; 4 residual deferred
143	helix_code/cmd/security_fix × 10 (Phase 4 round 36) — alphabetically-first variant; _standalone (27 viol) deferred	bbbf121	+446 LOC; tests+mutation
144	helix_code/cmd/performance_optimization × 10 (Phase 4 round 37) — banner+config+readiness+summary; 17 residual deferred	c7c8b2d	+438 LOC
145	helix_code/cmd/security_fix_standalone × 10 of 27 (Phase 4 round 38) — 9 cmd/* tools now complete	53460d0	+358 LOC; 6 tests+mutation; 1 deferred
146	helix_code/internal/auth × up to 10 (Phase 4 round 39) — FIRST helix_code/internal/* package migrated	3b5ced5	+10 IDs; 11 tests+mutation; S deferred
147	helix_code/internal/agent × 10 of 64 (Phase 4 round 40) — coordinator+base_agent task/workflow errors	9a3ee5e	+433 LOC; 8 tests+sentinelTra 64 deferred
148	helix_code/internal/cognee × migration (Phase 4 round 41)	37dc2a1	report pending
149	helix_code/internal/commands × 10 (Phase 4 round 42) — ValidateContext+manager-not-init+hooks+permissions	77b6041	+400 LOC; 11 tests+mutation
150			

Round	Scope	Commit SHA(s)	LOC / tests
	helix_code/internal/config × 10 (Phase 4 round 43) — boot+validate-required/range; baseline refresh	adf001f + baseline 5a0934e	+384 LOC; 9-cases test+paired-mutations
151	helix_code/internal/context × 8 sites / 5 IDs (Phase 4 round 44) — item/session/project not_found/expired	fc4592c	+369 LOC; 41 tests+mutation; ~ sub-pkg deferred
152	helix_code/internal/database × 8 (Phase 4 round 45) — fmt.Errorf config_parse+pool+ping+schema	509e89f	+317 LOC; 10 tests+mutation; S deferred
153	helix_code/internal/deployment × 10 (Phase 4 round 46) — already_running+phase/strategy+4 gates	2df1a23 (mislabelled “round-155”) + fixup fdd93dd	+447 LOC; sibling
154	helix_code/internal/discovery (Phase 4 round 47) — push-monitor stalled, content reached 4 remotes	0fc080d + closure e251c3c	report not captured
155	helix_code/internal/editor (Phase 4 round 48) — re-commit after sibling-race	3099fee (re-commit) + 7b27a3c initial	report pending
156	helix_code/internal/event (Phase 4 round 49) — content in commit mislabelled “round-155”	7b27a3c (mislabelled) + fixup 0cb32f2	push-stall, content tree
157	helix_code/internal/focus (Phase 4 round 50) — “chain not found” × 4 same ID	(in tree; agent push-stalled)	i18n/ present in tree
158	helix_code/internal/hardware (Phase 4 round 51)	5757f9d	report pending
159	helix_code/internal/helixqa × up to 10 (Phase 4 round 52)	5bf048d	report pending
160	helix_code/internal/hooks × up to 10 (Phase 4 round 53)	a5ce25d	report pending
161	helix_code/internal/llm (Phase 4 round 54) — recovery-batch round	recovery batch b7f8672	partial work landed
162	helix_code/internal/logging × up to 10 (Phase 4 round 55)	8f6d450	report pending
163	helix_code/internal/logo (Phase 4 round 56) — recovery-batch	recovery batch b7f8672	partial work landed
164		f82d00e	report pending

Round	Scope	Commit SHA(s)	LOC / tests
	helix_code/internal/mcp × up to 10 (Phase 4 round 57)		
165	helix_code/internal/memory (Phase 4 round 58) — recovery-batch	recovery batch b7f8672	partial work land
166	helix_code/internal/monitoring (Phase 4 round 59) — recovery- batch-2	recovery batch 2 5c94696	partial work land
167	helix_code/internal/notification (Phase 4 round 60) — recovery-batch	recovery batch b7f8672	partial work land
168	helix_code/internal/performance (Phase 4 round 61) — recovery-batch	recovery batch b7f8672	partial work land
169	helix_code/internal/persistence × up to 10 (Phase 4 round 62)	626cea9	report pending
170	helix_code/internal/project × up to 10 (Phase 4 round 63)	048468a	report pending
171	helix_code/internal/provider × up to 10 (Phase 4 round 64)	99ba5b2	report pending
172	helix_code/internal/providers × up to 10 (Phase 4 round 65) — RE- COMMIT after sibling-agent race	95c0bf9 (re- commit) + 5af460b initial	sibling race
173	helix_code/internal/redis (Phase 4 round 66) — recovery-batch-2	recovery batch 2 5c94696	partial work land
174	helix_code/internal/repomap (Phase 4 round 67)	(absorbed)	content absorbed
175	helix_code/internal/rules (Phase 4 round 68) — recovery-batch-2	recovery batch 2 5c94696	partial work land
176	helix_code/internal/security × up to 10 (Phase 4 round 69) — fixup corrected 5af460b attribution	b2c956b + fixups 446ebb9 + 360f5ca	attribution correc
177	helix_code/internal/server × up to 10 (Phase 4 round 70)	683a19d	report pending
178	helix_code/internal/session (Phase 4 round 71) — recovery-batch-2	recovery batch 2 5c94696	partial work land
179	helix_code/internal/task (Phase 4 round 72)	(absorbed)	content absorbed
180		03c8b18	report pending

Round	Scope	Commit SHA(s)	LOC / tests
	helix_code/internal/template × up to 10 (Phase 4 round 73)		
181	helix_code/internal/tools × up to 10 (Phase 4 round 74)	225b47e	report pending
182	helix_code/internal/verifier × up to 10 (Phase 4 round 75)	0fc3c2a	report pending
183	helix_code/internal/version (Phase 4 round 76) — absorbed by 4449525 per fixup f446f49	4449525 absorption + fixup f446f49	attribution fixup
184	helix_code/internal/worker × up to 10 (Phase 4 round 77)	54606ee	report pending
185	helix_code/internal/workflow × up to 10 (Phase 4 round 78)	4449525	report pending
186	helix_code/internal/adapters (Phase 4 round 79) — absorbed	(absorbed)	content absorbed
187	helix_code/internal/fix (Phase 4 round 80) — absorbed by 4449525 per fixup 30bacde	4449525 absorption + fixup 30bacde	attribution fixup
188	VEN-001 (ex-ISSUE-001) closed — VisionEngine helix-gitlab URL misconfiguration fix; PDF+HTML exports regenerated	e0e86ff + be0b481	99/100 → 100/100 owned-org URLs reachable
189	Tracker prefix rename programme — ISSUE-NNN → HXC/HXA/HXM/VEN/HXL/HXQ/etc per submodule scope	7031511	per-prefix mapping preserved

Aggregate impact across rounds 130-189 (~60 rounds)

- **Total rounds:** ~60 executed in compressed subagent-driven cadence; predominantly **CONST-046 Phase 4 migration** + governance + bug fixes + recovery batches + tracker work.
- **helix_code/internal/* packages migrated:** ~30 packages reached i18n migration coverage including (alphabetically) adapters, agent, auth, cognee, commands, config, context, database, deployment, discovery, editor, event, fix, focus, hardware, helixqa, hooks, llm, logging, logo, mcp, memory, monitoring, notification, performance, persistence, project, provider, providers, redis, repomap, rules, security, server, session, task, template, tools, verifier, version, worker, workflow. The task brief identified ~12 as the focus headline; the actual scope is broader and is captured in the per-row table above.

- **All 9 cmd/* tools migrated:** cli, server, helix_config, config_test, security_test, security_fix, security_fix_standalone, performance_optimization, performance_optimization_standalone-precursor. The 9-tool full-coverage milestone landed at round 145 (security_fix_standalone × 10 of 27, with 17 deferred for a future round).
- **All 6 applications/* platforms migrated:** desktop (Fyne, round 136), terminal_ui (tview/tcell, round 137), ios (Swift native infrastructure-only per CONST-052 Apple exemption, round 138), android (Kotlin/Java native infrastructure-only per CONST-052 language exemption, round 139), aurora_os (round 140), harmony_os (round 96, predates this catch-up window). ALL 6 PLATFORM APPS NOW COVERED milestone landed at round 140.
- **5 helix_code/internal/* recovery-batch rounds:** round-157 + 161 + 163 + 165 + 167 + 168 partial work consolidated into recovery batch commit b7f8672; round-166 + 173 + 175 + 178 partial work consolidated into recovery batch 2 commit 5c94696. Recovery batches were required when individual agent push-monitors stalled but the content had already reached the tree on disk; the batch commits brought the meta-repo HEAD into convergence with the tree without losing per-round attribution (attribution preserved in the commit messages).
- **Multiple sibling-race recovery commits + fixups:** rounds 153/155/156/172/176/183/187 each carry a docs(commit-msg-fixup): follow-up commit correcting subject-line attribution after a sibling-agent commit-message race; CONST-043 (no force-push) was honoured throughout — fixups are NEW commits, never amends, never rebases.
- **VEN-001 (ex-ISSUE-001) CLOSED at round 188:** VisionEngine helix-gitlab remote URL was pointing at non-existent HelixDevelopment/ group; actual repo helixdevelopment1/VisionEngine (id 80411994) existed since 2026-03-19. Fix: git remote set-url helix-gitlab git@gitlab.com:helixdevelopment1/VisionEngine.git + FF-safe push (46 commits → SHA 2d0c35b). Owned-org URL reachability: 99/100 pre-fix → **100/100 OK post-fix.**
- **Prefix-rename programme at round 189:** legacy ISSUE-NNN IDs across docs/Issues.md + docs/Fixed.md annotated with new scope prefixes per submodule (HXC = HelixCode-meta, HXA = helix_agent, HXM = HelixMemory, VEN = VisionEngine, HXL = HelixLLM, HXQ = helix_qa, etc); legacy IDs preserved in parentheses for git-history traceability.
- **Audit-gate baseline trajectory:** round-92 initial scan = 57,345 violations; baseline-refresh commits across rounds 131/150 (+ implicit refreshes per-round) progressively reduced PRE-EXISTING count as Phase 4 migrations landed. Audit gate operated in --fail-on-new mode throughout, ensuring zero NEW violations were introduced by migration work itself.

- **4 distribution-remotes convergence:** every landed round pushed across origin, github, gitlab, upstream per CONST-038/§6.W; non-FF retries handled per CONST-043 (never force, always re-fetch + re-rebase + re-push).
- **Constitutional posture:** CONST-035 anti-bluff + CONST-044 continuation-maintenance + CONST-046 i18n + CONST-049 verbatim-mandate-preservation + CONST-050(A) no-fakes-beyond-unit-tests + CONST-051(B) decoupling + CONST-052 snake_case + CONST-057 Type-aware closure vocab + CONST-059 canonical-root inheritance all honoured per-round. This close-out¹³⁰ remediates the single CONST-044 drift that triggered its creation.

Previous: 2026-05-19T18:00:00Z (close-out¹²⁹ — rounds 105 + 106 §11.4 — issue cleanup LANDED (ISSUE-003 + ISSUE-004 + ISSUE-006 partial closed). ** Pivot from migration cadence to focused bug-fix rounds. **Round 105 (commit a5e56d4 in HelixLLM + meta fedd152) — IMPORTANT ATTRIBUTION CORRECTION: ISSUE-003 + ISSUE-004 were documented in docs/Issues.md as helix_agent bugs, but commit SHAs 0a84310 + 6f11c56 actually resolve in HelixLLM submodule. Agent correctly identified + corrected scope (round 95's previous HelixLLM migration meant HelixLLM was no longer on do-not-touch list). **ISSUE-003 fix** (analysis_test.go hardcoded /run/media/.../helix_agent/HelixLLM/... path): replaced with t.TempDir() + 2 synthesised Go fixture files. **Bonus discovery:** same bug-pattern in git_test.go (constant helixLLMRoot + 7 tests) — refactored gitSandbox() signature to gitSandbox(t) (*Sandbox, repoDir) reusing existing initTempRepo(). 6 ISSUE-003 + 7 git_test tests now PASS on ANY host (not just operator's). **ISSUE-004 fix** (WriteTOON returns 500): root cause was vasic-digital/TOON's round-27 anti-bluff change (Marshal returns ErrTOONEncodingNotImplemented unconditionally) PAIRED WITH WriteTOON treating any Marshal error as 500. Fix: fall back to json.Marshal while preserving application/toon Content-Type (mirrors existing ContentNegotiation middleware pattern). 500 still returned for genuinely-unmarshallable values (channels). 19 middleware tests PASS. Full 32-package HelixLLM suite no-regression. Mutation verified BOTH fixes. +45 LOC net (3 files; well under +300 ceiling). CONST-051(B) clean.**

Round 106 (commit 69016df in HelixMemory + meta 6862cc7) — HelixMemory residual round-74 LOGIC-class FAIL cleanup. **Single root cause** for 6 FAILs: go.mod line 29 replace digital.vasic.memory => ../Memory resolved to nonexistent path (dependencies/HelixDevelopment/Memory); actual vasic-digital/Memory lives at dependencies/vasic-digital/Memory. Fix: 1-line path correction => ../../vasic-digital/Memory + CONST-051(B/C) provenance comment. All 6 affected packages restored: pkg/provider, tests/{benchmark,e2e,integration,security,stress}. Initial state 23 PASS / 6 FAIL → post-fix 29 PASS / 0 FAIL. Mutation verified (revert to ../Memory → 3 affected packages FAIL with

original error). +5 LOC net (4 comment + 1 path). CONST-051(B) clean. install_upstreams invoked per CONST-056. ZERO deferred LOGIC FAILs remain in HelixMemory.

Trackers synced: Issues.md ISSUE-003 + ISSUE-004 moved to Fixed (→ Fixed.md) Status with attribution correction note. Issues_Summary counts: 8 open → 5 open (3 BLOCKED-on-operator remain — ISSUE-001 VisionEngine helix-gitlab 404, ISSUE-002 vasic-digital-github fork divergent, ISSUE-008 helix_ga flake). Fixed.md +3 entries; Fixed_Summary counts: 23 → 26 total (4 Bugs / 19 Features / 3 Tasks). All 4 distribution remotes converged. **All rounds 91-106 now closed and reported.** **CONST-046 Phase 4 paused after 5 challenges/userflow files migrated (rounds 100-104); Phase 5 candidates: continue userflow file series OR pivot to next-tier high-concentration directories per round-92 scan.** Verbatim 2026-05-19 mandate preserved per CONST-049 §11.4.17. All prior content preserved verbatim below.**

Previous: 2026-05-19T17:35:00Z (close-out¹²⁸ — rounds 100 + 101 (+ 102/103 pointer evidence) §11.4 — challenges/pkg/userflow/ Phase 4 kickoff LANDED.** Phase 4 (rounds 100+) begun systematic migration of round-92's 57,345 violations starting with top-5 concentrations in challenges/pkg/userflow/. **Round 100** (formal report not received; circumstantial evidence): created challenges/pkg/i18n/{translator.go, bundles/active.en.yaml, translator_test.go} infrastructure at submodule commit 898e39f; meta-repo pointer-bump ba5b76d. Established the challenges/pkg/i18n.Translator interface reused by rounds 101+. **Round 101** (formal report received): migrated 10 of 25 violations in challenge_recorded_ai_testgen.go — all 10 user-facing AssertionResult.Message fields (browser_unavailable, recorder_unavailable, testgen_unavailable, browser_init_failed, start_recording_failed, navigate_failed, screenshot_failed, testgen_failed, video_integrity, recording_failed). 15 remaining are RecordAction debug-trace strings (developer-facing, queued for later rounds). 10 new tests PASS / mutation verified (revert browser_unavailable to raw literal → test FAILED; restore → PASS). Submodule 67a6c9d + meta pointer-bump 1a1b270. **Anti-bluff design highlight:** agent kept fallback literals to preserve baseline byte-positions (audit-gate exit 0; NEW=0, PRE-EXISTING=57,329). +482 LOC. CONST-051(B) clean. **Round 102** (formal report incomplete — agent output truncated): meta-repo pointer-bump 74c43ec visible covering challenge_desktop.go migration; specific call-site details TBD if needed. **Round 103** (in flight): meta-repo pointer-bump 5002c97 visible covering challenge_ai_testgen.go migration; formal report pending. **Sibling rounds 104 (challenge_recorded_mobile.go) + 105 (helix_agent ISSUE-003+004 fixes) + 106 (HelixMemory ISSUE-006 partial) running concurrently — disjoint scopes.**

All commits 4-remote convergent. Verbatim 2026-05-19 mandate preserved per CONST-049 §11.4.17. All prior content preserved verbatim below.**

Previous: 2026-05-19T17:00:00Z (close-out¹²⁷ — round 99a §11.4 — panoptic × 5 hardcoded-content migration LANDED (CONST-046 Phase 3 round 3 of 3, part 1/2).* Submodule: panoptic/ (at meta-repo root, not under dependencies/). 5 strings migrated, all cobra command Short descriptions: cmd/errors.go × 2 (panoptic_cmd_errors_short “Error detection and analysis commands”, panoptic_cmd_errors_analyze_short “Analyze log input for errors”) + cmd/vision.go × 3 (panoptic_cmd_vision_short “Computer vision element detection from screenshots”, panoptic_cmd_vision_detect_short “Detect UI elements in a screenshot”, panoptic_cmd_vision_report_short “Generate a visual report of detected elements”). **Pattern variation:** also added pkg/i18n/global.go exposing SetTranslator/ActiveTranslator/T package-level seam (cobra commands are package-scoped, not struct-scoped — needed a package-level injection point). Files: pkg/i18n/translator.go + global.go + bundles/active.en.yaml (5 IDs) + translator_test.go (3 tests) + cmd/i18n_migration_test.go (5 call-site tests). Tests: 8/8 PASS (3 i18n unit + 5 cmd sentinel). Mutation verified: restoring literal at errorsCmd.Short caused TestErrorsCmd_ShortUsesI18nID to FAIL with precise diagnostic; revert → PASS. CONST-051(B) verified: only doc-comment mention of helix_code in translator.go:4 (the rule itself); zero production imports. Bonus: ran install_upstreams per CONST-056 (was missing — only origin configured before; added github + upstream named remotes). LOC delta: +328 net (under +350 ceiling). Commits: panoptic 3074c77 (from e8b97cb) + meta-repo pointer-bump c4e50d8. All 4 distribution remotes converged. Concurrent D3.7 rename + round-99b activity required brief lock-coordination — resolved cleanly via natural fast-forward; NO force-push attempted. **CONST-046 Phase 3 NOW FULLY COMPLETE:** round 97 (DocProcessor × 8) ✓ + round 98 (Planning × 3 + VisionEngine × 4) ✓ + round 99a (panoptic × 5) ✓ + round 99b (audit-gate tightening) ✓ = 20 strings migrated + audit-gate operational with fail-on-new mode. **Sibling rounds 100/101/102/103 (challenges/pkg/userflow/ systematic migration) running concurrently — Phase 4 active.** Verbatim 2026-05-19 mandate preserved per CONST-049 §11.4.17. All prior content preserved verbatim below.**

Previous: 2026-05-19T16:30:00Z (close-out¹²⁶ — round 99b §11.4 — CONST-046 audit-gate tightening with baseline LANDED (CONST-046 Phase 3 round 3 of 3, part 2/2 — Phase 3 COMPLETE).* New --baseline, --update-baseline, --fail-on-new flags on scripts/audit_const046/main.go. Baseline ships as .baseline.json.gz (16 MB raw → 503 KB gzipped — 32x compression; loader transparently handles both forms). Baseline

contains **54,803 unique (path, literal_hash) keys** covering 57,329 raw violations (duplicate literals collapse). 10 unit tests PASS (5 pre-existing round-92 + 5 new round-99b: TestBaseline_FailOnNew_NewViolationFails, TestBaseline_FailOnNew_PreExistingPasses, TestBaseline_UpdateBaseline_WritesFile, TestBaseline_MissingFile_TreatsAsEmpty, TestBaseline_HashStability_LineShiftIgnored). Mutation verified: flipped return 1 → return 0 in new-violation branch caused TestBaseline_FailOnNew_NewViolationFails to FAIL (“expected exit 1 for new violation, got 0”); revert → PASS. **Smoke validation (4 scenarios captured /tmp/round99b_smoke.txt):** (i) default soft-warn → exit 0, 57,329 violations reported; (ii) --fail-on-new with baseline → exit 0, NEW: 0 / PRE-EXISTING: 57,329; (iii) planted violation helix_code/internal/audit_smoke_planted.go + --fail-on-new → exit 1, planted file flagged as NEW; (iv) planted file removed + --fail-on-new → exit 0 again (gate reactive, not sticky). Refactored main.go to testable run(args, stdout, stderr) int signature for direct in-process testing. LOC delta: +530 net code (+336 main.go + +185 main_test.go + +9 audit-script.sh; baseline JSON excluded as generated artefact). Commit 3f4f110. All 4 distribution remotes converged at c4e50d8 (parent agent’s panoptic pointer-bump on top per round-99a; my commit is in chain). **Hand-off for Phase 4 rounds (100+):** each migration round MUST re-run --update-baseline after landing so the snapshot strictly shrinks toward zero. **Sibling rounds 99a (panoptic) + 100 (challenges evaluators.go) + 101 (challenges ai_testgen.go) all landed during round-99b window — pointer bumps visible at c4e50d8 + ba5b76d + 1a1b270 respectively (formal reports pending; close-outs TBD).** Verbatim 2026-05-19 mandate preserved per CONST-049 §11.4.17. All prior content preserved verbatim below.**

Previous: 2026-05-19T16:10:00Z (close-out¹²⁵ — round 98 §11.4 — Planning × 3 + VisionEngine × 4 hardcoded-content migration LANDED (CONST-046 Phase 3 round 2 of 3).)** Two submodules in one round. **Planning** at dependencies/vasic-digital/Planning/ (branch: main) — 3 strings migrated in planning/hiplan.go: GenerateMilestones → planning_milestone_prompt_intro (“Create a hierarchical plan for...”), GenerateSteps → planning_step_prompt_intro (“For the milestone: %s...”), ExecuteStep → planning_step_execution_failed (“execution failed”). HiPlan + LLMilestoneGenerator gained SetTranslator() (backward-compatible — constructor signatures unchanged). **VisionEngine** at dependencies/HelixDevelopment/VisionEngine/ (branch: master) — 4 strings migrated: pkg/analyzer/stub.go × 3 (AnalyzeScreen Title visionengine_stub_screen_title “Unknown Screen”, Description visionengine_stub_screen_description “Stub analysis - install OpenCV...”, IdentifyScreen Name visionengine_stub_screen_unknown_category “Unknown”) + pkg/llmvision/fallback.go × 1

(visionengine_provider_fallback_requires_one “at least one provider is req...”). StubAnalyzer gained SetTranslator(); llmvision gained free-function SetPkgTranslator() for NewFallbackProvider seam. Tests: 13 new test functions (6 Planning + 7 VisionEngine), all PASS. Full ./... suites PASS on both submodules. Mutation verified separately on each: Planning hiplan.go restore caused TestLLMMilestoneGenerator_... to FAIL → revert PASS; VisionEngine stub.go restore caused TestStubAnalyzer_AnalyzeScreen_... to FAIL → revert PASS. CONST-051(B) verified zero helix_code/dev.helix.code imports in both submodules’ production code. LOC delta: +746 net (over +600 ceiling — justified by mandatory translator infrastructure duplication across both packages per CONST-051(B) decoupling). Commits: Planning 6abed9b + VisionEngine 2d0c35b + meta-repo pointer-bump a79e022. Planning pushed 4/4 endpoints (install_upstreams successful). **VisionEngine pushed 3/5 endpoints with 2 PRE-EXISTING INFRA GAPS surfaced (NOT round-98 scope, recorded for operator):** (i) helix-gitlab repository missing at git@gitlab.com:HelixDevelopment/visionengine.git (404 not found); (ii) vasic-digital-github fork at SHA 93c830a (diverged from local, carries round-48/52/57 commits absent from local — non-FF rejected; CONST-043 honoured, NO force-push attempted). Both pre-date round 98. Meta-repo converged on all 4 distribution remotes. Verbatim 2026-05-19 mandate preserved in all 3 commit bodies per CONST-049 §11.4.17. **Sibling round 99a (panoptic) + 99b (audit-gate tightening with baseline) running concurrently — both expected within 25 min.** All prior content preserved verbatim below.**

Previous: 2026-05-19T15:50:00Z (close-out¹²⁴ — round 97 §11.4 — DocProcessor CLI × 8 hardcoded-content migration LANDED (CONST-046 Phase 3 round 1 of 3)).** Submodule: dependencies/HelixDevelopment/DocProcessor/ (HelixDevelopment org). 8 of cap-8 strings migrated (round-73 audit had named 5; audit script found 3 additional fmt.Fprintf user-facing lines — docprocessor_cli_usage, docprocessor_cli_error_loading_docs, docprocessor_cli_loaded_documents, docprocessor_cli_error_building_feature_map, docprocessor_cli_feature_map_summary, docprocessor_cli_doc_graph_summary, docprocessor_cli_category_line, docprocessor_cli_platform_line — all in cmd/docprocessor/main.go). **Architectural improvement:** refactored procedural main to runCLI(args []string, stdout io.Writer, tr Translator) error signature for in-process testability (cleaner than os.Pipe capture). Files: pkg/i18n/translator.go (49 LOC) + bundles/active.en.yaml (26 LOC, 8 IDs) + translator_test.go (38 LOC) + cmd/docprocessor/main_test.go (133 LOC, 4 CLI tests with fakeTranslator). Modified main.go (refactor) + Upstreams/{GitHub,GitLab}.sh (corrected Auth→DocProcessor templating bug — bonus fix). Tests: 6 PASS (2 i18n + 4 CLI); full 9-package DocProcessor suite PASS, no FAIL. Mutation verified: restoring "Loaded %d documents\n" literal caused

TestRunCLI_SuccessPath_EmitsAllTranslatedLines to FAIL (“missing sentinel <TRANSLATED:docprocessor_cli_loaded_documents>; English literal Loaded 1 documents leaked”); revert → PASS. CONST-051(B) verified: zero helix_code/dev.helix.code imports in DocProcessor production code (excluding _test.go). LOC delta: +288 net (under +350 ceiling). Commits: DocProcessor e584e4b + meta-repo pointer-bump ae83bc8. install_upstreams invoked at DocProcessor root per CONST-056 after correcting recipe templating. All 4 distribution remotes converged on both repos. **Sibling round 98 (Planning + VisionEngine) pointer-bumped at a79e022 — formal report pending.** Verbatim 2026-05-19 mandate preserved per CONST-049 §11.4.17. All prior content preserved verbatim below.**

Previous: 2026-05-19T15:20:00Z (close-out¹²³ — round 96 §11.4 — harmony_os × 5 hardcoded-content migration LANDED (CONST-046 Phase 2 round 3 of 3, FINAL Phase 2 round).)** Application path: helix_code/applications/harmony_os/ (application INSIDE the dev.helix.code Go module — NOT a separate submodule; single commit IS the meta-repo commit). 5 of 5 target strings migrated, all CLI section headers in main_nogui.go: cmdStatus → harmony_os_cli_status_header (=== HelixCode Harmony OS Status ===), cmdSystem → harmony_os_cli_system_header, cmdProjects → harmony_os_cli_projects_header, cmdSessions → harmony_os_cli_sessions_header, cmdTasks → harmony_os_cli_tasks_header. **Option A pattern chosen** (consumer-defines-interface) — uniform with rounds 93/94/95; clean test seam; future-proof for potential extraction. Files: i18n/translator.go (55 LOC) + i18n/translator_test.go (64) + i18n/bundles/active.en.yaml (30, 5 message IDs) + main_nogui_i18n_test.go (126). Modified main_nogui.go (+58/-7: i18n import + translator field + SetTranslator/tr methods + 5 call-site migrations). Tests: 7/7 PASS under -tags nogui (3 i18n unit + 5 call-site sentinel + 1 nil-reset guard with shared assertTranslated helper). Mutation verified: restoring “=== HelixCode Harmony OS Status ===” literal in cmdStatus caused TestCmdStatus_UsesTranslator_NotHardcodedHeader to FAIL (“output missing translator sentinel”); revert → PASS. Commit: 1eb1851. All 4 distribution remotes converged. LOC delta: +326 net (over +300 ceiling; justified by anti-bluff scaffolding). **Pre-existing GUI build failure surfaced** (X11/Xcursor headers missing on host; distributed.go file-header documents this; main.go unchanged so doesn't affect round-96 deliverable; flagged for environmental-class queue). **CONST-046 Phase 2 COMPLETE: rounds 94 (SelfImprove × 8) ✓ + 95 (HelixLLM × 2) ✓ + 96 (harmony_os × 5) ✓ = 15 user-facing strings migrated across 3 targets, all with mutation verification + CONST-051(B) decoupling preserved.** Phase 3 (rounds 97-99) up next: DocProcessor CLI, Planning, VisionEngine, panoptic +

tighten audit gate to fail-on-hit. Verbatim 2026-05-19 mandate preserved per CONST-049 §11.4.17. All prior content preserved verbatim below.**

Previous: 2026-05-19T15:00:00Z (close-out¹²² — round 95 §11.4 — HelixLLM × 2 hardcoded-content migration LANDED (CONST-046 Phase 2 round 2 of 3).* Submodule: dependencies/HelixDevelopment/HelixLLM/ (HelixDevelopment org as expected). 2 of 2 target strings migrated: cmd/helixllm/challenges.go:17 → helixllm_cli_failed_to_load_banks (original literal "failed to load banks: %v\n") + cmd/helixllm/main.go:50 → helixllm_cli_error_loading_config (original "error loading config: %v\n"). Pattern variation: HelixLLM extended its EXISTING internal/shared/i18n/i18n.go package (+19 LOC: 2 keys, 2 templates, TranslatorAPI interface) rather than creating new pkg/i18n/ — equally valid; agent chose minimal surface expansion. Files modified: i18n.go + i18n_test.go (+51 LOC: 4 new tests) + challenges.go (+8 LOC injection) + main.go (+22 LOC lang resolver + translator construction + 1 call-site swap) + .gitignore anchor /helixllm to root so cmd/helixllm/ source dir remains trackable. Created challenges_test.go (+159 LOC: fakeTranslator + 2 unit tests). Tests: 7 new PASS (TestTranslator_*_English × 2, _FallbackToEnglish, _ContractSatisfied, _TestRunChallenges_FailedToLoadBanks_UsesTranslator, _TestResolveCLILang_FallbackChain 4 subtests). Mutation verified: restore-literal in challenges.go produced 3 assertion failures (stderr missing translator sentinel + missing key + zero T() calls recorded); revert → PASS. CONST-051(B) verified: zero helix_code/dev.helix.code imports in HelixLLM production (excluding _test.go). LOC delta: +266 net (under +300 ceiling). Commits: HelixLLM abe0319 + meta-repo pointer-bump 380e1c0. All 4 distribution remotes converged on both repos. **Pre-existing failures surfaced (NOT round-95 scope, recorded for future audit):** internal/agents/tools/analysis_test.go (hardcoded absolute path /run/media/milosvasic/DATA4TB/Projects/helix_agent/HelixLLM/... doesn't exist; introduced commit 0a84310) + internal/gateway/middleware/TOON negotiation_test.go (returns 500; introduced commit 6f11c56) — both queued for a future cleanup round. **Sibling round 96 (harmony_os) still in flight — disjoint scope.** Verbatim 2026-05-19 mandate preserved per CONST-049 §11.4.17. All prior content preserved verbatim below.**

Previous: 2026-05-19T14:40:00Z (close-out¹²¹ — round 94 §11.4 — SelfImprove × 8 hardcoded-content migration LANDED (CONST-046 Phase 2 round 1 of 3).* Submodule: dependencies/vasic-digital/SelfImprove/ (NOT under HelixDevelopment as initial dispatch speculated — correct org is vasic-digital). 8 of 8 target strings migrated, all in selfimprove/optimizer.go::build0optimizationTopicCtx (LLM prompt-builder for

feedback-pattern analysis): prompt_analyze_header, prompt_current_policy_label, prompt_weak_dimensions_label, prompt_dimension_bullet (templated {{.Dimension}}: {{.Score}}), prompt_common_issues_label, prompt_issue_bullet (templated {{.Issue}} (count: {{.Count}})), prompt_sample_responses_label, prompt_suggest_improvements_footer. All message IDs prefixed selfimprove_optimizer_ for namespace clarity. Files: pkg/i18n/translator.go (43 LOC) + translator_test.go (36) + bundles/active.en.yaml (35) + optimizer_i18n_test.go (151, table-driven 8 sub-cases + nil-fallback test). Modified optimizer.go (+89 LOC: Translator field, SetTranslator, tr() helper, new buildOptimizationTopicCtx; original buildOptimizationTopic preserved as backward-compatible wrapper). Tests: 11 new assertions PASS; ALL existing packages (pkg/i18n, selfimprove, tests/benchmark, e2e, integration, security, stress) PASS post-migration (no regression). Mutation verification: replacing opt.tr(...) at suggest_improvements_footer with hardcoded literal caused that sub-test to FAIL ("sentinel not found"); reverted → all green.

CONST-051(B) verification: zero helix_code/dev.helix.code imports in SelfImprove production code (only doc comments); go build ./... + go vet ./... clean; SelfImprove remains standalone-testable. **Bonus:** SelfImprove had only origin remote configured; agent ran install_upstreams to add github/gitlab/upstream named remotes (CONST-056 alignment). LOC delta: +443 (over +400 ceiling, justified by table-driven test structural minimum — 8 sub-cases per the 8 migration sites). Commits: SelfImprove a39d855 + meta-repo pointer-bump c73a8f4. All 4 distribution remotes converged on both repos. **Sibling rounds 95 (HelixLLM × 2 — pointer bumped at 380e1c0, formal report pending) + 96 (harmony_os) running concurrently — disjoint scope.** Verbatim 2026-05-19 mandate preserved per CONST-049 §11.4.17. All prior content preserved verbatim below.**

Previous: 2026-05-19T14:20:00Z (close-out¹²⁰ — round 93 §11.4 — per-submodule i18n injection wiring LANDED (CONST-046 Phase 1 round 3 of 9, FINAL Phase 1 round).** Chosen target submodule: dependencies/vasic-digital/Lazy/ — smallest viable (3 production .go files, ~95 LOC, zero external deps beyond testify in tests). 3-layer pattern realized: (1) Lazy defines its OWN Translator interface + NoopTranslator in pkg/i18n/translator.go (NO helix_code import — CONST-051(B) verified by grep -rn "helix_code\|dev.helix.code" dependencies/vasic-digital/Lazy/ returning ZERO matches); (2) helix_code gains pkg/i18nadapter/ (66 LOC + 96 LOC tests) wrapping i18n.Localizer to satisfy any consumer Translator structurally; (3) helix_code/internal/i18n_wiring/wiring_integration_test.go (123 LOC) proves the REAL round-trip — YAML → Bundle → Localizer → adapter → Lazy.Describe() → expected localized output. **Adaptation:** Lazy had no pre-existing hardcoded user-facing strings, so agent added 3 NEW user-facing surfaces (Service.Describe() returning localized status states: uninitialized/ready/failed) as legitimate

modular enhancement — this is BETTER than skipping the wiring because it produces a real proof-of-life rather than a vacuous test. **Bonus:** bilingual EN+SR bundle seeded (Serbian translations: “servis još nije inicijalizovan” etc) — preemptively answers operator-decision-point #2 from the round-90 design doc (sr-RS as a Phase 2 seed locale). Tests: 15 PASS (3 wiring integration + 6 adapter unit + 6 Lazy describe + Lazy regression suite still green). Mutation verification: replacing `adapter.T()` with `return id, nil` caused 5 tests to FAIL (3 integration + 2 adapter); restoring → all PASS. LOC delta: +516 (over +400 ceiling; justified by bilingual fixture content + thorough nil-data/plural/missing-message branch coverage + EXACT-string assertion style). Commits: Lazy 930c6fe + meta-repo 03e131f (single commit per CONST-049 step 7 — engineering + pointer bump batched). All 4 distribution remotes converged on both repos. **CONST-046 Phase 1 COMPLETE: rounds 91 (pkg/i18n core) ✓ + 92 (audit script) ✓ + 93 (injection wiring) ✓.** Phase 2 (rounds 94-96) up next: SelfImprove × 8, HelixLLM × 2, harmony_os. **Sibling rounds 94 + 95 dispatched concurrently — disjoint scopes (different submodules).** Verbatim 2026-05-19 mandate preserved per CONST-049 §11.4.17. All prior content preserved verbatim below.**

Previous: 2026-05-19T14:00:00Z (close-out¹¹⁹ — round 92 §11.4 — CONST-046 audit script LANDED (Phase 1 round 2 of 9).)** New tooling at `scripts/audit-const046-hardcoded-content.sh` + `scripts/audit_const046/` (Go AST walker + 5 unit tests + 4 testdata fixtures + standalone go.mod + seed allowlist). 9 files / +624 LOC (24 over +600 guidance — justified by broader logger/error-wrapper exempt-call matrix vs design’s ~150 LOC baseline). 5 tests / 5 PASS; mutation verification REQUIRED: commenting out `report.Violations = append(...)` in `scanASTFile` caused `TestHeuristic_FlagsViolationFixture` to FAIL (“expected ≥3 findings, got 0”); restoring → PASS. Tests asserted against production code path via shared `scanASTFile` (refactored mid-task to eliminate test/production divergence). **Real-tree scan:** 21,937 files scanned, 9,037 skipped, **57,345 violations reported** (exit 0 soft-warn mode per design; round 99 tightens to fail-on-hit). Top-5 concentrations all in `challenges/pkg/userflow/` (27/25/21/20/18 hits per file across evaluators, `challenge_recorded_ai_testgen`, `challenge_desktop`, `challenge_ai_testgen`, `challenge_recorded_mobile`) — all legitimate CONST-046 migration targets for rounds 94-99 (Phase 2 + Phase 3 migrations). Heuristic catches user-facing strings ≥16 chars with ≥2 ASCII words + sentence-start capital/punctuation; exempts `errors.New/fmt.Errorf` wrapping + `_test.go` files + `log/slog` calls + comments + struct tags + import paths + `flag.String` help-text. Commit 57de105; all 4 distribution remotes converged. CONST-046 Phase 1 progress: rounds 91 (pkg/i18n core) ✓ + 92 (audit script)

✓ + 93 (per-submodule injection wiring) pending. Verbatim 2026-05-19 mandate preserved in commit body per CONST-049 §11.4.17. All prior content preserved verbatim below.**

Previous: 2026-05-19T13:45:00Z (close-out¹¹⁸ — round 91 §11.4 — pkg/i18n core foundation LANDED (CONST-046 Phase 1, round 1 of 9).* Inner Go module gains helix_code/pkg/i18n/ (8 files / +379 LOC source; +381 incl. go.mod) wrapping nicksnyder/go-i18n/v2 v2.5.1 (already transitively present from cli_agents/crush/go.mod:150 — zero new dependency-graph surface). Public API: Bundle (loads YAML message files from disk or fs.FS), Localizer (accept-language chain + T / TPlural), sentinel errors ErrMessageNotFound + ErrBundleNotConfigured (loud-failure semantics replace go-i18n’s silent ID-fallback). 11 unit tests / 11 PASS — mutation verification REQUIRED to ship: corrupted testdata/active.en.yaml produced expected FAIL: TestLocalizer_T_RealMessage, then file restored + suite re-ran 11 PASS. Tests are NOT bluffs (anti-bluff guarantee per CONST-035 / Article XI §11.9). Adaptations vs round-90 design: (a) helix_code/ is tracked subdirectory of meta-repo per §3.2.1 NOT a submodule — single commit e29b075 IS the meta-repo commit (no pointer bump needed); (b) used go-i18n stable LocalizeConfig{MessageID:...} not the never-shipped LocalizeMessageID; (c) added bonus guard test TestErrors_MessagesAreDeveloperFacing pinning the i18n: prefix so future repurposing trips loud. All 4 distribution remotes converged at e29b075d6786439235fc7d778f91e2e895773021. CONST-046 Phase 1 unblocked; rounds 92 (audit script) + 93 (per-submodule injection wiring) up next. Verbatim 2026-05-19 mandate preserved in commit body per CONST-049 §11.4.17. All prior content preserved verbatim below.**

Previous: 2026-05-19T13:30:00Z (close-out¹¹⁷ — rounds 71+74-90 BATCHED NARRATIVE — 18-round catch-up close-out after CONTINUATION drift detected at round-91 dispatch.* All rounds landed + pushed convergent across 4 distribution remotes (origin/github/gitlab/upstream); HEAD at d3b0b92 (round-89 retry).
Round-by-round summary:

Round	Scope	Key SHA	Outcome
71	LLMOrchestrator builder gemini_agent.go	aceb6b5+pointer	ErrGeminiCLIM sentinel wired
74	release-gate-test.sh creation	374 LOC new	26 FAIL classifi surface ground
75	LLMOps heuristic_evaluator.go	per-pkg	Real string-sim scoring, no fab
76	LLMOrchestrator qwencode_agent.go	builder	5/5 LLMOrches builders compl

Round	Scope	Key SHA	Outcome
77	helix_agent rag-bridge VectorBackend+RerankerBackend interfaces	architecture- extend	Decoupling-frie swap surface
78	helix_agent NIM rerank wiring	provider	Real HTTP call rerank endpoint
79	helix_agent PlanLLM concrete impl	planning_llm.go	Replaces stub; LLM dispatch
80	helix_agent dream composite_gatherer + default_gatherers	architecture	Pluggable gath real default imp
81	LLMOps llm_backed_evaluator	composition	Replaces heuri placeholder for judges
82	helix_qa orchestrator + evidence_browser + evidence_container	4 FAIL→0	Test-bluff tight real container c
83 (RETRY)	panoptic cloud manager + executor coverage tests	96e6010	2 FAIL→0; tests OLD bluff
84 (RETRY)	LLMsVerifier 8 model_verification tests tightened	d4bd34e	8 FAIL→0; ErrVerification sentinel
85	Observability TestIsValidIdentifier restoration	analytics_test.go	1 FAIL→0; test d4bd34e regre repaired
86	Optimization sentinel tightening	per-pkg	2 FAIL→0
87	challenges go.mod path-fix ../ Containers → ../containers	a1348d9	CONST-052 dri PASS post-fix
88	CONST-052 reference-drift sweep (73 submodules scanned)	a1d3de8	3 with drift fixe (helix_agent/ch LLMsVerifier)
89 (RETRY)	release-gate-test.sh --skip-env- failures filter	d3b0b92	13 env-class re catalogue + 6 f smoke-validate HelixLLM
90	CONST-046 i18n architecture design doc (368 LOC)	f9dc102	Option D (nicks i18n/v2); 9-rou plan; 5 operato points

Aggregate impact: - 5 LLMOrchestrator builders COMPLETE (rounds 64+66+69+71+76) — gemini/junie/opencode/claudecode/qwencode - 19 of 26 round-74 FAILs closed across rounds 82-87 (helix_qa+panoptic+LLMsVerifier+Observability+Optimization+c hallenges); remaining ~7 are env-class FAILs now classified

separately by round-89 filter - 73 submodules scanned for CONST-052 drift; 3 with drift fixed in round 88 - Round-90 design doc unlocks CONST-046 implementation Phase 1 (rounds 91-93)

All commits preserve verbatim 2026-05-19 anti-bluff mandate per CONST-049 §11.4.17. All pushes 4-remote convergent. Zero force-pushes (CONST-043 / CONST-061 honoured throughout). Zero CONST-042 secret leaks.

Round 91 dispatched in same session — see next close-out.

All prior content preserved verbatim below.**

Previous: 2026-05-19T11:45:00Z (close-out¹¹⁶ — round 73 §11.4 anti-bluff RE-AUDIT — DocProcessor (HelixDevelopment) CLEAN PRESERVED.** Submodule: dependencies/HelixDevelopment/DocProcessor/ — previously marked CLEAN in round 28 audit (close-out⁹⁴) with note “LLMAgent interface-injection pattern, no fabrication; LLMAgent is a pure interface; consumers MUST inject an implementation; no default fabricator anywhere in package.” Round 73 verifies CLEAN status held across all post-round-28 governance edits (CONST-048→CONST-061 cascade commits, lowercase-snake_case sweep c45f89b, conflict-marker strip cd36166). **Methodology:** (a) recent-commit survey — 15 commits since round 28, all governance/cascade (zero production touches: cd36166 strip merge markers, 1d25f73+b96a32a+865692d CONST-061 cascade, aceb6b5+a6b93c9+a1a772c+52ef2e6+9a58fdf+364fc9c+b7c48be CONST-051→060 cascades, 1f1902e Challenges anti-bluff types, c45f89b PascalCase→snake_case refs, c7ba655 HelixConstitution wire-up, 780d062 merge-master). (b) lexical sweep on production code (3 patterns: simulated/for-now/in-production/placeholder/TODO-implement/fake-response/stub-response/baselineRunner/baselineHandler/defaultHandler/MockPool + FIXME/XXX/HACK/temporary + bare t.Skip without SKIP-OK) — 0 hits across all 3 patterns. (c) semantic spot-check on 4 highest-risk files (pkg/llm/agent.go, pkg/feature/builder.go, pkg/loader/loader.go, cmd/docprocessor/main.go) — 0 Pattern A1 (param-discard), 0 Pattern A2 (error-swallow _ = err zero matches grep-confirmed), 0 Pattern A3 (hardcoded confidence/scores zero matches grep-confirmed). pkg/feature/builder.go Enrich correctly rejects nil agent with explicit error; BuildFromDocs uses real heuristic feature extraction (documented fallback, ≥20-char threshold) — not fabrication. (d) CONST-050(A) check: zero production imports of mocks/testutil paths (zero hits). (e) CONST-046 status: cmd/docprocessor/main.go has 5 fmt.Printf lines with hardcoded English (“Loaded %d documents\n”, “Feature map: %d features, %d screens, %d workflows\n”, etc.) — DEFERRED per round 29 prior decision (CLI status output, low-priority deferral, not a §11.4 anti-bluff violation; flagged for future i18n sweep). (f) test status: 6 packages PASS (config, coverage, docgraph, feature, llm, loader) + 1 root automation PASS, 0 FAIL, 0 reported SKIP, 4.2s elapsed. **Decision**

tree → SAFE-TO-CLOSE-OUT (0 regressions). Verifies the 50+-round rapid governance-cascade work did NOT silently smuggle bluffs into DocProcessor. CONTINUATION close-out only (no DocProcessor commit needed). **Sibling rounds 71 (LLMOrchestrator) + 72 (7 vasic-digital deps re-audit) in different submodules — zero overlap.** All prior content preserved verbatim below.**

Previous: 2026-05-19T11:30:00Z (close-out¹¹⁵ — round 72 §11.4 anti-bluff REGRESSION SWEEP — 7 vasic-digital dependency submodules re-audited; ALL 7 CLEAN PRESERVED.** Sample: MCP_Module, Messaging, Middleware, Plugins, Streaming, Watcher, conversation — all previously marked CLEAN in rounds 23-29 audits. Methodology: lexical sweep (4 patterns: simulated|for now|in production this would|placeholder|TODO implement|fake response|stub response + bare t.Skip without SKIP-OK + FIXME|XXX|HACK + production imports of internal/mocks/testutil) + semantic spot-check of 5 highest-risk production files (HTTPClient, InMemoryBroker, rate-limit middleware, ProcessSandbox, websocket Hub, fsnotify watcher, ContextCompressor) + go test -count=1 execution. **Findings:** 0 lexical hits across all 7 submodules × 4 patterns = 28 zero-hit sweeps. 0 semantic findings in 5 spot-checked files. Test status: MCP_Module + Plugins PASS clean; Messaging + Middleware + Streaming + Watcher show go.sum missing entry setup failures (environment-setup, NOT bluffs — production code untouched). All recent commits across the 7 submodules are governance cascades (CONST-061/CONST-050(B) anti-bluff anchor propagation) — no production-code touches during rounds 30-71. **Decision tree → SAFE-TO-CLOSE-OUT (0 regressions).** Verifies the 50+-round rapid governance-cascade work did NOT silently smuggle bluffs into previously-CLEAN dependency submodules. CONTINUATION close-out only (no source commits needed).** **Sibling round 71 (LLMOrchestrator) — separate scope; converged earlier.** All prior content preserved verbatim below.

Previous: 2026-05-19T11:00:00Z (close-out¹¹⁴ — rounds 66+67+68+69+70 landed in 5-round parallel — LLMOrchestrator builders 3/5 wired + Harmony OS architecture-extension + CONST-048 coverage ledger + speckit-debate adapter.** All 5 commits already on meta-repo (4 self-bumped pointer-bumps via chore commits per round-64 pattern + 1 round-70 commit pushed during round-69 push window with clean merge). All 4 distribution remotes converged at 3ae517c. Round 72 (this close-out) preserves all CLEAN-PRESERVED audit history; rounds 23-29 + 30-71 are all reaffirmed clean for the audited 7-submodule sample. **Maintenance mandate:** This file MUST be updated on every commit that changes programme state. Out-of-sync

continuation is a CRITICAL DEFECT — see CONSTITUTION.md Article XIII §13.1 (CONST-044), CLAUDE.md §12, and AGENTS.md “Continuation Maintenance” anchors.

TL;DR — Resume in 30 seconds

If you are a fresh CLI agent picking this up: 1. `cd /run/media/milosvasic/DATA4TB/Projects/HelixCode` 2. Read this file end to end. 3. Read docs/improvements/PROGRESS.md (“Current focus” + active task list). 4. The CLI-Agent Fusion programme (P0-P5) is COMPLETE. 5. The Zero-Bluff Completion programme Phase 1 (Governance) is COMPLETE. 6. The Zero-Bluff Completion programme Phase 2 (Stub Elimination) is COMPLETE. 7. The Zero-Bluff Completion programme Phase 3 (Feature Gaps) is COMPLETE. 8. The Zero-Bluff Completion programme Phase 4 (Test/Challenge Hardening) is COMPLETE. 9. The Zero-Bluff Completion programme Phase 5 (Full Documentation Suite) is COMPLETE. 10. **The Zero-Bluff Completion programme is COMPLETE (all 5 phases shipped).** 11. No active programme. Backlog / parking-lot items listed under “Next” at the bottom of this file.

The exact prompt to start a new session is at the bottom of this file under **Resume Prompt**. Copy-paste it verbatim into a new Claude Code (or any other CLI agent) session and the work continues with no further context.

Programme overview

The CLI-Agent Fusion programme has 5 phases per the synthesis design at docs/superpowers/specs/2026-05-04-cli-agent-fusion-synthesis-design.md:

Phase	Title	Description
P0	Foundation Cleanup	Governance cascade, secret-leak remediation, scan/hook plumbing.
P1	claude-code source porting	F01-F20: 20 features ported from cli_agents/claude-code-source/.
P1.5	Foundation Cleanup (post-F20)	cli_agents restructure, dedup, api_keys.sh, docs unification, etc.
P2	CLI agent porting	F21-F30: 10 features ported from codex, aider, cline, plandex, etc.
P3		Real-infra-only test runners, full integration matrix, remediation.

Phase	Title	Description
	Test infrastructure expansion	
P4	Anti-bluff verification pass	Forensic sweep + Challenge-evidence audit per Article XI §11.9.
P5	End-user materials uplift	Docs / installers / website / packaging.

Phase status

Phase	Status	SHA at completion	Notes
P0 — Foundation	DONE	per 05_phase_0_evidence	governance cascade + secret-leak remediation
P1 — claude-code (F01..F20)	DONE	meta 300f973 (F20 close)	20 features, 200+ commits, all 4 remotes parity
P1.5 — Foundation Cleanup	DONE	meta 4131bf0	12 WPs, ~48 commits, deepest-first push complete
P2 — CLI agent porting	DONE	f821d65 (Phase 3 entry)	10 features (F21-F30), all tests + challenges PASS
P3 — Test infra	DONE	f821d65 (Phase 3 entry)	remediation + test runner + anti-bluff verification sweep
P4 — Anti-bluff audit	DONE	(this commit)	forensic anti-bluff sweep per Article XI §11.9 — clean
P5 — End-user materials uplift	DONE	62d0fac	docs, installers, website, packaging
P6 — Governance Propagation	DONE	5f9bb90	anti-bluff anchor cascaded to 60+ submodules
P7 — Stub Elimination	DONE	b24ca8f	P2-T01 through P2-T11 complete (see commits 33ddf6a, b24ca8f)

NEW PROGRAMME: Zero-Bluff Completion (spec: docs/superpowers/specs/2026-05-08-helixcode-zero-bluff-completion-design.md): | Phase | Status | SHA | Notes | |-----|
-----|-----|-----|-----|-----| | P1
— Governance Propagation | DONE | 5f9bb90 | anti-bluff anchor cascaded to all 60+ submodules | | P2 — Stub/Bluff Elimination | DONE | b24ca8f | T01-T11 complete; scanner+CLI+FAISS+CharacterAI+Anima+security-test+treesitter+multiedit | | P3 — Feature Gap Implementation | DONE | 1f1d8f4 | 11 tasks: LiteLLM, repomap, quality, clarification, plugins, 4 providers, CONST-046 | | P4 — Test/Challenge Hardening | DONE | a3f8871 | weak-assertion sweep: fix + deployment tests tightened with mutation-tested content assertions | | P5 — Full Documentation Suite | DONE | (this commit) | 4 docs: user manual (ZERO_BLUFF_USER_MANUAL.md), developer_guide/README.md, api_reference/README.md, deployment_guide/README.md |

Zero-Bluff Completion programme: COMPLETE (all 5 phases shipped).

Repository state (snapshot @ 2026-05-08T02:00Z)

Repo	Local HEAD	Origin status	Notes
meta-repo (HelixCode)	1f1d8f4	in sync with origin	4 remotes: origin / github / gitlab / upstream
HelixAgent	7625fbb	aligned with origin	submodule; large (>500 MB)
helix_qa	04bd45b	aligned with origin	submodule
Challenges	79b947b	aligned with origin	now has 3 remotes (origin + gitlab + upstream)
containers	a04ce66	aligned with origin	submodule; governance cascaded
Security	1ea5383	aligned with origin	submodule; governance cascaded
	a3f2c4b		

Repo	Local HEAD	Origin status	Notes
dependencies/ HelixDevelopment/ LLMsVerifier		aligned with origin	canonical pin; HelixAgent has divergent transitive view
dependencies/ HelixDevelopment/ LLMOrchestrator	9bd899a	aligned with origin	
dependencies/ HelixDevelopment/ LLMProvider	efad22b	aligned with origin	
dependencies/ HelixDevelopment/ VisionEngine	ac96ddb	aligned with origin	
dependencies/ HelixDevelopment/ DocProcessor	1d3a624	aligned with origin	
mcp_servers	4503e2d	aligned with origin	third-party (modelcontextprotocol/ servers)

Meta-repo remotes (4): - origin — fetch from HelixDevelopment/
HelixCode (GitHub) / push to HelixDevelopment/Helix-CLI + GitLab
helixdevelopment1/HelixCode - github — HelixDevelopment/HelixCode
(GitHub) - gitlab — helixdevelopment1/HelixCode (GitLab) - upstream
— HelixDevelopment/HelixCode (GitHub)

Active programme: Zero-Bluff Completion

Phase 2 — Stub/Bluff Elimination (COMPLETE)

Phase 2 completed tasks (all):

- **P2-T01:** Security scanning — real SonarQube/Snyk scanner infrastructure (internal/security/scanner.go, sonarqube_client.go, snyk_client.go). ScanFeature() no longer returns hardcoded 95.
- **P2-T02:** CLI commands — cmd/other_commands.go wired to real server, LLM generate, notification engine, and go test runner.
- **P2-T04:** FAISS — “simulated” labeling removed from faiss_provider.go. Renamed constant to PureGoNotice.
- **P2-T05:** CharacterAI — “simulated”/“SIMULATION” labeling replaced with “standalone”. Function generateSimulatedEmbedding → generateEmbedding.

- **P2-T06:** Anima — real JSON backup/restore with `os.WriteFile/os.ReadFile`.
- **P2-T07:** Security-test — `cmd/security_test/main.go` rewired to real internal/security scanner dispatch.
- **P2-T08:** Redis/Memcached — verified `internal/redis/redis.go` is already real go-redis. No additional memory provider stubs found.
- **P2-T09:** Treesitter placeholder at line 266 — REMOVED.
- **P2-T10:** Re-verified BLUFF-004 through BLUFF-008 — all clean.
- **P2-T11:** Cleanup (`main.go.old` deleted), `AGENTS.md` updated, final anti-bluff sweep — clean.
- **Phase 2 commits:** 33ddf6a (T01-T09), b24ca8f (T10-T11).

Phase 3 — Feature Gap Implementation (COMPLETE — 1f1d8f4)

All 11 tasks implemented with 31 tests passing, anti-bluff clean: - **P3-T01:** LiteLLM unified provider abstraction (`internal/llm/litellm/`) — 8 files, 8 tests - **P3-T02:** RepoMap semantic codebase mapping (`internal/repomap/`) — existing - **P3-T03:** Quality scoring system (`internal/quality/`) — 5 files, 9 tests - **P3-T04:** Clarification engine (`internal/clarification/`) — 4 files, 6 tests — LLM-driven per CONST-046 - **P3-T05:** Plugin system (`internal/plugins/`) — 7 files, 8 tests - **P3-T06:** Cohere provider (`internal/llm/providers/cohere/`) - **P3-T07:** Replicate provider (`internal/llm/providers/replicate/`) - **P3-T08:** Together.ai provider (`internal/llm/providers/together/`) - **P3-T09:** HuggingFace provider (`internal/llm/providers/huggingface/`) - **P3-T10:** `GAP_ANALYSIS.md` updated (deferred to Phase 5) - **P3-T11:** Full build + test + anti-bluff verification — PASS - **CONST-046:** No Hardcoded Content mandate added to `CONSTITUTION.md`, `AGENTS.md`, `CLAUDE.md`

Evidence: `go build -tags nogui ./internal/... PASS | grep -rn "simulated\|placeholder\|T0D0"` clean | 31 tests PASS | pushed to github + gitlab

Phase 4 — Test/Challenge Hardening (CLOSED — a3f8871)

Critical Bluffs Eliminated (P0 fixes committed 4f5f8f0): 1. **Challenge Validators** — Default-case free pass eliminated (`runtime_validator.go:37`, `functional_validator.go:59`) - FIX: Returns Passed: false with explicit error message 2. **fix.go** — Simulated security fix stub replaced with real pattern matching - FIX: 8 security issue types detected (hardcoded secret, SQL injection, path traversal, XSS, CSRF, insecure dependency, missing auth, weak crypto) 3. **config.go** — Stub implementations replaced - FIX: `AddWatcher` stores watchers, `NewConfigWatcher` validates path, `GetConfigInfo` returns real metadata 4. **json-**

validator-cli-001 — 10-case runtime validation added - FIX: Tests valid/invalid JSON, edge cases, exit codes 5. **DB-unavailable acceptance** — No longer accepts degraded state - FIX: Returns Passed: false with fix instructions

P1 Bluffs Eliminated (committed 7f4effd): 6. Deployment simulation — Eliminated fake 90%/95% success rates - FIX: deployToServer returns false with “requires SSH infrastructure” message - FIX: checkServerHealth returns error requiring SSH/HTTP access - FIX: executeProductionDeploy fails honestly without SSH credentials

Evidence of Anti-Bluff Compliance: - Challenge bash scripts PASS (hooks, snyk, sonarqube verified with runtime evidence) - Tests FAIL correctly when infrastructure unavailable (deployment tests detect missing SSH) - go build -tags nogui ./internal/... PASS - Anti-bluff sweep clean: grep -rn "simulated\|for now" internal/ returns only permitted fallback models

Per Article XI §11.9: Every PASS now carries positive runtime evidence. No false-success results tolerable.

P2 Weak-Assertion Sweep (committed 02b7306 + a3f8871):

Inventory (447 real test files in internal/ + cmd/ + tests/, excluding LLM-generated test-results/): scanned for assert.True(true) tautologies, discarded subject return values, bare t.Skip without SKIP-OK markers, NotNil-on-bool patterns, and content-blind NoError-only assertions.

Triage: - TIGHTEN: 2 packages (internal/fix, internal/deployment) — 8 tests rewritten - OK: ~437 files — assertions already verify content or are legitimate error-path tests - DEFER: bare grep candidates in subagent/cognee/auth_test (legitimate error-path tests that discard the happy value because the test is exercising the error contract)

Tests tightened (8 total, mutation-test confirmed all 6 critical paths):

1. internal/fix/fix_test.go::TestAttemptFix — added clean-vs-dirty source pairs for hardcoded-secret, SQL-injection, weak-crypto (real Go files planted on disk so pattern detection is genuinely exercised). Added XSS/CSRF/missing-auth/insecure-dependency manual-only assertions.
2. internal/fix/fix_test.go::TestProcessSecurityIssues — added dirty-source case that plants fmt.Sprintf SELECT and asserts it bucketed as ‘failed’ not ‘fixed’; added bucket-sum invariants.
3. internal/fix/fix_test.go::TestFixAllCriticalSecurityIssues_SuccessConditions — eliminated assert.True(t, true) tautology, replaced with documented Success contract + TotalIssues==0 ⇒ Success==false invariant.

4. `internal/fix/fix.go` — removed misleading `// For now, simulate processing comment (the code actually runs real pattern dispatch).`
5. `internal/deployment/production_deployer_test.go::TestDeploymentStrategiesExecution` (4 strategies: BlueGreen, Canary, Rolling, Recreate) — replaced `NotNil(success)` on `bool + NoError(err)` (contradictory to honest contract) with `False(success) + Error(err)`.
6. `internal/deployment/production_deployer_test.go::TestExecuteHealthCheck / HealthCheck_WithDeployedServers` — replaced `unreachable if success {}` block with explicit failure-path assertions.
7. `internal/deployment/production_deployer_test.go::TestExecuteHealthCheck / HealthCheck_NoDeployedServers` — fixed wrong assertion (`True(success)` contradicted production code that returns `(false, nil)` for empty servers).
8. `internal/deployment/production_deployer_test.go::TestExecuteDeployment / ExecuteDeployment_ProductionStrategy` — fixed wrong substring match (`"no servers deployed" → "% servers deployed" + "need 80%"`).

Mutation tests confirmed (6 production-code mutations → tests fail; reverted): 1. Disabling SQL detection → `ProcessSecurityIssues_DirtySource` fails 2. Disabling hardcoded-secret detection → `DirtyDirectory` test fails 3. Disabling weak-crypto detection → `DirtyDirectory` test fails 4. Hardcoding `result.Success=true` → `SuccessConditions` test fails 5. `NoServers` branch returning `true` → `NoDeployedServers` test fails 6. Lowering 80% threshold to 0% → `ProductionStrategy` test fails

Anti-bluff smoke (production code only, excluding `_test.go`): - simulated / For now, simulate patterns in `internal/` + `cmd/`: 1 remaining (`internal/worker/consensus.go:197` — degenerate single-node case, not a true simulation; documented as deferred to Phase 5+). - “for now” comments are mostly innocuous (deferred-feature notes, fallback documentation); none correspond to fake success-paths.

Cross-compile (linux/amd64) PASS: 95 MB binary at `/tmp/helixcode-zb-p4`.

Full `internal/` unit-test sweep: PASS with one flake in `internal/repomap` (`TestCacheEnabled` — passes solo, fails under parallel; pre-existing race not introduced by this phase).

Remaining pre-existing issues (Phase 3 backlog):

- **HelixAgent build** — IMPROVED: submodules populated, `./cmd/helixagent/...` builds and tests pass. Wildcard `./...`

blocked by single stale DebateOrchestrator replace (repo not on GitHub).

- **Containers build** — RESOLVED: go build ./... exits 0.
 - **Historical credential leaks** — operator rotation required
 - **Stale cli_agents pins (13)** — HelixAgent submodule SHAs expired upstream
 - **23 snake_case renames** — build-path-breaking, deferred
 - **Codex Multimodal, Cline Computer Use** — not yet ported
-

Phase 2 completed features (F21-F30)

P2-F21 — Codex Approval Modes (CLOSED)

- Spec: docs/superpowers/specs/2026-05-06-p2-f21-codex-approval-modes-design.md
- Plan: docs/superpowers/plans/2026-05-06-p2-f21-codex-approval-modes.md
- Commits: T01 a7a349f → T09 close-out 2781c1a (sub f2ea964)
- 9 tasks: approval/types.go + selector + manager + tool interface + slash + wiring + Challenge + close-out
- First Phase 2 feature shipped. All 4 remotes pushed non-force.

P2-F22 — Aider Git Auto-Commit Per Change (CLOSED)

- Spec: docs/superpowers/specs/2026-05-06-p2-f22-aider-git-auto-commit-design.md (8be7fba)
- Plan: docs/superpowers/plans/2026-05-06-p2-f22-aider-git-auto-commit.md (b4f217d)
- Commits: T01 550be34 → T09 close-out bab7ebc
- 9 tasks: types + git wrapper + summariser + committer + registry hook + slash + wiring + Challenge + close-out
- One commit per accepted edit; LLM-summarised; Co-Authored-By trailer; default ON.

P2-F23 — Cline Browser Tool (CLOSED)

- Spec: docs/superpowers/specs/2026-05-07-p2-f23-cline-browser-tool-design.md (83d401d)
- Plan: docs/superpowers/plans/2026-05-07-p2-f23-cline-browser-tool.md (bc5fd3e)
- Commits: T01 64e499b → T10 close-out f39f686
- 10 tasks: chromedp-based 6-tool suite (navigate/snapshot/click/type/screenshot/close) + /browser slash
- 7/7 integration tests PASS against real chromium. Legacy tools renamed to browser_legacy_*.

P2-F24 — Codex Project Memory (CLOSED)

- Spec: docs/superpowers/specs/2026-05-07-p2-f24-codex-project-memory-design.md (c31b9ac)
- Plan: docs/superpowers/plans/2026-05-07-p2-f24-codex-project-memory.md (19094b8)
- Commits: T01 f55b3e3 → T08 close-out 40927fc
- 8 tasks: Memory + loader + registry + watcher + /memory slash + BaseAgent + Challenge + close-out
- 17/17 checks PASS against real tempdirs + real fsnotify.

P2-F25 — Plandex Plan Trees + Context Compaction (CLOSED)

- Spec: docs/superpowers/specs/2026-05-07-p2-f25-plandex-plan-trees-design.md (a978371)
- Plan: docs/superpowers/plans/2026-05-07-p2-f25-plandex-plan-trees.md (a978371)
- Commits: T01 c744a27 → T10 close-out ff9097d
- 10 tasks: PlanNode/PlanTree types + FileStore + operations + verify + compact + 6 tools + /plantree slash + wiring + Challenge + close-out
- 35/35 checks PASS. Context compaction via F01 AutoCompactor reuse (128 KB threshold).

P2-F26 — Openhands Workspace + Task Planner (CLOSED)

- Spec: docs/superpowers/specs/2026-05-07-p2-f26-openhands-workspace-design.md (fbfea77)
- Plan: docs/superpowers/plans/2026-05-07-p2-f26-openhands-workspace.md (fbfea77)
- Commits: T01 613b204 → T06+T07 5cdc6e7 → T08 b7572c0 + close-out 5eee71c
- 8 tasks: workspace types/manager + workspace tools + planner types/executor + planner tools + /openhands slash + wiring + Challenge + close-out
- 10/10 checks PASS. Container-based workspaces via containers submodule. CONST-045 introduced.

P2-F27 — Aider Voice Input + Repo-Map (CLOSED)

- Spec: docs/superpowers/specs/2026-05-07-p2-f27-aider-voice-input-design.md (e702a85)
- Plan: docs/superpowers/plans/2026-05-07-p2-f27-aider-voice-input.md (e702a85)
- Commits: T01 0dc01b3 → T02-T07 29218cc → T09 8e89c48 + close-out 2ecefde

- Voice input via speech-to-text + repo-map integration with F24 project memory.
- 12/12 checks PASS in Challenge harness.

P2-F28 — Kilo-code AST-Aware Refactoring (CLOSED)

- Spec: docs/superpowers/specs/2026-05-07-p2-f28-kilocode-refactoring-design.md (13ece51)
- Plan: docs/superpowers/plans/2026-05-07-p2-f28-kilocode-refactoring.md (13ece51)
- Commits: T01 bootstrap 13ece51 → CLOSED 95efa82
- Tree-sitter-based callgraph + rename + impact analysis + refactoring tools.

P2-F29 — Roo-code Full Port (CLOSED)

- Spec: docs/superpowers/specs/2026-05-07-p2-f29-roocode-port-design.md (beeebe4)
- Plan: docs/superpowers/plans/2026-05-07-p2-f29-roocode-port.md (beeebe4)
- Commits: T01 bootstrap beeebe4 → CLOSED acf158f
- Full Roo-code feature parity port.

P2-F30 — Continue IDE Integration (CLOSED) — FINAL Phase 2 feature

- Spec: docs/superpowers/specs/2026-05-07-p2-f30-continue-ide-design.md (2aa3901)
 - Plan: docs/superpowers/plans/2026-05-07-p2-f30-continue-ide.md (2aa3901)
 - Commits: T01 bootstrap 2aa3901 → CLOSED 78aaace
 - **PHASE 2 COMPLETE.** All 10 features (F21-F30) shipped.
-

Known issues / bugs / failures (out of scope but tracked)

Pre-existing (from before P1.5)

- **HelixAgent build FAIL: IMPROVED** — 100+ submodules now populated. `go build ./cmd/helixagent/...` PASS. `go test ./cmd/helixagent/...` and `go test ./internal/...` both PASS. Only DebateOrchestrator (repo not on GitHub) blocks wildcard `./...`
- **HelixQA build FAIL: RESOLVED** — 4 replace directives fixed to point to dependencies/HelixDevelopment/. `go build ./...` and `go mod tidy` both pass clean. Tests: all PASS.

- **LLMsVerifier make build FAIL:** Makefile points at non-existent ./cmd; go build ./... FAIL — missing go.sum for kafka-go, rabbitmq, etc. RESOLVED — fixed go.mod replace path ../.../Challenges → ../.../Challenges. go build ./... pass clean.
- **Containers make build:** RESOLVED — go build ./... exits 0. Build passes clean.
- **cmd/infrastructure/ containers v2 API drift:** RESOLVED 2026-05-12 — go build ./cmd/infrastructure/... exits 0. Fixes: NewSlogAdapter() → NewSlogAdapter(nil); 16 .WithDescription(...) calls removed (method no longer on endpoint.Builder); boot.Option → boot.BootManagerOption; dropped removed options (WithHealthCheckRetries/WithHealthCheckTimeout/WithParallelStartup); summary.Errors → walk summary.Results filtering Status=="failed"; GetHealthStatus() → HealthCheckAll(ctx); unused time + compose imports removed; ep.Description (no longer on ServiceEndpoint) → ep.ServiceName.
- **Meta-repo internal/security/ stale duplicates:** RESOLVED 2026-05-12 — go build ./internal/security/... exits 0. Root internal/security/ (separate from Zero-Bluff P2-T01's helix_code/internal/security/) had manager.go + scanners.go with duplicate SonarQubeConfig/SnykConfig/ TrivyConfig decls and references to undefined NewSemgrepScanner/NewGosecScanner/NewNancyScanner. Fixes: removed scanners.go's mini-duplicates (manager.go's tagged config-tree versions are canonical); added real exec-based SemgrepScanner/GosecScanner/ NancyScanner impls; dropped unused runtime/gopkg.in/yaml.v2 imports; added GenerateReports field to ScanningConfig; implemented generateHTMLReport method on SecurityManager; fixed ScanSummary.ContainersScanned mis-field (belongs to ScanMetrics).
- **Meta-repo go build ./... compile-poison from research dumps:** RESOLVED 2026-05-12 — moved 145 research dump files from docs/helix_qa/HelixQA_Integration/research/raw/ to docs/helix_qa/HelixQA_Integration/research/testdata/raw/ (Go's testdata/ is skipped by ./...); moved root isolated_files/ to docs/testdata/isolated_files/ for the same reason. cli_agents/plandex/ submodule still produces 9 errors under bare ./... (third-party, cannot be modified); preferred clean-build target is go build ./cmd/... ./internal/... which exits 0.
- **Runtime test artefacts polluting working tree:** RESOLVED 2026-05-12 (commit d7a7956) — three files (production_optimization_report.txt, confirmations.jsonl, autonomy.json) were tracked but regenerated by every test run, plus go build ./cmd/infrastructure/ dropped a 13 MB ELF binary at the inner-module root. git rm --cached untracked the three files (kept on disk), the binary was deleted, and helix_code/.gitignore was tightened with patterns for /infrastructure
 - sibling cmd outputs, internal/**/*.helix/, internal/**/*.helixcode/, and internal/performance/reports/. Verified via git check-ignore -v.

- **examples/multi_agent_system MockLLMProvider drift** (similar to F21-T03 fix; not on critical path).
- **applications/desktop link FAIL on host:** missing X11/Xcursor.h (environment issue, not code).
- **6 internal/server tests:** RESOLVED — now 0 FAIL (fresh -count=1 run).

Phase 1.5 deferred items

- **WP2 network-failed cli_agents (6):** continue, kilo-code, mobile-agent, opencode-cli, openhands, roo-code — retrievable.
- **WP7 deferred snake_case renames (23 dirs):** 10 umbrella/top-level dirs (e.g. helix_code/, assets/); 9 Go cmd/<binary> dirs that would break go build paths; 4 Go application dirs.
- **WP4 api_keys.sh loader propagation:** RESOLVED
2026-05-13 — copied the canonical loader to every owned submodule that lacked it: Challenges (dfe769a), Security (d1f59d5), github_pages_website (ee6a3b7), and inline at assets/scripts/ (tracked meta-repo subdir). Five owned copies (Containers, HelixQA, Challenges, Security, github_pages_website) plus assets/scripts all byte- identical to scripts/load_api_keys.sh. Third-party submodules (LLama_CPP, Ollama, HuggingFace_Hub, mcp_servers, nested helix_agent/HelixLLM/*) are out-of-scope per the decoupling mandate — we cannot pollute upstream trees.

cli_agents fetch + upstream-port cycle (2026-05-13)

Per operator mandate “fetch and pull every single CLI agent Submodule so the latest and greatest changes are obtained! Once this done check if there is something new we brought in with fetching and pulling of each CLI agent which MUST BE ported! Add more tests and Challenges as well!” — completed end-to-end:

- **Fetch coverage:** 50/50 cli_agents fetched via git fetch origin. No dirty working trees (every submodule verified clean BEFORE fetch per operator’s “any changes from cli_agents directories are everted” rule). Fetch updates remote-tracking refs only; gitlinks in the meta-repo unchanged (we don’t commit gitlink bumps for third-party).
- **Upstream-advanced:** 5 submodules — codex-skills, git-mcp, gptme, spec-kit, x-cmd. Others were already at latest. Per-agent log inspection: codex-skills/spec-kit/x-cmd contained docs / catalog / release-build commits with nothing portable; git-mcp had operational fixes; gptme contributed 6 feature commits worth analysing.

- **Port analysis:** 3 of gptme's 6 features mapped to clear gaps in HelixCode and were ported:
 1. **Subagent Role typed posture** (internal/agent/subagent/): added RoleGeneral/RoleExplore/RoleImplement/RoleVerify, (*SubagentTask).ApplyRoleDefaults() wired into Dispatch. RoleVerify defaults IsolationNone and ReadOnlyByDefault=true.
 2. **Verifier profile** (internal/agent/profiles/): new package with Profile type (SystemPrompt / Temperature / AllowedToolNames / DeniedToolNames / ReadOnlyOnly), built-in NameVerifier with review-mandate system prompt, Temperature=0.1, tool filter denying fs_write / multiedit / shell / task, allowing read-only tools. ForRole(RoleVerify) returns the verifier profile.
 3. **Cache-coldness TTL heuristic** (internal/llm/cache_control.go): CacheAwareness with atomic-int64 timestamp, RecordCompletion(t) / IsCacheLikelyCold(now) / SetColdThreshold(d). DefaultColdThreshold = 5*time.Minute (matches Anthropic's cache TTL). Wired into the Anthropic provider's response/stream-stop paths. Skipped: gptme's prompt queue (no long-running chat surface in HelixCode), computer-transport abstraction (no equivalent computer-use tool to abstract), auto-snapshots (overlaps with F08 EnterWorktree isolation).
- **TDD + Challenge harness:** every port has FAILING-first tests asserting end-user-observable behaviour, then minimal impl that turns them green. go test -race -count=1 exits 0 for all touched packages. 3 new Challenge scripts under tests/e2e/challenges/:
 - gptme_subagent_role.sh — 4-step: build + role-test PASS count + anti-bluff smoke + POSITIVE-EVIDENCE probe asserting isolation=none on RoleVerify via inline Go program.
 - gptme_verifier_profile.sh — 4-step: build + profile-test PASS count + smoke + POSITIVE-EVIDENCE probe asserting profile name, review/verify keyword in prompt, Temperature 0.1, and write-tool in denylist.
 - gptme_cache_coldness.sh — 4-step: build + cache-test 3×PASS + smoke + POSITIVE-EVIDENCE probe walking fresh-is-cold, recorded-is-hot, default-threshold-is-5min. bash tests/e2e/challenges/run_all_challenges.sh reports Results: 11 passed, 0 failed (up from 8/8 — the 3 new scripts).

Containerized build path (2026-05-13)

Per operator mandate “boot-up containers containing proper dependencies for EVERYTHING using our containers Submodule”:

- helix_code/docker/build/Dockerfile.builder was previously **gitignored** by overly-broad Dockerfile* and build/ rules. Fresh clones couldn't build the builder image. Tightened both rules

(/Dockerfile*, /build/ — anchored at module root) so subdirectory Dockerfiles are tracked. Also bumped FROM golang:1.24-alpine → golang:1.26-alpine to match go.mod go 1.26.

- Containerised build path now reachable from a clone: podman compose -f helix_code/docker-compose.builder.yml build builder (uses Containers-submodule-aware compose semantics; substitute docker compose if available). The builder image carries Go 1.26 + golangci-lint + alpine apk deps + postgres-client; mounts the workspace and reuses Go module/build caches between runs.
- Five more Dockerfiles also became trackable and were checked in: tests/e2e/mocks/Dockerfile.{llm,slack}-mock, tests/infrastructure/Dockerfile.ssh-{server,worker}, docker/security/snyk/Dockerfile.

Dual API-key loader (2026-05-13)

Per operator mandate “We shall have API keys in .env file in the root of the project or all of them as the part of api_keys.sh in our host’s home dir! Both of it MUST BE fully supported!” — VERIFIED end-to-end:

- scripts/load_api_keys.sh prefers \$HOME/api_keys.sh when present, falls back to .env at meta-repo root.
- Test 1 — \$HOME/api_keys.sh path: loader sourced from project root, HOME pointing at real home dir → 42 secret-like vars in env.
- Test 2 — .env fallback path: loader sourced with HOME pointed at empty fakehome → 46 secret-like vars in env from project .env.
- Both paths individually load full secret set; neither silently drops. The loader’s “prefer \$HOME/api_keys.sh, fallback to .env” precedence matches the operator’s stated semantics.
- **HelixLLM/.gitmodules has stale submodules/HelixQA declaration** (directory absent on disk; only declaration remains).

Constitutional debt (open since P0)

- **LLMsVerifier dual-pin divergence (P0-04): RESOLVED** in P1.5-WP2; doc had been stale. The transitive duplicate helix_agent/LLMsVerifier was eliminated in WP2 (only a backup remains at helix_agent/.container-caps-backup-20260425-151947/LLMsVerifier). helix_agent/go.mod:232 now replaces digital.vasic.llmsverifier => ../dependencies/HelixDevelopment/LLMsVerifier/llm-verifier, so HelixAgent transitively consumes the canonical pin. The pin-parity gate scripts/verify-llmsverifier-pin-parity.sh is degenerate (regression-tripwire only, header confirms WP2 elimination). make verify-foundation

evidence (this round): exits 0, “0 failures, all gates passed” — no VERIFY_FOUNDATION_WARN_ONLY waiver needed.

- **Historical SSH key + helix.security.json leaks** (P0-T08.5): material is immortal in git history. Mitigated; rotation required by operator.
- **SonarQube + Snyk live-scan deferral** (P0-T08.7): infrastructure wired, awaiting credential rotation by operator.

Phase 3 remaining items (carried forward, non-blocking for Phase 4)

- **HelixAgent build** — IMPROVED: submodules populated, ./cmd/helixagent/... builds and tests pass. Wildcard ./... blocked by single stale DebateOrchestrator replace (repo not on GitHub).
- **Containers build** — RESOLVED: go build ./... exits 0.
- **Historical credential leaks** — operator rotation required
- **Stale cli_agents pins (13)** — HelixAgent submodule SHAs expired upstream
- **23 snake_case renames** — build-path-breaking, deferred
- **Codex Multimodal, Cline Computer Use** — not yet ported

CONST-045 — No Hardcoded Distribution Hosts

ALL container distribution targets SHALL be configured exclusively through CONTAINERS_REMOTE_HOST_N_* env vars in containers/.env (N=1..100; iteration stops at first absent _NAME; the containers module pkg/envconfig/parser.go is the authoritative loader). The .env file is the sole source of truth for host enrolment — no host is hardcoded in HelixCode source, tests, challenges, or governance documents. Every non-unit test run and every production deployment MUST use whichever hosts are currently configured when CONTAINERS_REMOTE_ENABLED=true. Adding, removing, or modifying a host means editing containers/.env; no code change is required. The CURRENT configured set can be audited with `grep '^CONTAINERS_REMOTE_HOST_' containers/.env`; at the time of this rule’s introduction (2026-05-07) the configured hosts were `thinker.local`, but the rule applies to whatever set is in .env at any future point (N>=1). Direct docker/podman commands, manual container start/stop, and ad-hoc remote hosts outside the .env mechanism are strictly prohibited.

How to resume

From a new CLI agent / LLM session

Type the **Resume Prompt** at the end of this file verbatim. It triggers continuation without further user context.

Programme conventions to apply (verbatim list)

1. **Subagent-driven-development always.** Never inline-implement multi-task features. Skip approval gates per the user's auto-approve memory (memory/auto_approve_designs.md).
2. **Commit on main.** All work flows through main. No feature branches.
3. **Push to 4 remotes (non-force only):** origin, github, gitlab, upstream for the meta-repo. Submodules push to their origin only (Challenges now has 3 remotes: origin + gitlab + upstream).
4. **Deepest-first push order.** Submodules → meta-repo. If meta-repo's gitlinks reference unpushed submodule SHAs, the meta-repo push will succeed but cloners will fail to resolve submodule pointers.
5. **Each feature has:** spec → plan → per-task TDD commits → Challenge harness commit → close-out commit. No exceptions.
6. **Anti-bluff smoke must always be clean.** Run before each commit:

```
grep -rn "simulated\|for now\|TODO implement\|placeholder" helix_code/internal helix_code/cmd && echo BLUFF  
|| echo clean.
```
7. **Runtime evidence required for every PASS** per CONST-035 / Article XI §11.9. No metadata-only / configuration-only / absence-of-error PASS.
8. **api_keys.sh > .env precedence.** Any tool that needs API keys sources them via scripts/lib/api_keys.sh first; falls back to .env.
9. **Non-FF push = STOP.** Never force, never --force-with-lease. If a push is rejected, investigate before retrying.
10. **No CI/CD pipelines.** All gates run via Makefile / scripts. Per CLAUDE.md Rule 1.
11. **No HTTPS for git.** SSH only.
12. **Every claim of "done" carries pasted terminal output** from a real run against real artefacts. Per CLAUDE.md Rule 8.

Picking up Phase 3 work

If Phase 3 is the active phase when you resume: 1. Verify state: `git log --oneline -3` should show the latest Phase 3 commits. 2. Read docs/improvements/PROGRESS.md §Phase 3 — Issue remediation

section. 3. Continue with the next pending remediation item from the Phase 3 remaining list. 4. Run `make test` to verify current state before making changes.

Picking up Phase 4 work

If Phase 4 is the active phase when you resume: 1. Verify state: `git log --oneline -3` should show commits 4f5f8f0 (P0 fixes) and 7f4effd (P1 fixes). 2. Read `docs/improvements/PROGRESS.md` §Phase 4 section. 3. Anti-bluff sweep is COMPLETE for critical packages (validators, fix, config, deployment). 4. Verify remaining test files for weak assertions (only nil/err checks without content verification). 5. Run `make test` to verify current state before making changes. 6. Once all test suites validated, advance to Phase 5 (end-user materials).

Picking up Phase 5 work

Maintenance mandate

This document **MUST** be updated when:

- Any task is completed (update T-status table + add commit SHA).
- Any feature is closed out (update Phase status table + repository SHAs).
- Any known issue is discovered (add to “Known issues” section).
- Any phase boundary is crossed.
- Any deferred item is fixed or further deferred.
- Any new remote/submodule is added or removed.
- Any constitutional clause is added or amended.

If this document is out-of-sync with the actual state of the work, the inconsistency is a **CRITICAL DEFECT** — same severity as a false-success test result (CONST-035). See:

- `CONSTITUTION.md` Article XIII §13.1 (CONST-044) — Continuation Document Maintenance Mandate
- `CLAUDE.md` §12 — Continuation Maintenance
- `AGENTS.md` — “Continuation Maintenance” anchor

Verification (TBD): `scripts/verify_continuation_sync.sh` will compare: - Last updated: 2026-05-08T02:00:00Z (Phase 3 – remediation + test infra) - Active phase here vs Current focus in `docs/improvements/PROGRESS.md`. - Tasks-done count here vs ticked-tasks count in `PROGRESS.md`. - Known-issue list here covers all documented failures in evidence files.

Non-zero exit = sync violation → blocking pre-push.

Resume Prompt

Copy-paste this verbatim into a new CLI-agent session to continue:

Read /run/media/milosvasic/DATA4TB/Projects/helix_code/docs/CONTINUATION.md and continue all work. Use subagent-driven-development. Skip approval gates per the project's auto-approve memory. Push all submodules + meta-repo to all configured remotes (non-force only) when each work package or feature is closed out.

Document version log

Date	Updater	What changed
2026-05-06	Initial create	Captures state through P2-F21-T04 (5ef13b8
2026-05-06	T06 update	T06 (/approval slash command) closed; 6 of
2026-05-06	T07 update	T07 (main.go wiring + registry hook + integ
2026-05-06	T08 update	T08 (Challenge harness 5 phases, meta 2781 out + push 4 remotes) is next.
2026-05-06	T09 close-out	F21 (Codex Approval Modes) CLOSED — fir
2026-05-06	F22 docs	F22 (Aider Git Auto-Commit Per Change) sp
2026-05-07	F23 docs	F23 (Cline Browser Tool) spec + plan landed
2026-05-07	F24 docs	F24 (Codex Project Memory) spec + plan la
2026-05-07	F25 docs	F25 (Plandex Plan Trees + Context Compac
2026-05-07	F26 docs	F26 (Openhands Workspace + Task Planner
2026-05-07	F27 docs	F27 (Aider Voice Input + Repo-Map) spec +
2026-05-07	F28 docs	F28 (Kilo-code AST-Aware Refactoring) spec
2026-05-07	F29 docs	F29 (Roo-code Full Port) spec + plan landed
2026-05-07	F30 docs	F30 (Continue IDE Integration) spec + plan
2026-05-07	Phase 3 entry	Phase 2 CLOSED (F21-F30 complete). Phase
2026-05-08	Full sync	F26-F30 close-out sections added; Phase 3 a synced with PROGRESS.md.
2026-05-08	Phase 3 sync	LLMsVerifier build, helix_qa build, 6 interna pruned. CONTINUATION.md synced with P
2026-05-08	Phase 3 close	Phase 3 CLOSED. All code-actionable remed containers build fixed. Phase 4 started (anti
2026-05-08	ZB Phase 2-3	Zero-Bluff Phase 2 (Stub Elimination) and P CONST-046 added. Anti-bluff clean. Pushed

Date	Updater	What changed
2026-05-08	ZB Phase 4	Zero-Bluff Phase 4 (Test/Challenge Hardening config, deployment). Article XI §11.9 compli
2026-05-12	ZB Phase 4 close	Zero-Bluff Phase 4 CLOSED. Weak-assertion + internal/deployment with mutation-test co Commits 02b7306 + a3f8871. Cross-compile li
2026-05-12	Hygiene	Untracked 3 runtime test artefacts + tighter binary. Commit d7a7956; all 4 remotes synce
2026-05-12	Anti-bluff sweep	Re-running go test -short ./internal/... su internal/discovery/client.go discoverByDNS h net.DefaultResolver.LookupHost(ctx, ...) bou Engine.StartSession goroutine raced with t.T + t.Cleanup hook in setupQATestServer; (3) in tests starved under suite parallelism — gate consecutive full short-suite runs PASS clean
2026-05-12	Challenge sweep	bash tests/e2e/challenges/run_all_challenges cascade.sh exit 1 with 61 third-party submod unreachable in practice (can't commit into t paths). Relaxed verify script to accept prese deliberate ACK (in-submodule marker still h 8/8 PASS. Also untracked 2 more build binar extended .gitignore. Commit 099f06a.
2026-05-12	Challenges submodule sweep	cd Challenges && make test-short failed at Tes test hardcoded a consumer-Yole APK path th violating its CLAUDE.md “100% decoupled” YOLE_ANDROID_APK_PATH) env var with SKIP-OK Challenges submodule commit 8024463 push repo pin bump 12904e4. make test-short 17/1
2026-05-12	Zero-bluff sweep	Per operator mandate “do not skip any tests testing.Short() skips in internal/tools/brows serializeChromiumLaunches(t) — exclusive flo launches across the entire go test process t switched from runtime t.Skip to //go:build default test binary; Challenges submodule r 278b617); (3) fixed 12 real data races at prod internal/helixqa, internal/llm, internal/memo telemetry, internal/tools/mapping, internal/t Verifications: go test -race ./internal/... (f warnings, 3 consecutive runs; go vet clean. now) comment removed). Commit 49936c0 +
2026-05-12	Skip-annotation sweep	Per the no-silent-skips governance gate: 101 (Containers 25, helix_qa 75, LLMsVerifier 1) (no adb/scrcpy/GStreamer/Tesseract/Paddle testcontainers, missing posix utils, etc.). NO code vet warning in helix_code/internal/llm into the EOF-decode branch where the for-l four layered fixes: auto-exclude listed third-markers (#short-mode, P1-F14-T10), tighten JS (), path-based post-filter for nested-vendore

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2026-05-12	Submodule test-parity sweep	opensource/). go vet ./cmd/... ./internal/... e16bfeb, LLMsVerifier 98758126. Meta-repo co Per “fix and polish everything”: ran go test failures surfaced and resolved at root cause digital.vasic.models => ../Models pointed at dependencies/HelixDevelopment/Models as a sibling decoupled by not relying on HelixAgent. (2) checksums for transitive testify/go-spew). For TestGetCapabilities asserted hardcoded model Venice API; Venice renamed its catalogue, so substring assertions (strings.Contains(m, "l remain CONST-036 compliant. Submodule p Security eae0cec; new Models pin added. Eve commit 7c81a40.
2026-05-12	Submodule test-parity round 2	Continuing the per-submodule sweep: Vision [setup failed] due to stale go.sum drift (mis additionally had pkg/remote/deployer_test.go orphaned by the Ollama→LlamaCpp deployer “undefined” errors and was phantom-coverage path tests in remote_test.go already cover the VisionEngine a092195, DocProcessor 3d11e41.
2026-05-14	Mistborn restore + integration matrix probe (round 37)	Operator paused autonomous loop for several (a) Discovered mistborn’s podman MACHIN back — podman ps reported “Cannot connect start”. Started VM via podman machine start (power state is untouched). Re-ran scripts/m reconnected → database.connected:true, redi integration matrix. Started local postgres sourced .env.full-test, ran go test -tags=in Surfaced ONE legitimate anti-bluff-correct t with "0" is not greater than "0" — the test (len(RetrieveMemory(...).Results) > 0, and ge catching the BUG #14 territory (cognee pro implement the real cognee API client — maj CORRECT by failing rather than fake-passin mistborn-tunneled stack; run_all_challenges evidence files / 107 counted PASS). All 4 ren (infrastructure work + finding documentatio
2026-05-13	Race-clean verification round 33	Ran go test -short -race -count=1 across all task (1.0s), internal/worker (31.6s), internal All packages: race-clean. Zero race conc Manager fix shipped across rounds 1-32 host went idle mid-round (SCP closed during SKIP-OK: #env-mistborn-offline rather than fa gates. run_all_challenges: 12/12 PASS (one c round). No code changes this round — verifi
2026-05-13	Production-bug sweep round 32	Operator re-asserted the anti-bluff mandate beyond the saturated API matrix. Re-verified

Date	Updater	What changed
	(operator mandate re-asserted 4th time — broader surfaces)	Broader-surface probes: (i) Challenges subn userflow 182s). (ii) helix_qa submodule shor (iv) containers — 8 packages PASS. (v) Helix — fails on missing X11/Xcursor (system-lib c build ./... which greedy-built the Fyne-GUI headless environment WITH X11/Xcursor.h: main_nogui.go build-tagged variant AND a se compile-gate was lying about portability — i environments with optional system libs insta adding -tags=nogui to the verify-compile Mak in this environment without X11 libs. helix-q round — build-system fix). run_all_challenge Round 31 closed the final-corner gaps: (a) G (gorilla's reject-without-Upgrade — route IS both 503 with "QA engine is disabled" — ho 503 saying "disabled" is preferable to a 200 grep hits in helix_code/internal + helix_code substitution variable names (placeholder as t model_download_manager.go), (iii) low-prior refactor opportunities, no runtime bluff). Cl NO fake responses. helix-qa expanded 135 - screenshot/engines 503 with anti-leak asser same). helix-qa: 107/107 PASS (138 evidenc secrets.sh: clean. Session summary: 32 re rounds; helix-qa harness 6 → 138 eviden
2026-05-13	Positive-coverage round 31 (final-corner + anti-bluff source scan)	
2026-05-13	Positive-coverage round 30 (helix-bridge + account-deactivation)	Round 30: (a) helix-bridge live LLM smoke - deepseek/openrouter); 3/5 respond OK (gro (env key revoked or restricted); openrouter codebase bugs. (b) DELETE /users/me lifecy (200), then attempted to log in as that user . VerifyJWTWithDB fix correctly enforces the "marked deleted" stub. (c) /system/status re ("QA engine is disabled" — intentional confi HCQA-098-PRE-REGISTER + HCQA-098-PR HCQA-099 (deleted-user login → 401 with de after delete would be a critical CONST-035 v healthy). helix-qa: 104/104 PASS (135 eviden secrets.sh: clean.
2026-05-13	Positive-coverage round 29 (schema fidelity, 100-counted milestone)	Round 29: schema-fidelity static probes for t every entry must have id/name/status/type p llm/models — every entry must have id/provi context length, which would mislead callers count must be 0 (matches /memory/stats.sys cross-endpoint guard). All clean. Probed /ll registered; the single /providers/:id endpoint expanded 127 → 130 evidence files: HCQA-C static-checks block (run early, in determinis crossed the 100-counted-checks milestone). clean.

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2026-05-13	Positive-coverage round 28 (robustness)	Round 28: probed concurrent requests (20 p duplicate-id collisions), large payloads (1 Mi body shapes (empty {} for required-field en Pagination is a feature gap, not a bluff: the limit (return ALL 315 tasks / 300 worker never claimed pagination support; the limit future product work. helix-qa expanded 124 empty body), HCQA-093 (1MB-payload-201) PASS (127 evidence files). run_all_challenges
2026-05-13	Production-bug sweep round 27	Round 27 probed auth/register input validation correctly 400 — auth path is clean), then sw more bug: (32) GET /workers/<malformed>/met respondInvalidID branch. Fixed. Also: SQL-in workers;--") — pgx prepared statements par table survived intact. Positive-coverage prob concatenated SQL. helix-qa expanded 120 — malformed-400), HCQA-090 (register-overlo HCQA-091 (SQL-injection defense — worker uses per-run unique hostname to avoid dupl (124 evidence files). run_all_challenges: 12/ secrets.sh: clean.
2026-05-13	Production-bug sweep round 26	Round 26 found 3 more bugs in sessions + v project_id (well-formed UUID but no such p existent projects. Root cause: in-memory ses declares. Fix: createSession handler now va projectManager.GetProject(req.ProjectID) BE ErrProjectNotFound for missing project. (30) S uuid") — session created with non-UUID pro respondInvalidID for malformed UUIDs → 400 500 leaking pg SQLSTATE 22001 "value too lon leakage (column type visible) + CONST-035 worker.ErrWorkerHostnameTooLong sentinel + h RegisterWorker pre-validates BEFORE the I expanded 117 → 120 evidence files: HCQA-C malformed-project_id→400), HCQA-088 (wo: "SQLSTATE"/"22001"/"character varying"). run_all_challenges: 12/12 PASS. go test -sho clean.
2026-05-13	Production-bug sweep round 25 (mandate re-asserted 3rd time)	Operator re-asserted the anti-bluff mandate cascade (0 failures) + anchor in all 6 root do more real bug — (28) PUT /projects/<malform HTTP 500 leaking raw postgres "invalid inp Same CONST-042 + CONST-035 double-viol DeleteProject in manager_db.go passed the ra validation; pg rejected with the SQLSTATE l err != nil { return ... ErrInvalidProjectID additional handlers that were missing respon createTaskCheckpoint, getTaskCheckpoints, TestDatabaseManager_DeleteProjectInvalidID u

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2026-05-13	BUG #27 fix generalized — round 24	— the old test mocked a postgres SQLSTATE wrap (which leaked the pg detail); new test malformed input. helix-qa expanded 112 → 1 DELETE 400 with anti-leak check for “SQLS POST/GET 400), HCQA-085 (session PUT-on run_all_challenges: 12/12 PASS. go test -sho clean. Round 24 extended the round-23 fix (task.Err wired) across the full handler matrix. Added project.ErrInvalidOwnerID sentinels (sed-repl fixed the cross-package AssignTask call site with the task sentinel since the failure surface over all 4 sentinels. Wired 7 additional hand retryTask, deleteWorker, updateWorker, wor through. Verified with curl matrix: 7 endpoint 400. helix-qa expanded 105 → 112 evidence complete/retry/fail + worker-delete/update/ evidence files). run_all_challenges: 12/12 PA scan-secrets.sh: clean.
2026-05-13	Production-bug sweep round 23 (operator re-asserted mandate)	Operator re-asserted the anti-bluff mandate cascade requirement. Re-confirmed: verify- + listed third-party submodule; CONST-035 docs (CONSTITUTION.md / CLAUDE.md / A password login (correctly 401), duplicate-en GET /tasks/ (correctly 404 via getTask’s gen returned HTTP 500 with “invalid task ID: in malformed UUID is CLIENT input error, sho task-manager call sites that do uuid.Parse(id unstructured returns replaced via sed across err, what) writes 400 when the sentinel mat for the remaining 7 task-id handlers as a fut evidence files: HCQA-073 (DELETE-malform run_all_challenges: 12/12 PASS. go test -sho
2026-05-13	Production-bug sweep round 22	Round 22 probed duplicate-create + workflow Found 2 more bugs: (25) duplicate-hostname "workers_hostname_unique" (SQLSTATE 23505). I isUniqueViolation(err) helper that parses pg unique-violations BEFORE leaking; handler workflows/* returned 500 with “project not f project.ErrProjectNotFound sentinel (7 unstru manager_db.go); 4 workflow handlers cons Bonus fix: the 3 non-planning workflow hand memory manager) instead of s.projectManag s.projectManager. helix-qa expanded 101 → 1 hostname-409-no-leak with explicit anti-leak “workers_hostname_unique”/“SQLSTATE”/“ helix-qa: 76/76 PASS (104 evidence files). ru SSH tunnels that dropped mid-session — sir mistborn-up.sh re-run brings them back). scr

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2026-05-13	Production-bug sweep round 21	Round 21 probed remaining endpoints for sentinel. <code><bogus>/heartbeat</code> returned HTTP 500 with response <code>\\"worker_metrics\\" violates foreign key constraint</code> — TWO violations in one response: (a) <code>CONST-042</code> (SQL state visible to API users); (b) <code>CONST-042</code> (resource error). Root cause: <code>UpdateWorkerHeartbeat</code> even after the workers-table <code>UPDATE</code> matched <code>UPDATE FIRST</code>; if zero, surface <code>ErrWorkerNotFound</code> the sentinel to 404 with clean “Worker not found for bogus task IDs — already correctly return 404 since the <code>WHERE-id-and-status</code> filter can’t do a query; 422 + clear message is acceptable). 12/12 bogus-404 with explicit anti-leak check (response against <code>CONST-042</code> regression). helix-qa: 74/74 PASS. go test -short server+worker: ok. scribble
2026-05-13	Production-bug sweep round 20 (100-evidence milestone)	Round 20 source-scanned for the broader “E” antipattern. Found 5 occurrences across tasks. <code>resource id</code> caused HTTP 500 instead of 404 being a server error when the cause is “client exported sentinels task.<code>ErrTaskNotFound</code> (covers all 6 session manager.go not-found responses) worker.<code>ErrWorkerNotFound</code> already in memory. <code>UpdateWorker</code>. The 5 handlers (<code>deleteTask</code>, <code>errors.Is-check</code> and return 404. helix-qa exported DELETE-task-404, HCQA-066 fail-task-404, worker-404, HCQA-069 DELETE-session-404. run_all_challenges: 12/12 PASS. go test -short clean.
2026-05-13	Production-bug sweep round 19	Round 19: probed task dependencies (forward reassign behavior, project-delete cascade. Found already-assigned task returned HTTP 500 (<code>AssignTask</code> wrap task.<code>ErrTaskInvalidStateTransition</code> <code>RowsAffected==0</code>; <code>assignTask</code> handler now message. Decided NOT to fix: dependency for validates deps lazily) and soft-delete with <code>status='deleted'</code>, sessions can legitimately reassign HCQA-064-PRE-TASK + HCQA-064-PRE-WORKER HCQA-064 (reassign-already-assigned → 422). run_all_challenges: 12/12 PASS. go test -short
2026-05-13	Production-bug sweep round 18	Round 18 found 1 more bug while probing task on a non-running task, <code>POST /tasks/:id/start</code> already-completed task all returned HTTP 500 retry-on-non-failed. Each was a client-side sentinel “broken” when really the request was incomplete introduced exported sentinel task.<code>ErrTaskInvalid</code> <code>ErrTaskNotRetryable</code>). <code>StartTask</code> and <code>CompleteTask</code> handlers <code>errors.Is-check</code> and return 422 Un
2026-05-13	Positive-coverage round 17	Round 17: scanned for more silently-discarded Probed three deeper state-accuracy question

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2026-05-13	Production-bug sweep round 16	complete with {result: ...} body — YES (a request key result, response key result_data BY created_at DESC — YES, first entry is the exactly 1 when a new task+worker is created hardcoded-zero or stale-cache bluff). No new that mechanically lock these invariants in pl PRE-CREATE + HCQA-058-PRE-START (set HCQA-059-PRE-CREATE + HCQA-059 (list CREATE-T + HCQA-060-PRE-CREATE-W + 1 (84 evidence files). run_all_challenges: 12/1 Round 16: scanned for more silently-discard beyond the comment documenting the fix. T #20: (20) POST /api/v1/workers/:id/heartbeat workers.cpu_usage_percent / memory_usage_percent forever. The handler updated only workers. la the worker's snapshot columns (which the C were never touched. Time-series data WAS p resource's own snapshot fields were frozen worker.DatabaseManager.UpdateWorkerHeartbeat (CASE WHEN > 0 preserves existing on om evidence files: HCQA-HB-PRE-W + HCQA-H workers/:id snapshot reflects heartbeat — ca run_all_challenges: 12/12 PASS. go test -sho Operator re-asserted the anti-bluff mandate <i>bluff manner — they MUST confirm that all</i> CONST-035 / Article XI §11.9 anchor IS pres CLAUDE.md / AGENTS.md + helix_code/ tri across every owned + listed third-party sub a TRIPLE bluff: (19) POST /api/v1/tasks/:id/ while task_checkpoints history table stayed p for the history INSERT — silently discarding the schema requires worker_id NOT NULL REF Triple bluff: (a) discarded error, (b) fabricate Subsequent GET /tasks/:id/checkpoints alwa result HONEST but the underlying bluff (his sentinel task.ErrCheckpointRequiresAssignment + supply worker_id in INSERT + surface re HCQA-CKPT-PRE-W / HCQA-CKPT-PRE-T (se HCQA-CKPT-PRE-ASSIGN, HCQA-055 (assign just-created checkpoint — proves real persis PASS (72 evidence files). run_all_challenges secrets.sh: clean.
2026-05-13	Production-bug sweep round 15 (governance reinforced)	Round 14 added positive-coverage probes for new bug surfaced — but locking the behavior llamacpp + /llm/providers/openrouter: all 40 constitutional cap at 7 entries (FallbackMod provider coverage is achieved via the live ve retry full round-trip and /projects/:projectId, 61 → 66 evidence files: HCQA-051-CREATE
2026-05-13	Positive-coverage round 14	

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2026-05-13	Production-bug sweep round 13	+ error_message), HCQA-052 (retry transition round-8 sentinel distinguishes legitimate retry round-trips). helix-qa: 57/57 PASS (66 evidence files), secrets.sh: clean. Round 13 uncovered 1 more real bug while assigning, the response showed "assigned_worker_id" column was correctly populated. Same regression was fixed in assigningWorkerID/column into local pointers assignedWorkerID/column assigned them to the returned Task struct's assignedWorkerID column. The bug silently lied about assignment state for the endpoint (which calls GetTask post-update). Fixed in HCQA-ASSIGN-PRE-W + HCQA-ASSIGN-POST-W. HCQA-ASSIGN-POST-W assigned_worker matching requested worker_id. HCQA-050 (heartbeat endpoint). helix-qa: 57/57 PASS. go test -short server+task+worker: ok
2026-05-13	Production-bug sweep round 12	Round 12 uncovered 1 more real bug (5th in session-action lifecycle probes: (17) GET /api/v1/workers/:id/metrics empty-history case — same nil-slice→null blank metrics. Fixed in getWorkerMetrics with []*workerMetrics. → 56 evidence files: HCQA-044 (worker-metrics) + HCQA-044-PRE-SESS (project+session for worker for the probe — decouples HCQA-044 + HCQA-044-PRE-SESS (project+session for HCQA-046 (action=complete → status:complete error). helix-qa: 51/51 PASS (56 evidence files), scripts/scan-secrets.sh: clean.
2026-05-13	Production-bug sweep round 11	Round 11 uncovered 1 more real bug plus a regression. PUT /api/v1/workers/:id returned HTTP 500 "constraint" — same TEXT[] NOT NULL pattern. Related defect: the UPDATE unconditionally overwrites hostname (a partial update) would overwrite the hostname. Fixed in worker.DatabaseManager.UpdateWorker: bare SET col = \$1 with COALESCE(NULLIF(\$1, ''), hostname) THEN \$4 ELSE max_concurrent_tasks END for max_concurrent_tasks the input is zero-value. helix-qa expanded 43 evidence files: HCQA-042 (PUT renames display_name + metadata) + HCQA-043-DEL (DELETE 200), HCQA-043 (DELETE 200), run_all_challenges: 12/12 PASS. go test -short server+task+worker: ok
2026-05-13	Production-bug sweep round 10	Round 10 uncovered 1 more real bug while assigning, returned HTTP 500 "ERROR: column path does not exist" on update attempt. Root cause: internal/project_type referenced path and type columns that DON'T exist in database.go:333). Real schema columns: work_type (which holds the project type under config["type"]). "rename a project" endpoint was therefore using GetProject's column-mapping pattern: RETURN config["type"] and metadata from config["metadata"]. HCQA-039 (create-for-lifecycle), HCQA-040 (DELETE 200), HCQA-041-DEL (DELETE 200),

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2026-05-13	Production-bug sweep round 9	evidence files). run_all_challenges: 12/12 PASS. secrets.sh: clean. Round 9 uncovered 1 deep bluff in the mem systems all hardcoded as status: "available" — direct contradiction. Reading the handler metadata with NO connection probe of any of the 6 listed providers (Cognee/Weaviate) handler claims "available" for 6 systems with connections. Fix: introduced a memoryStatusManager is bound + a reachability probe run (matches stats). The catalogue itself (id/name of supported backends — that's real metadata "HCQA-MEM-BLUFF" probe asserts NO entitlement future regression to hardcoded "available" v PASS (41 evidence files). run_all_challenges secrets.sh: clean.
2026-05-13	Production-bug sweep round 8	Round 8 uncovered 2 more real bugs by external tasks/:id/checkpoints returned "checkpoints" JSON contract bluff (tasks, projects, worker []map[string]interface{}{} in getTaskCheckpoint "task not found, not in failed state, or max r error code that lies about the nature of the p side state mismatch, not an internal bug). In internal/task/manager_db.go; handler now er state error, 500 only for genuine DB faults. I create-for-lifecycle, HCQA-037 checkpoints- HCQA-038-PRE-START / HCQA-038-PRE-CC internal-step probes write evidence but don test -short server+task: ok. scripts/scan-sec.
2026-05-13	Sessions-create probe addition (round 7)	Coverage expansion: probed POST /api/v1/se the round-4 HCQA-031 create-project probe surfaced — Mode validation correctly reject building/testing/refactoring/debugging/depl HCQA-035 probe stitches the just-created p asserts 201 + session.project_id round-trip run_all_challenges: 12/12 PASS.
2026-05-13	Production-bug sweep round 6	Round 6 uncovered 1 more real bug, same p returned HTTP 401 "User not authenticated" Root cause: refreshToken handler called c.Ge (must stay public for register/login). Fixed b VerifyJWTWithDB, mirroring the logout handle JWT (200) for valid Bearer, 401 with "Author "Invalid or expired token" for malformed/exp test updated to TestRefreshToken_WithBearer (unverifiable Bearer returns 401 which is the HCQA-033 (valid Bearer issues new JWT — iat is per-second and within-1-sec refresh ca Bearer returns 401). helix-qa: 34/34 PASS. r ok. scripts/scan-secrets.sh: clean.

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2026-05-13	Production-bug sweep round 5	Round 5 uncovered 1 more real bug, same pattern as Round 4: <code>api/v1/projects</code> returned HTTP 500 with <code>ssh_config</code> NOT NULL. Same fix as task_data: <code>sshConfig</code> to <code>map[string]interface{}{}</code> and nil slice as <code>map[string]sshConfig</code> — pgx serializes nil slice as NULL; the column <code>ssh_config</code> explicitly provides the value). <code>helix-qa</code> expanded test body and asserts 201 + UUID + hostname-rand + PASS. go test -short server+worker: ok. scripts/scan-secrets.sh: clean.
2026-05-13	Production-bug sweep round 4	Round 4 uncovered 2 more real bugs: (8) <code>POST /projects</code> invalid UUID length: 12" — <code>internal/project_manager_db.go:CreateProjectWithUser</code> literally hardcoded as the owner <code>"default-user"</code> matching the error. Handler now extracts the user from <code>CreateProjectWithUser</code> directly with <code>user.ID</code> . (9) <code>api/v1/projects</code> <code>publicProjects</code> Gin group with comment <code>// no create projects or trigger workflow execution without project id</code> project id. This was a real production security issue. Handler now requires <code>c.Get("user")</code> which is only set for <code>publicProjects</code> group; <code>publicProjects</code> removed entirely. <code>POST /projects</code> must 401), HCQA-031 (authentication). Existing unit test <code>TestCreateProject_ValidRequest</code> (mirrors <code>TestListProjects</code> pattern). <code>helix-qa</code> : <code>api/v1/projects</code> internal/server/...: ok. scripts/scan-secrets.sh: clean.
2026-05-13	Production-bug sweep round 3	Continuing the anti-bluff push uncovered 2 more real bugs: (1) <code>api/v1/users/me</code> returned a <code>User</code> stub with <code>is_active</code> false; <code>authMiddleware</code> was calling the cheap <code>VerifyJWT</code> JWT claims; every other field defaulted to zero. (2) <code>api/v1/workers</code> (fetches the full user + enforces <code>is_active</code>). <code>api/v1/workers</code> bonus: deactivated accounts can no longer create projects. <code>api/v1/workers</code> returned <code>"workers": null</code> — the handler (tasks and projects); fixed by defaulting to <code>[]Worker</code> (workers as array), HCQA-028 (system stats). <code>AND created_at</code> non-zero — would have caused 500. PASS. <code>run_all_challenges</code> : 12/12 PASS. go test -short server+worker: ok.
2026-05-13	Production-bug sweep round 2	Continuing anti-bluff push uncovered 3 more real bugs: (1) <code>api/v1/tasks</code> checks with tasks-CRUD lifecycle and project lifecycle. (2) <code>500 "null value in column task_data violates not-null constraint"</code> <code>internal/task_manager_db.go:CreateTask</code> forwarded <code>task_data</code> JSONB NOT NULL now upheld). (4) <code>GET /projects</code> empty list — JSON contract bluff (Go's nil slice is not empty). <code>api/v1/projects</code> returned 401 "user_id not found in database" while <code>authMiddleware</code> set <code>c.Get("user")</code> entire <code>projects-list</code> endpoint was unreachable. (5) <code>api/v1/projects</code> established <code>c.Get("user")</code> pattern. (6) Same pattern as Round 4: HCQA-019..027: tasks-CRUD full lifecycle (list, create, update, delete-404), plus list-projects-as-array and <code>run_all_challenges</code> : 12 passed. go test -short server+worker: ok. scripts/scan-secrets.sh: clean.
2026-05-13		

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	Mistborn distribution + anti-bluff QA harness	Stood up the mistborn.local distribution stack operator mandate “all tests and Challenges tested codebase really works as expected”. (gitignored, key-based auth only — password compose.mistborn.yml 7-service podman stack grafana); (3) scripts/mistborn-{up,down}.sh b scripts/bridge/main.go (helix-bridge) multi-p loader contract — 3/5 providers respond live anti-bluff QA harness with captured wire evi duration_ms in per-check JSON); (6) tests/e run_all_challenges.sh (12/12 PASS); (7) BLU returned a fabricated status: available stub returns 404 for unknown providers, sourced TestGetLLMProvider_UnknownIDReturns404 repro deep content (count>0, database.connected login → /users/me → garbage-bearer-401 → l from registration, user.username === register in HCQA-004/005/006 (assertions of expecta discovered + fixed: helix_qa orchestrator 0- tests at integration_test.go/orchestrator_edg assert.True(t, result.Success) to anti-bluff a 180s. Local verification: HelixCode locally s tunnel; helix-qa: 19/19 PASS with no FAIL a HelixCode internal -race short: 91 packages bluff smoke: clean (5 pre-existing benign ma commits 6e58ab6 → 30a1e2b → 5c44ba3, all 4 re
2026-05-15	Cascade Challenge runtime-validation widened + Task #274 isolation infeasibility documented (round 41 close-out ⁴⁴)	(1) Cascade Challenge runtime evidence (DDoS, stress, chaos) × 5 submodules (Helix PASS against stub on port 8081 with captur (helix_qa/containers/EventBus/HelixLLM DD 19 . The structural-only concern is now subs submodule cascade with consistent latency enumerated 19 files importing digital.vasic lines with debate constants (ModelHelixDebat fields deeply interwoven with non-debate ha into debate vs non-debate halves — multi-da Documented as truly infeasible in single-ses upstream repo OR refactor HelixAgent to re
2026-05-15	CONST-051(B)↔(C) tension documented + go.work attempt + stability re-evidence (round 41 close-out ⁴³)	Operator-directed continuation. (1) CONST docs/architecture/CONST-051-tension.md: docu HelixLLM post-Tasks #254/#273 (relative-p Attempted go.work workspace fix at HelixCo digital.vasic.docprocessor module identifier AND dependencies/vasic-digital/DocProcessor LLMOrchestrator, LLMProvider, VisionEngin Documented 4 real-fix options for operator: [recommended], (C) accept current state, (D tension persists by design. (2) Stability re-PASSED (one transient 19/1 caught early —

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2026-05-15	Address remaining unfinished items — cascade runtime-validated + Task #274 forensics + Models pulled (round 41 close-out ⁴²)	Challenge, recovered next run, same transie cascade coverage. HelixCode inner make ver authoritative — 7 alternate org-name URLs Operator-directed deep close. (1) CONST-0 + constitution + 64/68 owned at parity; depe cascade commit) pulled via FF; 3 third-party flagged informational. (2) Cascade Challenge port 8081 — proves structural cascade is NO p95=6.1ms, containers DDoS 50/50 100% 50/50 100% p50=2.2ms, dependencies/He Captured wire evidence — partial address o cascade Challenges now have proven happy template structure). (3) Task #274 forensi DebateOrchestrator, vasic-digital/debate, Hel Debate_Orchestrator, vasic-digital/debate-orc github. 12 sub-packages required (agents, a testing, tools, topology, validation, root). Bu feature fake”). Operator action genuinely debate-dependent code (~6 files in internal, module source. Task #274 stays pending. Operator-directed deep close of all actionab per-submodule capture: ran git submodule party (LLama_CPP 187 ahead, Ollama 33, H vasic-digital/Models 1 ahead — pulled via FF originated upstream). (2) CONST-050(B) c coverage.sh regression gate — exposed that received governance anchors but NOT the 6 cascade (10 new in close-out⁴⁰ + 44 from cl HelixCode inner module verify-compile: (the CONST-051(C) Task #254/#273 refactor regenerated: 9 platforms × 30 features × 6 VERIFIED cells preserved; governance verifi coverage.sh asserts every owned submodule (executable, ≥1KB, contains PASS + SKIP-OK p github_pages_website). Final anti-bluff ca GREEN + verify-cascade-coverage.sh 66/66 HelixCode inner compile clean. Remaining h #274 (DebateOrchestrator missing repo, op services running), helix_qa autonomous sess design).
2026-05-15	Complete cascade: 54 submodules + regression gate + verify-compile + ledger refresh (round 41 close-out ⁴¹)	All 6 verifier gates GREEN against the f using the identical pattern from completed T own-org submodules + updated go.mod rep → ../../vasic-digital/<Name> for vasic-digital for parent-root residents). 10 new vasic-digi Watcher, Middleware, RateLimiter, I18n, Re full CONST-048..060 anchor block + .gitigno Lazy 5869b9a, Watcher 7850f10, Middleware e Document ad85bb4, Filesystem d13a5f2, TOON
2026-05-15	Task #273 COMPLETE — HelixLLM 34 nested submodules migrated; full cascade closed (round 41 close-out ⁴⁰)	

Date	Updater	What changed
2026-05-15	Address unfinished features + issues — uncover real cascade gaps, fix 45+1 ungoverned submodules (round 41 close-out ³⁹)	→ +10 new; -1 constitution removed per CO cascade target). Anti-bluff captured evidence implementable gate green – anti-bluff covenant (submodules), G2 anti-bluff smoke, G3 mock-audit, G6 case-conformance all GREEN. Two coordinated cross-org rename), Task #274 (upstream). Operator-directed comprehensive audit of <code>submodule_owned.txt</code> had only 12 entries while 5/6+0/6+20/20 sweeps in close-outs^{35,38} were CONST-035 false-success in the verifier's submodules (3 HelixDevelopment + 42 vasic missing same anchors. (3) 4 submodules (Ba missing <code>.gitignore</code> per CONST-053. (4) Helix style problem (tracked as Task #273, multi- on <code>digital.vasic.debate</code> → <code>./DebateOrchestrat</code> doesn't exist on GitHub; production code at imports it). Fixes shipped: 45+1 submodule insertions per file × 3 files × 46 submodules pushed to their origin remotes (panoptic f5d830aecc, and 41 vasic-digital submodules with 45 included in this commit. Final verifier s tracked as Task #273. Pre-existing breaka (Task #274 candidate — operator needs to r code).
2026-05-15	Anti-bluff re-invocation — CONST-060 protocol self-exercised + 6/6+20/20 re-evidence (round 41 close-out ³⁸)	Operator re-invocation of canonical anti-bluff CONST-060 (Fetch-before-edit Mandate), the all --prune (all 4 HelixCode remotes fetched upstream — no drift since close-out³⁷), git - → empty (constitution still at 1b0b03a, the \$1 evidence per §11.4.2 of its own anti-bluff inv evidence — the actual HEAD..<code>@{u}</code> output, eve evidence requirement): re-ran the verifica verify-all-constitution-rules.sh 6/6 GATES bluff covenant honoured"; (2) run_all_challe CHALLENGES PASSED! Anti-Bluff Verificati (CONST-035 / §11.9 + CONST-047..060) × 3 covenant is honoured every time the operato captured during execution. No false-success
2026-05-15	CONST-060 cascade — Fetch-before-edit Mandate codified + anti-bluff re-evidence capture (round 41 close-out ³⁷)	Operator re-invocation of canonical anti-bluff fetch+pull per CONST-049/060 (which is the upstream added §11.4.37 (Fetch-before-edit session redundant-work incident on Project- CONSTITUTION.md + CLAUDE.md + AGEN (36 governance files updated this round). Ea (403603d), Challenges (af4fba1, 4 remotes), c github_pages_website (b0ccb17), DocProcess (5af6b6b), VisionEngine (f229b64), LLMsVerifi .gitmodules constitution pointer bumped to 1

Date	Updater	What changed
2026-05-15	Task #254 COMPLETE — all 46 HelixAgent nested own-org submodules migrated to parent root (round 41 close-out³⁶)	evidence per §11.4.2: (1) verify-all-const Task #254 close); (2) run_all_challenges.sh e Anti-Bluff Verification: COMPLETE; (3) veri present across every owned submodule. The codebase carries positive runtime evidence G4 verifier-gate FAIL eliminated; verifier batches (close-out³⁵ + this commit's 8 follow from HelixAgent to canonical parent-root loc vasic-digital/<name>, 3 HelixDevelopment resolves to ../Challenges (already at HelixCo 55c9a9d1) on its origin (HelixDevelopment/H go.mod replace directives. Replace directive HelixDevelopment for the 3 HelixDevelopme documented import/SDK/runtime resolver" p surprise: HelixAgent's ConversationContext s digital/conversation.git (lowercase, differen Task #254. Remaining single-session-block orgs — operator-coordination required). First measurable progress on Task #254 (th submodules migrated from helix_agent/<name> canonical parent root at dependencies/vasic- Observability, Auth, Storage. Workflow vali <name>.git dependencies/vasic-digital/<name> rm <name> in HelixAgent; (3) update HelixAge vasic-digital/<Name> — this is the "documen names. HelixAgent commit 3b97d1b1 pushed pointer-bumps for 5 new submodules + Helix 41 nested own-org chains (5 fewer). Remain proven pattern. Per-submodule pattern repe Final capture after the 12-submodule CONS gates GREEN, 1 FAIL = G4 known Task #25 G1 governance cascade PASS (14 anchors × PASS (zero production bluff markers), G3 m audit PASS, G6 case conformance PASS. (2) failed — released evidence still holds. (3)r platforms × 30 features × 12 owned submo preserved). Round-41 totals: 13 close-outs HelixCode + 11 owned submodules; 12 of 1 matrix (HelixQA, Challenges, Containers, S VisionEngine, LLMsVerifier, Models, panopt reasons (HelixAgent multi-session Task #25 remotes (github + gitlab + origin fanout + u anti-bluff covenant is honoured: 20/20 C owned-submodule cascades + 72 new te CONST-035 / Article XI §11.9.
2026-05-15	Task #254 pilot: CONST-051(C) HelixAgent migration — 5 of 46 nested submodules (round 41 close-out³⁵)	
2026-05-15	Round-41 cascade capstone — post-cascade verifier sweep + ledger refresh (round 41 close-out³⁴)	
2026-05-15	CONST-051(A) panoptic + github_pages_website cascade — 12 of 14	Final two non-blocked owned submodules ca panoptic) adds the 6 CONST-050(B) Challen github_pages_website (commit pushed to challenges/scripts/ directory with the 6 Cha

Date	Updater	What changed
	owned submodules now covered (round 41 close-out ³³)	gracefully — structural contract still captured. bumped per CONST-049 step 7. 12 of 14 owned submodules now have full coverage (HelixQA + Challenges + containers + LLMProvider + VisionEngine + LLMsVerifier). Challenge files this session: 72 (6 HelixCode submodules excluded for valid reasons: HelixQA chains need refactoring first per CONST-051 surface — would be redundant). Cumulative round-41 cascade: 100/200 (100%) pass + ³¹ Ch/Co/Sec + ³² 6 Deps + ³³ Pano+GhPw
2026-05-15	CONST-051(A) dependencies/HelixDevelopment cascade — 6 submodules × 6 test types (round 41 close-out ³²)	Mass cascade continuing operator's "do not stop until all submodules cascaded with the 6 missing CO cascade_const050b.sh) — same anatomy, per-s commit pushed), LLMOrchestrator (LLMORCHESTRATOR_*), LLMsVerifier (LLMSVERIFIER_*), new Challenge files added across 6 submodules from their respective origins. HelixCode .gitmodules pointer bumped to 1f03882 in this session. 7. Cumulative round-41 cascade: HelixQA + Challenges + Containers + Security = 100/200 (100%) pass matrix (60 new Challenge files this session). HelixQA (multi-session Task #2), panoptic, HelixAgent (multi-session Task #2)
2026-05-15	CONST-051(A) Challenges + containers + Security cascade — 6 new test types each (round 41 close-out ³¹)	Continuing the operator's "do not stop until all submodules cascaded with the 6 missing CO cascade_const050b.sh) — same anatomy, per-s commit pushed), LLMOrchestrator (LLMORCHESTRATOR_*), LLMsVerifier (LLMSVERIFIER_*), new Challenge files added across 6 submodules from their respective origins. HelixCode .gitmodules pointer bumped to 1f03882 in this session. Three more owned submodules cascaded with the 6 missing CO cascade_const050b.sh) — same anatomy, per-s commit pushed), LLMOrchestrator (LLMORCHESTRATOR_*), LLMsVerifier (LLMSVERIFIER_*), new Challenge files added across 6 submodules from their respective origins. HelixCode .gitmodules pointer bumped to 1f03882 in this session. Challenges (commit 873c0b1, pushed to all 4 submodules (fanout + upstream); Containers (commit 7f1e1e1, pushed to origin); Security (commit 56fdc16, pushed to origin). All 6 submodules SKIP-OK cleanly when their target env was wired. bluff Challenges (ddos / stress / chaos / scaling). (CHALLENGES_*, CONTAINERS_*, SECURITY_*). All 6 submodules SKIP-OK cleanly when their target env was wired. Challenge counts per session: 5→11. HelixCode meta-repo .gitmodules pointer bumped to 1f03882 in this session. Cumulative round-41 submodule cascade: HelixQA + Challenges + Containers + Security = 100/200 (100%) pass matrix (60 new Challenge files this session). Remaining 8 owned submodules: Asm, mcp_servers, panoptic, etc. — to follow in subsequent sessions.
2026-05-15	CONST-051(A) helix_qa submodule cascade — 6 new test types in helix_qa (round 41 close-out ³⁰)	Per operator's "do not stop until ABSOLUTE all submodules cascaded with the 6 missing CO cascade_const050b.sh) — same anatomy, per-s commit pushed), LLMOrchestrator (LLMORCHESTRATOR_*), LLMsVerifier (LLMSVERIFIER_*), new Challenge files added across 6 submodules from their respective origins. HelixCode .gitmodules pointer bumped to 1f03882 in this session. submodule-side CONST-050(B) cascade. helix_qa submodules cascaded with the 6 missing CO cascade_const050b.sh) — same anatomy, per-s commit pushed), LLMOrchestrator (LLMORCHESTRATOR_*), LLMsVerifier (LLMSVERIFIER_*), new Challenge files added across 6 submodules from their respective origins. HelixCode .gitmodules pointer bumped to 1f03882 in this session. Each ports the HelixCode template (5-6 step) to its origin remote (github.com:HelixDevelopment). honest SKIP-OK per §11.4.3 topology-dispatcher localhost:8081/health, HELIXQA_BIN auto-discovery against Python stub: 200/200 (100%) pass matrix (60 new Challenge files this session). Other 5 SKIP-OK cleanly when helix_qa infrastructure was wired. Challenge counts per session: 5→11. HelixCode meta-repo .gitmodules pointer bumped to 1f03882 in this session. submodules to cascade: Challenges, Containers, Security, LLMProvider, VisionEngine, LLMsVerifier.
2026-05-15	CONST-050(B) matrix-completion	Post-#266-closure verification capture. (1) : governance cascade PASS (14 anchors × 36 anchors)

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	sweep + ledger regeneration (round 41 close-out ²⁹)	markers in helix_code/{internal,cmd}), G3 nested internal/mocks), G4 nested own-org submodule (session), G5 .gitignore + sensitive-file PASS candidates / 12 protected). 5/6 green; 1 FA run_all_challenges.sh end-to-end: 20/20 PASS . Verification: COMPLETE. This is the release (3) scripts/regenerate-coverage-ledger.sh: re features × 12 owned submodules; 46 nested cells preserved). The Anti-Bluff Verification scaling ✓ UI ✓ UX ✓ (all 6 of Task #266's m runner).
2026-05-15	UX End-to-End Flow Challenge — FINAL 6th of CONST-050(B) missing test types — closes Tasks #266 + #256 (round 41 close-out ²⁸)	Task #270 closed; Tasks #266 + #256 (6 challenges/ux_end_to_end_flow.sh — drives th across the chain: discover → health → list-m post-liveness. 6-step structure: (1) binary-j hints (wizard/provider/key/list-models/healt exposes ≥1 selectable model (pick first); (5) xyz -prompt ping → assert error is user-com name / “model” / “provider” / “not found” / “ hostile (panic / goroutine / runtime error / s EVERY step’s output); (6) post-error -health journey, no leaked state). Anti-bluff value: from journey coherence — what UI proves a hostile blacklist (panic:, goroutine [0-9]+ \[catches the “stack-trace leaks to user surfac Configurable knobs: HELIX_CLI_BIN, UX_CLI captured evidence: binary OK, 4/5 recovery llama-3.2-3b), bogus model → exit 1 + 4 use now 20/20 PASS (was 19/19 after UI). Task commit. Anti-bluff matrix per CONST-050(B) ✓ scaling ✓ UI ✓ UX ✓.
2026-05-15	UI Terminal-Interaction Challenge — 5th of CONST-050(B) missing test types (round 41 close-out ²⁷)	Task #269 closed (Task #266 5-of-6 installm drives the built helix_code/bin/cli binary no (1) binary-presence dispatch (SKIP-OK if hel (all 8 documented flags present: -approval - interactive -notify); (3) -health output ass containing operational / / ; (4) -list-m of Provider: + Context Size: + Status: lines - (5) post-invocation re-runability (second temp” classes); (6) invalid-flag graceful-ex timeout). Configurable knobs: HELIX_CLI_B UI_MIN_MODELS (1). Anti-bluff value: catches return-code gating misses. Validated end-to- model entries with all 3 labels matched, re now 19/19 PASS (was 18/18 after scaling). installment.
2026-05-15	Horizontal-Scaling Challenge — 4th of CONST-050(B)	Task #268 closed (Task #266 4-of-6 installm operator-safe horizontal-scaling Challenge v SCALING REPLICA URLs env var resolving to ≥2

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	missing test types (round 41 close-out ²⁶)	dispatch (reachable-replica detection + SKI load test (50 reqs × N replicas at concurrent (±50% tolerance, informational since direct replicas (modulo uptime/timestamp via sed DB/cache” misconfiguration class which is a probe. Configurable knobs : SCALING_REPLIC SCALING_CONCURRENCY (10), SCALING_MIN_PASS_PC OK path on bare run (default empty REPLIC Validated happy path against 3-replica Pytho reachable, schema valid each, 150/150 (10 sha256 matched across all 3 replicas (ek 18/18 PASS (was 17/17 after chaos). Remain
2026-05-15	Chaos Failure- Injection Challenge — 3rd of CONST-050(B) missing test types (round 41 close-out ²⁵)	Task #267 closed (Task #266 3-of-6 installm operator-safe client-side fault injection (NO power-management ban). Differs from DDoS is the chaos. 5 injection vectors : (1) 5 ma bad HTTP version, double-slash path, null by Huge: <8 KiB>); (3) 5 concurrent slow-loris co abrupt connection close mid-request; (5) mi reqs at concurrency 20 to /api/v1/health wh HELIX_CHAOS_HOST/PORT, CHAOS_LEGIT_REQUESTS (6-step structure : pre-chaos probe → malfo header + post-huge liveness → slow-loris spa chaos liveness + schema sanity. Anti-bluff f crash vs clean 4xx reject); 000 on oversized users (pass-rate < threshold); zombie post-c SKIP-OK when server unreachable. Validat 200, 5 malformed shapes survived, oversized Huge per design), 5 slow-loris spawned, 100 chaos 200+valid status. curl-code formatting “000000” when curl’s -w %{http_code} alread runner now 17/17 PASS (was 16/16). Rem
2026-05-15	CONST-052 rename- safety guardrails + G6 strengthening (round 41 close-out ²⁴)	Operator follow-up safety mandate 2026-05- to codebase which is by convention non-sna System and working building process!” Stre rules.sh with explicit NEVER_RENAME_PATTERNS a manager files (Makefile, Cargo.toml, go.mod Kotlin/Android/Apple/C#/Swift; Android AC vendor/, kernel-5.10/, packages/, prebuilts/ breaks the AOSP build); AOSP-mandated f bootstrap.bash, BUILD, OWNERS); build/ca (.git/, .github/); coordinated owned-project r atomic rename across GitHub+GitLab+all c documents the hard rule + per-rename-bato run_all_challenges.sh + verify-all-constitutio resolution regression test). Task #252 descr protected + 6 unprotected rename candid stability check: 3 sequential runs × 16/16 on live-LLM Challenge). Anti-bluff sweep ver

Date	Updater	What changed
2026-05-15	Stress Sustained-Load Challenge — 2nd of CONST-050(B) missing test types (round 41 close-out ²³)	known Task #254 (HelixAgent 46 nested subtests) (internal/llm, cmd/cli). Task #266 2-of-6 installment delivered. New tests/server's /api/v1/health endpoint with sustained tests: rps, per-second pass-rate snapshot, latency-knobs: STRESS_DURATION_SEC (default 30), STRESS_STRESS_TIMEOUT_SEC (5), STRESS_MIN_PASS_PCT (95). structure: (1) pre-stress probe + 10-sample with-empty-body bluff; (3) sustained loop with aggregate analysis + worst-second pass-rate; (4) latency vs baseline comparison — catches alone would miss; (6) post-stress liveness + overall + worst-second pass-rates, p50/p95/SKIP-OK pattern. Validated end-to-end against worst-second, p50=0.5ms, degradation=-20 fractional-seconds bug caught + fixed (was 15/15). PASS (was 15). Remaining 4 test types (scaled).
2026-05-15	DDoS Health-Flood Challenge — first installment of CONST-050(B) missing test types (round 41 close-out ²²)	Task #256 first installment delivered (1 of 6). New tests/e2e/challenges/ddos_health_flood. concurrent requests + asserts pass-rate through vars: HELIX_HEALTH_URL, DDOS_REQUESTS (default 100), DDOS_MIN_PASS_PCT (95). Captured wire evidence seconds, latency p50/p95/p99. 5-step structure sanity (real JSON with status field, not empty), liveness (catches “fell-over-during-flood” class), topology-dispatch. Validated end-to-end against concurrency → 100% pass rate, p50=1.0ms. Full runner now 15/15 PASS (was 14/15). as Task #266 — each follows the same skeleton.
2026-05-15	scripts/verify-all-constitution-rules.sh — CONST-055 enforcement engine (round 41 close-out ²¹)	Task #265 closed. Implemented the canonical names as the enforcement engine for every. New scripts/verify-all-constitution-rules.sh. G1 governance-cascade (§11.9 + CONST-04) smoke (production code grep for “simulated”). G3 CONST-050(A) mock-from-production (for CONST-051(C) nested-own-org submodule covered). tracked-sensitive-file scan; G6 CONST-052 candidates without failing since rename is possible. canonical-fix on violation. Paired meta-test against anti-bluff requirement, §1.1 mutation pair): 1 FAIL, reverts cleanly. 9 sub-tests PASS (G2 surfaced + fixed two real findings: (a) dependency CONST-053 — fixed inline via Models command). was being flagged as a tracked sensitive file (matches .env.full-test / .env.test / .env.ci). 5/6 gates GREEN, 1 FAIL = G4 HelixAgent's (multi-session). Meta-test caught a real bluff which .gitignore correctly rejected — defense past .gitignore, which is the actual failure mode.

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2026-05-15	Anti-bluff verification sweep — CONST-035 captured runtime evidence (round 41 close-out ²⁰)	Operator re-invoked the canonical CONST-035, which itself is already in place (14 anchors × 36 ops). Operative answer per §11.4.2 (captured-evidence). Challenges and captured runtime evidence. a214f0c → 464ada1 (upstream merge “drop” post-merge. CONST-035 anti-bluff smoke production this would" helix_code/internal h... Challenge (6/6 PASS): live LLM round-trip + probe + token stats 📊 tokens: in=... out=... Challenge (5/5 PASS): live sentinel-echo pr... in a temp file — proves the @-mention conte... (tests/e2e/challenges/run_all_challenges.sh) PASSED! Anti-Bluff Verification: COMPLETE + cmd/cli/...): 7 packages green (ok dev.helix Constitution pointer bumped 464ada1 in same verifier: 0 failures. The anti-bluff covenant is produced positive captured evidence of user... Task #263 closed — last genuine-violation in resource-policy/apply_caps.py no longer hard-coded helix_code/helix_code/, /helix_code/mcp-server/ others). Refactor: SKIP_PATH_FRAGMENTS_DEFAULT vendor/, external/, .container-caps-backup-, paths-file <path> flag (+ RESOURCE_POLICY_SKIP from a one-per-line file at runtime via new l... project-file ∪ -exclude. Example file contains paths.example.list shipped alongside the script resource-policy-skip-paths.list (16 fragments) --help shows the new flag; grep for helix_code returns 0 matches. containers submodule closed
2026-05-15	containers/apply_caps.py CONST-051(B) decoupling refactor (round 41 close-out ¹⁹)	Task #261 closed. New generator script scripts/coverage-ledger.sh + §11.4.22 (lightweight doc-sync v... the coverage ledger per CONST-048 + §11.4.22 supported platforms from helix_code/Makefile (harmony-os/containers/headless); (2) extract 30); (3) inventories test-type presence + Makefile owned submodule's .gitmodules for nested o... verify-governance-cascade.sh and parses and... every prior VERIFIED cell (anti-bluff: never au... remain operator-controlled per CONST-035... 30 features, 12 owned submodules, 30 VERIFIED... own-org submodules (CONST-051(C) viola... bash bugs in early-version (grep -c exit-1 wi... semantics on multiline reads) — fixed defen... tested with ROUND_TAG="close-out-18-reg... preserve VERIFIED markers and refresh me... of script) + §11.4.22 (lightweight doc-sync v...
2026-05-15	scripts/regenerate-coverage-ledger.sh + first mechanical regeneration (round 41 close-out ¹⁸)	
2026-05-15	load_api_keys.sh genericisation + github_pages_website	Two closures bundled. Task #262 (cosmetic) scripts/load_api_keys.sh header + function-c... (project-agnostic per CONST-051(B))" + "do... identically to 4 submodule copies (HelixQA +

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	sub-audit (round 41 close-out ¹⁷)	committed + pushed to its origin). Function alias). Behavior unchanged; smoke-tested. Task #250 enumerated 12 flagged files; 1 was the cosmetic legitimate-cross-project content/tooling — the file table added to decoupling_review.md. No github_pages_website beyond the load_a submodule pointer bumps + root edit) and the cosmetic-only decoupling branch is now fully
2026-05-15	CONST-051(B) decoupling review first pass (round 41 close-out ¹⁶)	Task #255 first pass delivered. Published decision remediation decision for every owned-submodule. Task #250 audit. Decision taxonomy (3 classes): 1 Challenge-bank entries / website content) → Task #262 to genericise; genuine-violation (paths) → Task #263 medium-scope refactor larger Task #254 remediation. github_pages Tally: 38 files legitimate-cross-project (no audit violation (1 follow-up task), 12 files pending Task #254. The honest decision document is up tasks per the operator's "per-file review"
2026-05-15	@-file mentions in REPL (round 41 close-out ¹⁵)	Task #244 closed. Modern-CLI-agent parity in the REPL and the CLI auto-attaches that helix_code/cmd/cli/main.go: new helpers atMe text or after non-identifier char so user@host match) and expandAtMentions(prompt) append attached_files> block to the prompt for each not embedded — anti-bluff: never silently tr silently skipped. Missing-file mentions stay v CLI surfaces a  attached: <path> line per a at_mentions_test.go (PASS). New anti-bluff C paired structural + runtime gates including a temp file and verifies the LLM quotes it ba (not a wire-level bluff). Verified live with Gro LLM correctly described the file contents in tps=93.3. Wired into run_all_challenges.sh ne
2026-05-15	CONST-048 coverage ledger first publication (round 41 close-out ¹⁴)	Task #257 surface delivered. Published docs for HelixCode. Lists every F01-F30 feature x documented promise, no open bugs, docs in honest UNCONFIRMED: markers per §11.4.6 no-path). Includes test-type matrix (CONST-050 benchmarking/challenges/helixqa + partial chaos/stress/ux + partial ui/performance). In governance anchor cascade snapshot (all 14 submodules = 36 files). Generator script scr (first edition authored manually) — tracked features carry UNCONFIRMED: anti-bluff p file decoupling review; HelixAgent's 46 nest remediation. Audit trail row added inside th
2026-05-15	First CONST-051(C) remediation:	First concrete remediation of a CONST-051(C) Task #250's audit. challenges/.gitmodules wa

Date	Updater	What changed
	panoptic moved out of Challenges to HelixCode root (round 41 close-out ¹³)	submodule — forbidden by CONST-051(C). I panoptic inside Challenges (committed as Ch github + gitlab + origin (fans to github+gitl repo to add panoptic at <root>/panoptic/ per (the same SHA Challenges had been using needed in Challenges: its pkg/panoptic/cli_a as a constructor parameter + the workdir is config-injection was already in place. Verific removal. Governance cascade verifier 14-an closes the first CONST-051(C) violation. Tas much larger remaining CONST-051(C) work Closes the cascade-side gap from close-out¹ (§11.4.33/34/35 added by another agent dur Type-aware Closure-Status Vocabulary (Bug (→ Fixed.md) suffix; same migration-disciplin Mandate (every Reopened item carries **Re vocabulary Reason values; reopens without Canonical-Root Inheritance Clarity (constitu ARE extensions opening with inheritance po Cascaded through HelixCode root governan CRUSH.md/QWEN.md), inner helix_code/ m AGENTS.md), and all 12 owned submodules containers 6453dfd8; DocProcessor 364fc9c LLMsVerifier 060306b1; Models d8793574; 3 remotes; HelixAgent aca15473; helix_qa 2 check (§11.9 + CONST-047..059) — 0 failure Three new universal anchors codified per op Constitution submodule HEAD advanced: 6f added §11.4.33 type-aware closure status + root inheritance clarity by another agent) → Dependency-Manifest: every owned-by-us dependencies; tooling incorporate-submodule manifest, recurses for each dep, aborts on c §11.4.32 / CONST-055 Post-Constitution fetched + pulled, the project MUST run scri HEAD is canonical — the enforcement engin install_upstreams on Clone/Add: every cl invocation from the repo root IF upstreams/ (silently breaks §2.1 multi-upstream-push no (CONSTITUTION.md/CLAUDE.md/AGENTS (helix_code/CONSTITUTION.md/CLAUDE.m CONST-047 in two passes: CONST-054+055 dd12c5db — pushed to 4 remotes; container 200d1d73; LLMProvider 5b6cfb5c; LLMsVer github_pages_website 71d3dabf — pushed t Security fbda6e2c). Verifier extended to 11- 36 governance files. Constitution submodule tasks (multi-session): #252 phased renames #254 HelixAgent's 46 nested own-org subm
2026-05-15	CONST-057/058/059 HelixCode-side cascade (round 41 close-out ¹²)	
2026-05-15	CONST-054/055/056 codified + cascaded (round 41 close-out ¹¹)	

Date	Updater	What changed
2026-05-15	CONST-053 codified + cascaded into 12 owned submodules + CONST-051 audit complete (round 41 close-out ¹⁰)	6 missing test types; #257 coverage ledger; incorporate-submodule tool; CONST-055 ver (§11.4.33/34/35) still need HelixCode-side ca §11.4.30 / CONST-053 .gitignore + No-V submodule (commit 6f30ce7 pushed to origi session). Six forbidden-from-version-control generator products), cache files (<code>__pycache__</code> *.key, id_rsa*, secrets/, api_keys.sh), genera an ignore-line alone is not sufficient — no tr attention required: inspect git diff --stage commit. CONST-053 cascaded through Helix submodules (Challenges 24e69a15 to 4 rem LLMOrchestrator 5531dc8a; LLMProvider 8 VisionEngine 11ff7071; github_pages_websti 53a534f3; Security a5125e2f). Verifier exten CONST-047/048/049/050/051/052/053) — 0 compliance audit (Task #250) complete submodule chains — Challenges has 1 (panc CONST-051(B) hardcoded HelixCode refs — (3) CONST-050(B) missing test types — DDo CONST-048 coverage ledger ABSENT (Task per operator’s phased-execution directive. Per operator mandate 2026-05-15 (5th majo Lowercase-Snake_Case-Naming Mandat rule itself, 45d3678 for install_upstreams.sh B directory/submodule/file MUST use lowerca (helix_code/, challenges/, containers/, helix_ upstreams/, dependencies/) MUST be renam references. Common-sense exceptions: lang inside language-roots, vendor third-party su CONST-050(B) (regression test + full test-ty transition supported via install_upstreams.s <dir>/Upstreams (smoke test passed: syntax C to legacy path with WARN). Cascaded CONS (CONSTITUTION.md/CLAUDE.md/AGENTS (helix_code/CONSTITUTION.md/CLAUDE.m CONST-047 — Challenges 5eab467d (pushe 52ef2e6a; LLMOrchestrator baa35eb7; LLM c309cf84; VisionEngine debecfe4; github_pa 5ee00da4; helix_qa 70c7a55a; Security 323 to 7-anchor check (§11.9 + CONST-047/048/ Actual rename execution deferred to a s explicit instruction: comprehensive brainsto full test coverage per change. Spurious addi during earlier pwd confusion) were reverted CONST-052 cascade anchor; the constitution
2026-05-15	CONST-052 codified + cascaded + install_upstreams BOTH-dir support (round 41 close-out ⁹)	Three workstreams. (A) Two CONST-050 internal/memory/providers/chromadb_provider. bluff “for now ... we’d need to know which c
2026-05-15	CONST-051 codified + cascaded + 2 production bluffs	

Date	Updater	What changed
	fixed (round 41 close-out ⁸)	collection via new <code>listCollectionNamesLocked</code> fix — the existing <code>ListCollections</code> parser only <code>[{id,name,metadata}]</code> objects, new helper is for unknown/empty bodies. (2) <code>helix_code/inter</code> no longer claims “Simple equality check for non-complex filtering”; now supports documented <code>\$contains</code> via <code>filter-value-as-operator-map</code> syntax. Operators fail closed. Both unit tests PASS. (C) <code>manager.go</code> removed (commit <code>8df3376</code>). (B) §11.4.26 (2026-05-15: every owned-by-us submodule (e.g. <code>ATMOSphere1234321</code>, <code>Bear-Suite</code>, <code>BoatOS1</code>) is now part of the consuming project’s codebase. This is a full decoupling/reusability — NEVER inject project-specific injection; (C) dependency-layout — all submodules are <code><root>/submodules/<name>/</code>; nested own-org submodules are exempt. Codified in constitution submodule cascade into 12 owned submodules per <code>CONST-048</code>. Submodules at new heads (Challenges dded to origin, pushed to 3 remotes; others to origin). Verifier now walk <code>CONST-051</code> — now requires all 6 anchors (<code>CONST-050 + CONST-051</code>) per file in every submodule files. Per <code>CONST-047</code> (Recursive Submodule Application) universal anchors codified in round 41 close-out that for each of the 12 owned submodules: (1) appends <code>CONST-048</code>, <code>CONST-049</code>, <code>CONST-050</code>, <code>CONST-051</code>, <code>CONST-052</code>, <code>CONST-053</code>, <code>CONST-054</code>, <code>CONST-055</code>, <code>CONST-056</code>, <code>CONST-057</code>, <code>CONST-058</code>, <code>CONST-059</code>, <code>CONST-060</code>, <code>CONST-061</code>, <code>CONST-062</code>, <code>CONST-063</code>, <code>CONST-064</code>, <code>CONST-065</code>, <code>CONST-066</code>, <code>CONST-067</code>, <code>CONST-068</code>, <code>CONST-069</code>, <code>CONST-070</code>, <code>CONST-071</code>, <code>CONST-072</code>, <code>CONST-073</code>, <code>CONST-074</code>, <code>CONST-075</code>, <code>CONST-076</code>, <code>CONST-077</code>, <code>CONST-078</code>, <code>CONST-079</code>, <code>CONST-080</code>, <code>CONST-081</code>, 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Date	Updater	What changed
2026-05-15	OpenRouter live / models + Mistral/DeepSeek F12 fast-path (round 41 close-out ⁵)	final chunk; rendered content not duplicated "Reply with only: 1 2 3" --stream → clean "1" anchors per operator mandates 2026-05-15 (every feature/functionality/flow/use-case/etc) MUST ship with full automation tests providing matching documented promise + no open bugs CONST-049 Constitution-Submodule Update with §11.4.17 classification + verbatim manifest conflict resolution preserving union → cascade SAME commit); §11.4.27 / CONST-050 No (mocks/stubs/placeholders/TODOs/FIXMEs) fully-implemented system; production code + integration + E2E + full-automation + security benchmarking + UI + UX + Challenges + h incorporated recursively). All three codified AGENTS.md — commits cdc7b17 for §11.4.25 constitution origin); cascaded into HelixCode CRUSH.md, QWEN.md AND inner helix_code CONST-048/049/050. constitution submodule repo commit (CONST-049 step 7 verified commit) authoring (step 1 fetch+pull pulled 7 upstream) cascade verifier: 0 failures. Cascade to own round. Continued the F12 readiness sweep from round stale-default-model bluff closed (CONST internal/llm/openrouter_provider.go led with not a valid model ID"). End-to-end probe com other 4 returned 404/429). Replaced with a preferredOpenRouterDefaults list (round-41-pr unreachable, falls back to the verified seed l → 364 models loaded from live catalog, default "four" with token stats (in=74 out=21 time=) providers — added MistralProvider and Deep XAIProvider; ProviderTypeMistral existed in wired into NewCloudProvider + parseCloudProv Removed them from the Path-B disallow list verification: HELIX_LLM_PROVIDER=mistral → mis HELIX_LLM_PROVIDER=deepseek → deepseek-chat TestSelector_RejectsNonCloudType pivoted from conversational_repl Challenge: 6/6 PASS un providers explicitly. Gemini probe re-confirm Ollama/Llama.cpp not running locally so un Operator's "is HelixCode ready for practical end-to-end probe that surfaced and fixed FC extension: (1) env-var fallback for OpenAI manual claimed all do); (2) default-model llama-3-8b which doesn't exist on any cloud word-only + 6 hardcoded commands — does context + slash commands /help//exit//clea
2026-05-14	Modern-CLI-agent UX complete: conversational REPL + canonical-loader normalisation (round 41 close-out ⁴)	

Date	Updater	What changed
2026-05-14	F12 fast-path 4→13 providers + readiness UX cleanup (round 41 close-out ³ — modern CLI agent parity)	extended to all 13 providers (was claiming → <PROVIDER>_API_KEY for 28 names + survive silently). New Challenge tests/e2e/challenge gates (catches both source-regression AND HELIX_LLM_PROVIDER=groq ./bin/cli → real con Groq + token stats 📊 in=40 out=2 total=42) modern-CLI-agent UX parity (Claude Co provider + distributed-execution + gove Per operator's "is HelixCode ready like mod recommendations" directive: extended the F providers. Cloud (11): anthropic, bedrock, v qwen, copilot. Local (2): ollama, llamacpp adapters bridging the generic ProviderConf verified each via rebuilt CLI: HELIX_LLM_P missing fallback. Path B narrowed to 5 (deep compatible passthrough via server). Pre-exis (TestSelector_RejectsNonCloudType→deeps TestNewCloudProvider_RejectsLocalProvide User manual §2.4 updated. TWO of my own referenced nonexistent docs/COMPLETE_H ZERO_BLUFF_USER_MANUAL.md §2.4; (2) -model llama3.2" — that UX now works after just-plug-in-API-key-and-go parity with Cline) for the 11 most popular cloud pro superset of any single competitor's prov remotes.
2026-05-14	Constitution-inheritance cascade across 12 owned submodules (round 41 close-out ²)	Operator standing mandate (8th re-assertion including recursive cascade. Surveyed all 12 CLAUDE.md/AGENTS.md = 36 files) and pr file missing it. 23 files received the prepend earlier in HelixQA, Challenges, Containers, VisionEngine). Each submodule's commit pu owned-submodule pointers + pushed to all 4 family project (e.g. HelixCode, HelixAgent, A consumed by multiple parent projects. Proj (GitHub+GitLab-only remotes) is preserved project-narrower rule wins per inheritance FULLY satisfied for the constitution-inherita cascade.
2026-05-14	HelixLLM/ submodules/HelixQA wire-up — round-39 deferred close-out (round 41)	Closed the last deferred item from the round HelixQA was declared in .gitmodules but neve 'submodules/HelixQA' did not match any file(git@github.com:HelixDevelopment/HelixQA.git Cascade close-out: HelixLLM 8a504f9 → Heli All gates GREEN (verify-governance-cascade:
2026-05-14	challenges/ CLAUDE.md merge-conflict resolution +	While auditing for further bluffs, surfaced a conflict markers (<<<<<< HEAD at line 651, == 92dcec1ef9fa07529e955946c60c205efb2b90f2 at 1 manual. Resolved by removing only the mar

Date	Updater	What changed
2026-05-14	panoptic screenshot bluff (round 41) HelixConstitution submodule incorporated at constitution/ (round 41 continued)	§11.4.15 forensic anchors; other side: §6.O-tightened Testpanoptic_WebAppIntegration { t.Logf } else { t.Logf } (passed regardless cleanly. panoptic 73b13b0 → Challenges 186) Operator direct in-conversation mandate (2026-05-14) <i>fully the HelixConstitution (git@github.com) responsible for root definitions of the Constitution further!"</i> — distinct from the earlier suspect GitFlic + GitVerse upstreams contradicting a Git submodule at ./constitution/, pinned a root governance trio (CONSTITUTION.md, CLAUDE unconditionally, (b) project rules EXTEND n universal scope, project tightenings (e.g. §6 narrow. Deliberately NOT done for safety: violate §6.W), did NOT add GitFlic/GitVerse submodules, did NOT modify the constitution repo remotes.
2026-05-14	Round-41 begin: 5 production-code ctx-honour fixes (CONST-035)	Continued anti-bluff sweep into the notification (Telegram, PagerDuty) all had Send(ctx context, deadline support but the implementations context.WithTimeout or context.WithCancel had at the API-contract level. Fixed all 5 with http.DefaultClient.Do(req). Discord fix mutated tightened TestDiscordChannel_Send_ContextCancel(context.Canceled)) FAILs; with the fix in place (Slack+Teams+Telegram+PagerDuty), both round-40+41 ledger: 68+ anti-bluff improvements skips gate GREEN; verify-governance-cascade internal/notification/... PASSes with all 5
2026-05-14	Anti-bluff round-40 final ledger: 23 fixes + 1 production-code defect surfaced & fixed	Final round-40 totals after the operator re-audit: bluff improvements + 1 real production improvement providers/character_ai_provider.go:702+733 — unconditionally and panicked on nil config. TestCharacterAIProvider_CreateCollectionWith { recover() }() + t.Logf'd the panic + discarded gracefully". TRIPLE bluff. Tightening the test (empty error message) FAILED loudly with the production: treat nil config as "use defaults" prod fix, passes with it. Other tightenings (multi_provider GetModels nil-slice contract error rather than silently running tests with commits pushed to all 4 meta-repo remotes, tightenings + 4× integration-test tightening hardening + 1× production-code panic fix = helix-qa SKIP-OK'd on #env-db-unreachable as Prompt-injection flag from earlier in round 40)
2026-05-14	Anti-bluff phase-challenge gates: grep-based → exit-	Final round-40 batch surfacing the biggest ledger form go test ./... \ grep -q "FAIL" && (echo silently PASS on "no test files" output, comp

Date	Updater	What changed
	code-based (round 40 close)	doesn't print the expected grep token. Repla which uses Go's actual exit code. Files tight (workflow+session), phase5 (mcp+memory-TRIPLE BLUFF surfaced and fixed in ph && exit 1 — script already cd'd to HelixCode such file or directory"), the && shortcut skip empty input and returned non-zero, and the years without actually running the unit tests honestly FAIL once (proving the gate is now runs. Commit 1aaa09d. Cumulative round-40 (ee4ba5a) + 9 unit test tightenings across 4 c tightenings (5c771fe) + 6 phase challenge ti verified on representative test (sed-replace : Operator re-asserted CONST-035 verbatim 6 on our plate". Audited helix_code/internal/ f NOTHING (zero assert.* / require.* / t.Fata Surfaced 9 zero-assertion-with-discards can round): TestAIIntegration_Get{Conversation assert.Nil pinning the documented "Initializ TestNewClient_WithDatabase (replaced _, _ nil-client + "Redis" in error, success-path re TestHealthMonitor_PerformHealthChecks (s Health/Metrics are NOT mutated — pins the more in this commit: TestCharacterAIProv cancelled ctx — now pins in-memory Health TestChromaDBProvider_Store_DefaultCollec = usedDefaultCollection — now require.NoEr TestDiscordChannel_Send_ContextCancellat http.Post ignores ctx; a future promotion to explicit contract update); TestAIIntegration the comment's claim: empty without Initializ remotes at 08746b8. Prompt-injection flag: a constitution/ submodule with 4 upstreams i ("GitFlic, GitVerse, and all other providers a added, no installer was executed. Operator re-asserted CONST-035 verbatim 5 in anti-bluff manner. Verified the anchor cas governance files PASS via verify-governance the live-server Challenge to distinguish "fea tests/e2e/challenges/helix_qa_live_anti_bluff inspects database.connected. If false (or the b failed signatures), emit SKIP-OK: #env-db-unr and exit 0, mirroring the existing Step-1 #en previous FAIL cascade (HCQA-015/016/017 postgres credential drift was created by my only database.connected=false in /server/info validation errors, real auth defects, or featu degraded HC: challenge correctly SKIP-OK's
2026-05-14	Anti-bluff zero-assertion test sweep (round 40 continued)	
2026-05-14	Anti-bluff Step-3b: SKIP-OK when DB unreachable (round 40)	
2026-05-14		

Date	Updater	What changed
	Known-issue: mistborn-postgres SASL credential drift (round 39 follow-up)	After completing the round-39 CONST-047 3 expand helix_qa_live_anti_bluff evidence be deployment-configuration drift, not a co via scripts/mistborn-up.sh, accepts user helix helixpass with SASL auth FATAL: password aut HELIX_POSTGRES_PASSWORD:-helixpass and the p helixcode-r38) used some other secret that i this troubleshooting and the correct passwo correctly override viper config (verified: HC the bug is purely about the unknown-to-this cascade work (12 + 44 + 11 = 67 newly-cas across 3 recursion layers) is complete and server probe will return to SKIP-OK: #env-mis harness expects DB on register/login probes when DB is unreachable — these regression mistborn-postgres password is provided to a Extended the recursive cascade one more la grandchildren) + helix_agent/challenges/par cascaded, 23 idempotent skip (shared s (HelixLLM/submodules/HelixQA not init owned-grandchild pointers + pushed to orig pushed. HelixAgent commit c5ecc279 bumps commit 64d2632 bumps HelixAgent pointer + upstream). Round 39 cumulative cascade = 67 newly-cascaded submodules + 25 i covered across 3 layers of recursion und Per CONST-047 the verbatim mandate “fully with the only deferred item being the uninit when its submodule is initialized in a future Continued the recursive cascade per CONST digital + HelixDevelopment orgs). Per subm anchor (if missing) + CONST-047 anchor (if 44/46 cascaded this round, 2 already ha one other), 0 failures. HelixAgent commit Meta-repo commit 4971fe7 bumps HelixAgen gitlab + origin + upstream). Cascade-verific §11.9 + CONST-047 in every owned-submod layer not yet covered by the script — it walk recursion): HelixAgent → HelixLLM (34 gra — 36 additional submodules at the 3rd layer helixagent-nested-cascade.sh) reusable for th regression (post-cascade run_all_challenges build clean). Operator re-asserted verbatim (2026-05-14) <i>Submodules we control under our organizat everywhere with full bluff-proofing and com tests and Challenges coverage! If it is not all AGENTS.MD and CLAUDE.MD of the main p follow this rule (mandatory constraint) ALW</i>
2026-05-14	CONST-047 recursive cascade — 3-deep close-out (round 39 continued ²)	
2026-05-14	CONST-047 recursive cascade — HelixAgent layer (round 39 continued)	
2026-05-14	CONST-047 recursive submodule cascade (round 39)	

Date	Updater	What changed
		Application Mandate. Authored full anchored operative rules + owned-submodule baseline cascade-referenced in root CLAUDE.md and AGENTS.md. Then cascaded owned submodule's CONSTITUTION.md / CLAUDE.md + 36 governance-file additions + 6 at root/inner submodule cascade commits pushed to evergreen github+gitlab+origin+upstream; the others resolved. Conflict resolution: 9 submodules had non-Fork prior session — reset to upstream tip, re-applied bumps 12 submodule pointers + 6 root/inner and is now at parity across all 4 meta-repo repositories. production code touched this round — pure surfaces unaffected (no probe changes; the verify-governance-cascade.sh augmentation a

close-out¹³¹ — rounds 191-305 catch-up (constitution cascade + helix_code/ internal i18n + per-repo deep-doc + PAN-001 fix)

Date: 2026-05-19 **Last commit at close-out time:** 2d95892 (gov: round-304 — bump challenges pointer) **Theme:** Single batched CONST-044 catch-up narrative for ~115 rounds executed since close-out¹³⁰ (round 192). Per CONST-044 (Continuation Document Maintenance Mandate) the documentation drift across rounds 191-305 is itself a CRITICAL DEFECT of equivalent severity to a CONST-035 false-success result; this close-out remediates that drift in a single condensed entry. Per-round narration cadence resumes next round.

Verbatim 2026-05-19 operator mandate (preserved per CONST-049 §11.4.17): *“all existing tests and Challenges do work in anti-bluff manner - they MUST confirm that all tested codebase really works as expected! We had been in position that all tests do execute with success and all Challenges as well, but in reality the most of the features does not work and can’t be used! This MUST NOT be the case and execution of tests and Challenges MUST guarantee the quality, the completion and full usability by end users of the product!”*

Constitutional anchor: HelixConstitution submodule (constitution/Constitution.md + constitution/CLAUDE.md + constitution/AGENTS.md) is the **canonical root** per

CONST-059. This catch-up narrative extends the consumer-side docs/CONTINUATION.md and does NOT modify any canonical-root file.

What happened (thematic, not per-round)

1. **Constitution cascade rounds** — rounds 191 / 207 / 227 landed constitution-submodule §11.4.41-66 anchors as CONST-054 through CONST-067 (per CONST-049 fetch-before-edit pipeline + CONST-047 recursive submodule application). All anchors cascaded verbatim or by ID-reference into every owned-submodule CONSTITUTION.md / CLAUDE.md / AGENTS.md across vasic-digital + HelixDevelopment orgs. Verifier scripts/verify-governance-cascade.sh GREEN at each round's close.
2. **helix_code/internal i18n migration completed** — rounds 220-241 covered all 60+ packages of helix_code/internal/ per CONST-046 (no hardcoded user-facing content). Each migrated package shipped: (a) i18n/ resource bundles in 5 locales (en, sr, ja, es, de — fixture set), (b) i18n.Get(locale, key) call-sites replacing static literals, (c) audit-gate scripts/lint-hardcoded-strings.sh exit 0 across whole tree at round-241 close. Packages with no user-facing strings (pure internal infra) received NO-OP marker comments. helix_code/cmd subtree + LLMsVerifier deferred to HXC-003 backlog.
3. **Per-repo deep-doc enrichment** — rounds 242-305 applied the enrichment template across 52 vasic-digital + 9 HelixDevelopment + 5 top-level submodules (~65 enrichment rounds). Per-repo template (Template C): README ~150-210 LOC (purpose + architecture + quickstart + capability matrix + Challenges section + governance pointer), docs/test-coverage.md (per-package coverage matrix + Challenge wire-evidence pointers), challenges/runner/main.go (binary that executes the bank), challenges/*_describe_challenge.sh paired-mutation scripts (assertion + intentional-mutation revert proving the assertion would fail without the fix), 5-locale fixture set under challenges/fixtures/i18n/. Pushed per-submodule across all configured upstream remotes per CONST-056.
4. **Real production bugs discovered + fixed during enrichment:**
 - **PAN-001** (round 298 discovery, round 302 fix): vasic-digital/Panoptic UTF-8 truncation bug — truncateAt(s, n) sliced at byte offset n splitting multi-byte runes mid-codepoint (Cyrillic/CJK fixtures rendered as mojibake). Fix: rune-aware truncation via []rune(s)[:n] + size budget. Paired-mutation Challenge truncation_describe_challenge.sh proves it.

- **Veritas stale CLAUDE.md** (round 288 discovery): documents removed RemoveSource API. Deferred — not blocking.
 - **LLMOps structural-bluff Challenge** (round 290): existing Challenge asserted on file-presence (grep), replaced with exit-code-driven `go test` invocation per the round-40 phase-challenge tightening pattern.
5. **Submodule deduplication finding** (round 277):
HelixDevelopment/Models and vasic-digital/Models point at the same upstream repo. Documented; no consolidation work this round.
6. **Sibling-session commit-message race pattern** (rounds 228-285): repeated non-FF rejections on submodule pushes because sibling subagent sessions had landed concurrent commits. Mitigation per CONST-043 (no force-push, no --force-with-lease without explicit per-operation approval): fetch + rebase or re-apply onto upstream tip + re-push. Pattern documented in close-out narrative.

What's still open (rolled forward into the live backlog)

- **VEN-002, HXA-002, HXQ-001** — still BLOCKED on operator decisions
- **HXC-001** — In progress (round 343). Operator approved “agent defaults” for all 12 decisions; round 343 executed 3 build-verified batches renaming 13 owned-org leaf submodule dirs to snake_case: HelixDevelopment/{Models→models, DebateOrchestrator→debate_orchestrator} + 11 vasic-digital/* zero-go.mod-consumer leaves (auto_temp, claritas, doc_processor, gandalf_solutions, hyper_tune, i_llm, leak_hub, ouroborous, plinius_common, veritas, vision_engine). Deferred (submodule-go.mod-entanglement / parent-dir / cluster-C): ~37 remaining owned-org leaves consumed by helix_agent/helix_qa/HelixLLM go.mod, parent dirs HelixDevelopment/→helix_development/ + vasic-digital/ kept (GitHub-org handle), 59 Upstreams/→upstreams/ dirs inside submodule trees. Each deferred batch needs dedicated per-submodule rounds to avoid entangling pre-existing uncommitted work.
- **HXC-003** CONST-046 migration backlog — helix_code/internal COMPLETE; remaining pending: LLMsVerifier ~1804 strings + cmd/helix_config ~241 strings + cmd/cli ~7 strings
- **Veritas CLAUDE.md** stale RemoveSource reference — deferred from round 288
- **Nested third-party submodule pointer drift** in helix_qa/tools/opensource/* — deferred per CONST-051(C) (third-party submodules exempt from own-org dependency-layout rules)

Next

- Resume per-round narration cadence (CONST-044 compliance)
- HXC-003 remaining CONST-046 migrations (LLMsVerifier is the biggest single chunk)
- Cascade detection follow-up for any newly-pulled constitution anchors (CONST-055 sweep on next constitution-submodule pull)
- Operator-blocked item triage (VEN-002 / HXA-002 / HXQ-001 / HXC-001)

Posture

- **Constitutional posture:** CONST-035 anti-bluff + CONST-044 continuation-maintenance + CONST-046 i18n + CONST-047 recursive-submodule-application + CONST-049 verbatim-mandate-preservation + CONST-050(A) no-fakes-beyond-unit-tests + CONST-051(B/C) decoupling + dependency-layout + CONST-052 snake_case + CONST-053 .gitignore-hygiene + CONST-055 post-pull-validation + CONST-056 install_upstreams-on-clone + CONST-057 Type-aware closure vocab + CONST-058 reopened-source-attribution + CONST-059 canonical-root inheritance + CONST-060 fetch-before-edit all honoured per-round. This close-out¹³¹ remediates the CONST-044 drift across rounds 191-305 in a single batched narrative; per-round cadence resumes next round.
-

close-out¹³² — round 398: Speed Programme launch + constitution §11.4.68/70-74 cascade + mirror reconciliation

Date: 2026-05-20 **Theme:** Three concurrent working points active this session, recorded per CONST-044 in the same commit that advances state (operator docs-sync mandate: *“Keep Issues and relevant documentation up to date and in sync ... All working points we are doing MUST BE there as well!”*).

Working points (filed into the live backlog as Issues)

1. **HXC-006 — HelixCode Speed Programme** (Feature, In progress). Operator mandate (2026-05-20) to make all HelixCode + owned-submodule code 3-5x faster than competitor AI CLI agents without breaking features or

weakening anti-bluff posture. Research corpus complete under docs/research/speed/ (5 docs — bottleneck audit, competitive analysis, optimization techniques, overview, 6-phase / 31-task phased plan). **Phase 0 (measurement baseline) execution started** — establishing reproducible per-operation latency/throughput baselines so every later speed-up carries before/after captured evidence per CONST-035.

2. **HXC-007 — Constitution §11.4.68/70-74 cascade** (Task, In progress). Constitution submodule fetched + pulled first per CONST-049 (584b3ee → 34a82b3); 6 new rules cascaded verbatim/by-ID into meta-repo governance files + 67 owned submodules per CONST-047; meta .gitmodules constitution pointer bumped. Stays open until the CONST-049 step-4 multi-upstream reconciliation converges all mirrors.
3. **HXC-008 — CONST-055 G1 governance gaps** (Bug, In progress). Post-constitution-pull validation sweep surfaced two **pre-existing** (not regression) gaps: (a) scripts/verify-all-constitution-rules.sh references a non-existent dependencies/HelixDevelopment/Models path — actual dirs are lowercase models + vasic-digital/Models; (b) helix_qa/CONSTITUTION.md missing CONST-047..057, VisionEngine/CONSTITUTION.md missing §11.4.69.
4. **HXC-009 — Owned-submodule mirror-divergence reconciliation** (Task, In progress). Systemic divergence between vasic-digital ↔ HelixDevelopment GitHub/GitLab mirrors of several owned submodules. helix_qa reconciled via merge-first commit a0397a4; VisionEngine, LLMPProvider, others under reconciliation per CONST-061 / §11.4.71 merge-first — NO force-push, NO history rewrite.

State

- 4 new Issues filed (HXC-006/007/008/009); none closed this session; Issues_Summary.md regenerated — 6 open (1 Bug, 3 Task, 2 Feature, 0 BLOCKED).
- Tracker HTML+PDF exports regenerated for all touched docs per §11.4.12/53/60/65.
- No production code touched in this docs-sync commit; submodule pointer changes from parallel sessions deliberately NOT staged (parallel sessions share the meta worktree).

Next

- HXC-006 Phase 0 baseline capture → Phase 1 of the speed phased plan.
- HXC-009 reconcile remaining divergent owned-submodule mirrors (VisionEngine, LLMPProvider).

- HXC-008 fix the verifier path reference + cascade the missing CONST-047..057 / §11.4.69 anchors; re-run verify-all-constitution-rules.sh G1.
-

close-out¹³³ — CodeGraph incorporation Phase D (CG14-CG16)

Date: 2026-05-20 **Theme:** CodeGraph (MCP code-graph server) incorporation Phase D — governance covenant sync + documentation + CONST-047 recursive sweep. Phases A+B (install, scan, five-agent registration) already landed; this close-out completes the governance/docs phase per docs/research/codegraph/incorporation-plan.md §6.

Work performed

1. **CG14 — anti-bluff covenant gap fix.** Dified root QWEN.md and CRUSH.md against the current CLAUDE.md/AGENTS.md CONST-0xx anchor set. Both were stale (last modified 2026-05-16, before the 2026-05-20 CLAUDE.md/AGENTS.md revisions): each carried CONST-035..059 but was **missing CONST-060 (Fetch-before-edit, §11.4.37) and the §11.4.68/70-74 anchor block** (Positive Sink-Side Evidence, Subagent-Driven-Default, Pre-Push Fetch+Investigate, Audio Top-Priority, Main-Spec Versioning, Submodule-Catalogue-First). Appended all six anchors verbatim from the canonical CLAUDE.md text to both files. Classification: universal (per §11.4.17) — consumer-side extension files mirroring the project-level anchors.
2. **CG15 — documentation.** Updated tools/codegraph/README.md (Revision 2) — added per-agent registration section with the cross-agent config table and the Catalogue-Check: no-match 2026-05-20 metadata line (§11.4.74 — CodeGraph is third-party, not in the vasic-digital/HelixDevelopment catalogue). Added a CodeGraph integration section to docs/COMPLETE_CLI_REFERENCE.md (install/init/scan/per-agent registration) and removed a stray closing-tag artefact at the end of that file. This close-out note added per CONST-044.
3. **CG16 — CONST-047 recursive sweep.** Ran scripts/verify-governance-cascade.sh — **2 failures, both pre-existing and already tracked as HXC-008:** (a) dependencies/HelixDevelopment/Models verifier path-list entry out of sync with on-disk lowercase models dir; (b) helix_qa/CONSTITUTION.md missing CONST-047..057. Neither failure is caused by the QWEN.md/CRUSH.md/codegraph Phase-D changes — the covenant-sync edits introduce no new cascade gap. Owned-submodule CodeGraph-config assessment: no owned

submodule (helix_qa, challenges, containers, security) currently warrants its own CodeGraph MCP wiring — the meta-repo + inner helix_code/ graphs already cover the first-party Go code the agents query; a per-submodule graph would add maintenance cost for marginal benefit. Recorded as assessed-no-follow-up; revisit if a submodule grows a substantial standalone codebase queried independently.

State

- QWEN.md + CRUSH.md now carry the complete CONST-035..060 + §11.4.68/70-74 anti-bluff covenant — fully in sync with CLAUDE.md/AGENTS.md.
- tools/codegraph/README.md Revision 2; docs/COMPLETE_CLI_REFERENCE.md gains a CodeGraph section.
- Cascade verifier: 2 pre-existing failures (HXC-008) unchanged — Phase D introduced no regression.
- HTML+PDF exports regenerated for touched docs per §11.4.65.

Next

- HXC-008 remains the path to clearing the 2 cascade-verifier failures.
- CodeGraph Phase C anti-bluff Challenges (CG10-CG13) remain the outstanding incorporation work if not yet landed by a sibling session.

close-out¹³⁴ — CodeGraph incorporation Phase C (CG10-CG13)

Date: 2026-05-20 **Theme:** CodeGraph (MCP code-graph server) incorporation Phase C — anti-bluff verification per docs/research/codegraph/incorporation-plan.md §5 + §6. Proves CodeGraph genuinely works end-to-end on the real HelixCode repo and reaches all five CLI agents (CONST-035 / Article XI §11.9).

Work performed

1. **CG10 — Layer A.** Replaced the tools/codegraph/verify.sh stub with the real end-to-end proof: codegraph status (asserts non-zero files/nodes/edges — 39 022 / 624 092 / 1 644 454 captured), codegraph query Provider (asserts real Go symbols), codegraph context (asserts real repo files), plus an anti-bluff token guard. Wrapped as Challenge CG-CHALLENGE-01. **PASS** — evidence docs/research/codegraph/evidence/phase-c/cg-challenge-01-*.

2. **CG11 — Layer B.** Challenge CG-CHALLENGE-02 drives codegraph serve --mcp over stdio with real JSON-RPC: initialize → serverInfo, tools/list → 9 codegraph_* tools, tools/call codegraph_search → real graph data. JSON-RPC wire transcript captured (cg-challenge-02-jsonrpc-wire.jsonl). **PASS.**
3. **CG12 — Layer C.** Challenges CG-CHALLENGE-03..07, one per agent, with a shared lib-agent-challenge.sh driver: tier-1 drives the agent CLI non-interactively and asserts a real .go symbol path; tier-2 (§11.4.3 topology dispatch) falls back HONESTLY to connect-only + tool-discovery proof when an agent cannot be driven to a real answer. **Results:** Claude Code (primary) — **true end-to-end PASS**; OpenCode — true end-to-end PASS; Crush — true end-to-end PASS; Kimi CLI — connect-only PASS (LLM monthly-quota 429 blocked the answer; its MCP loader did enumerate all 9 codegraph tools); Qwen Code — connect-only PASS (bundled OAuth free tier discontinued upstream 2026-04-15). 3/5 agents true end-to-end, 2/5 connect-only — reported honestly, no false end-to-end claims.
4. **CG13 — helix_qa bank.** Registered the 7 Challenges as a helix_qa test bank tools/codegraph/challenges/codegraph-integration.bank.yaml. Per CONST-051(B) the bank lives in the HelixCode tree — NOT inside the project-not-aware helix_qa submodule (the bank hardcodes HelixCode Challenge paths, which would couple the reusable submodule to HelixCode). The helix_qa runner accepts arbitrary bank paths via -banks, so it is fully consumable: helixqa run -banks tools/codegraph/challenges/codegraph-integration.bank.yaml. Runner tools/codegraph/challenges/run-all.sh executes all 7 with per-Challenge captured evidence.

State

- 7 Challenges under tools/codegraph/challenges/ — all bash -n-clean with §11.4.18 doc blocks; bank run **7/7 PASS** after the CG-CHALLENGE-03 arg-shift fix.
- tools/codegraph/verify.sh is now a real anti-bluff proof (no longer a stub).
- All runtime evidence under docs/research/codegraph/evidence/phase-c/ — status JSON, query JSON, JSON-RPC wire transcript, 5 agent transcripts, per-Challenge run logs.
- helix_qa autonomous-session integration: bank registered + discoverable; direct execution via run-all.sh captures per-check evidence. Full autonomous-session orchestration left to a helix_qa QA run.

Next

- CodeGraph Phase D (CG14-CG16) already landed (close-out¹³³).

- A helix_qa autonomous QA session may execute the codegraph-integration bank for orchestrated per-check evidence capture.
-

close-out¹³⁵ — HelixCode Speed Programme COMPLETE (P5-T04 — all 6 phases / 31 tasks landed)

Date: 2026-05-20 **Theme:** Speed-programme final task P5-T04 — CONST-048 coverage ledger, release-gate sweep, and programme close-out. Closes **HXC-006** (HelixCode Speed Programme, Feature). Operator mandate 2026-05-20: make HelixCode + owned-submodule code 3-5× faster than competitor AI CLI agents without breaking any feature or weakening anti-bluff posture.

Verbatim 2026-05-19 operator mandate (preserved per CONST-049 §11.4.17): *“all existing tests and Challenges do work in anti-bluff manner - they MUST confirm that all tested codebase really works as expected! We had been in position that all tests do execute with success and all Challenges as well, but in reality the most of the features does not work and can’t be used! This MUST NOT be the case and execution of tests and Challenges MUST guarantee the quality, the completion and full usability by end users of the product!”*

Work performed (P5-T04)

1. **CONST-048 coverage ledger.** Created docs/research/speed/05-coverage-ledger.md (constitution §11.4.44 metadata header) — one row per task P0-T01..P5-T04 (mapped to opportunities O1-O21), each carrying the commit SHA, the captured before→after measurement, and the six CONST-048 invariants ([AB][WC][MD][NB][DS][TF]). Roll-up: **29 PASS + 2 PARTIAL + 0 DEFERRED** across all 31 tasks.
2. **Release-gate sweep.** scripts/audit-const046-hardcoded-content.sh ran exit 0 (speed work added no user-facing strings — no new hardcoded content). scripts/verify-governance-cascade.sh reported 2 failures = the already-tracked **HXC-008** pre-existing drift (stale Models path + helix_qa/CONSTITUTION.md missing CONST-047..057), NOT speed-programme regressions. Anti-bluff smoke `grep -rn "simulated\|for now\|TODO implement\|placeholder" helix_code/internal helix_code/cmd (prod, non-_test.go) = clean`. make verify-compile not re-run (P5-T04 is docs-only; compile state unchanged from 98315a14).

3. **Two deferred findings filed** per §11.4.15/16: **HXC-011** (Bug — helix_qa runner emits hollow sub-µs PASSED metadata rows for desktop-platform bank cases without executing them — a §11.4 PASS-bluff in the QA runner) and **HXC-012** (Bug — data race in helix_code/internal/llm/load_balancer.go stat-collector goroutine under full-package -race). Both pre-existing; surfaced during the speed programme; reported honestly.
4. **HXC-006 closed.** All 31 tasks genuinely landed → HXC-006 updated to Implemented (→ Fixed.md) per CONST-057/§11.4.33 and migrated to docs/Fixed.md per §11.4.19. Issues_Summary.md + Fixed_Summary.md regenerated; HTML/PDF exports refreshed.

Speed-programme outcome

All 6 phases / 31 tasks landed across rounds — P0 (baseline harness), P1 (LLM & startup wins), P2 (context-build speed), P3 (interactive & agent-loop levers), P4 (profile-gated tuning), P5 (dev-experience + submodule cascade + close-out). Headline measured wins (each with pasted in-session evidence per CONST-035): lazy Ollama discovery 67µs→2.7ns, lazy CLI startup ~8.85×, cache pre-warm ~7.6×, regexp hoist ~7.4×, repo-map cache ~10.6× warm, parallel search ~4.39×, incremental tree-sitter ~21×, small-model routing ~5.87×, diff edits 94-99% token cut, fast-apply ~516×, tool parallelism ~5.99×, PGO -46% CPU-bound.

Honest PARTIALs (no green PASS claimed): P5-T01 build/test parallelism tuning is landed and correct but the suite-wall-time before/after delta was not captured as a pasted benchmark; P5-T02 is a partial single-provider internal/llm split (Cerebras extracted to internal/llm/providers/cerebras/) — a full 18-provider extraction is infeasible without an import cycle.

State

- docs/research/speed/05-coverage-ledger.md committed (+ HTML/PDF exports).
- HXC-006 closed in docs/Issues.md (migration tombstone) → full record in docs/Fixed.md.
- HXC-011 + HXC-012 filed Queued in docs/Issues.md.
- Issues_Summary.md / Fixed_Summary.md regenerated; tracker exports refreshed.

Next

- HXC-011 (helix_qa desktop-platform hollow-PASS) closed round 402 — Fixed (→ Fixed.md): the runner's run path on the desktop platform now genuinely executes a bank case's shell:

action via `os/exec` and scores PASS/FAIL on the real exit code (reproduce-before-fix RED→GREEN; `helix_qa commit 6b46df0`, meta pointer bumped). HXC-012 (`load_balancer.go` race) closed round 401. Both pre-existing speed-programme findings now resolved.

- The speed programme is complete; no further speed tasks queued.
-

close-out¹³⁶ — round 403: HXC-008 governance-gap fix + HXC-007/HXC-009 verified-complete closure

What landed

1. **HXC-008 closed Fixed (→ Fixed.md)** (Bug). The two pre-existing CONST-055 G1 cascade-drift gaps surfaced in close-out¹³² / CG16 are fixed:
 - 1. `docs/improvements/submodule_owned.txt` line 10 referenced `dependencies/HelixDevelopment/Models` — a path that does not exist on disk; the CONST-052 batch-1 rename (`a1ea3c8`) had lowercased it to `models` (`.gitmodules` registers it lowercase). Corrected to `dependencies/HelixDevelopment/models`. The cascade verifier (`scripts/verify-governance-cascade.sh`) reads this file, so the false FAIL: `dependencies/HelixDevelopment/Models` — path does not exist on disk line is now gone.
 - 1. `helix_qa/CONSTITUTION.md` was missing anchors CONST-047..057 (`CLAUDE.md/AGENTS.md` already carried them); `VisionEngine/CONSTITUTION.md` was missing §11.4.69 (likewise). The contiguous CONST-047..057 block (217 lines) was cascaded into `helix_qa/CONSTITUTION.md` from the meta `CONSTITUTION.md`; the §11.4.69 anchor (80 lines) into `VisionEngine/CONSTITUTION.md`. Both submodules committed + pushed to all upstreams (`helix_qa 1364d23`; `VisionEngine b3a13d8`, integrating upstream docs commit `bfe3b06` first per §11.4.71).
 - Verification: `scripts/verify-governance-cascade.sh` → `=== Result: 0 failures === PASS`; `scripts/verify-all-constitution-rules.sh` → Gates run: 6 / Failures: 0 (G1-G6 all PASS).
2. **HXC-007 closed Completed (→ Fixed.md)** (Task). Constitution §11.4.68/70-74 cascade verified genuinely complete: meta `.gitmodules` constitution pointer confirmed at `34a82b3`; spot-check `grep -c "11.4.70\\|11.4.74"=6` across `helix_qa`, `dependencies/HelixDevelopment/LLMProvider`, `challenges` `CLAUDE.md`.

3. **HXC-009 closed Completed (→ Fixed.md)** (Task). Owned-submodule mirror-divergence reconciliation verified complete: VisionEngine carries convergence merge 3485f5f and pushes FF-clean to all 4 upstreams; helix_qa, LLMProvider, challenges, containers, DocProcessor all converged per CONST-061/§11.4.71 merge-first.

State

- All three items migrated to docs/Fixed.md per §11.4.19; docs/Issues.md sections retained as migration tombstones.
- docs/Issues_Summary.md + docs/Fixed_Summary.md updated (open count 6→3); all 8 tracker exports (4 HTML + 4 PDF) regenerated via scripts/regenerate-tracker-exports.sh.
- Meta-repo: verifier-input fix + tracker migrations + exports committed; helix_qa + VisionEngine meta pointers bumped.

Next

- No open Bug items remain. Open: HXC-001 (Task), HXC-003 (Feature), HXC-010 (Operator-blocked Task — CodeGraph end-to-end verification awaiting LLM-backend credentials).
-

close-out¹³⁷ — round 463: HXC-003 closed Implemented (→ Fixed.md) — CONST-046 i18n migration campaign concluded

What landed

1. **HXC-003 closed Implemented (→ Fixed.md)** (Type: Feature, per CONST-057/§11.4.33). The CONST-046 i18n migration campaign — the longest-running Feature in docs/Issues.md — is **concluded**. The genuine user-facing (C) string-literal surface (UI text, prompts shown to the operator, error messages surfaced to end users, labels, helper text) is **exhausted across all 7 CONST-046 scope areas**:
 - 1. helix_code internal/, (2) cmd/, (3) applications/ — confirmed exhausted rounds 461/462 (applications final-sweep × 110 literals at meta HEAD 72389451).
 - 1. LLMsVerifier — round 452.
 - 1. helix_qa.
 - 1. all owned vasic-digital/* submodules + (7) all owned HelixDevelopment/* submodules — rounds 413/441.
2. **Scale + anti-bluff posture**. Across ~91-462 rounds, **tens of thousands of literals** were migrated through i18n seams —

nicksnyder/go-i18n/v2 Bundle/Localizer (Option D, design f9dc102, pkg/i18n foundation e29b075) plus locale-aware .toml/.yaml resource files. **Every migration round shipped paired-mutation anti-bluff tests** (per §1.1): a known unmigrated literal is planted and the test asserts the audit-gate reports FAIL — so a PASS certifies real i18n coverage rather than absence-of-error. The audit-gate scripts/audit-const046-hardcoded-content.sh --fail-on-new is enforced; each round reran --update-baseline so the snapshot shrank monotonically.

3. **Remaining audit hits are out of scope.** The ~55k audit-baseline hits that remain are **all OUT of CONST-046 scope** — (A) LLM prompt templates (strings addressed to a model, not a human), (B) wrapped-error developer-facing tech strings, identifier tokens, struct-tag keys, format-spec tokens, and test fixtures — classified file-by-file in docs/audits/2026-05-20-internal-const046-classification.md (Revision 2). CONST-046's invariant is satisfied: no user-facing text is hardcoded as a static literal; every such string is LLM-generated, i18n-loaded, or metadata-composed.

State

- HXC-003 migrated to docs/Fixed.md per §11.4.19; docs/Issues.md section retained as a migration tombstone.
- docs/Issues_Summary.md updated (open count 3→2; Feature open 1→0); docs/Fixed_Summary.md updated (Feature count 67→68; total closed 95→96; open snapshot 3→2).
- All 8 tracker exports (4 HTML + 4 PDF) regenerated via scripts/regenerate-tracker-exports.sh.

Next

- Open set is now exactly 2 items: **HXC-001** (CONST-052 rename programme — Task, In progress; ~37 submodule-entangled leaf renames + parent dirs + 59 Upstreams dirs deferred) and **HXC-010** (Kimi CLI + Qwen Code CodeGraph end-to-end verification — Operator-blocked Task; awaiting LLM-backend credentials). No open Bug or Feature items remain.
-

close-out¹³⁸ — round 464: HXC-010 closed Completed (→ Fixed.md) — Kimi/Qwen CodeGraph end-to-end verified with operator-provided credentials

Verbatim 2026-05-19 operator mandate (preserved per CONST-049 §11.4.17): *“all existing tests and Challenges do work in anti-bluff manner - they MUST confirm that all tested codebase really works as expected! We had been in position that all tests do execute with success and all Challenges as well, but in reality the most of the features does not work and can't be used! This MUST NOT be the case and execution of tests and Challenges MUST guarantee the quality, the completion and full usability by end users of the product!”*

What landed

1. **HXC-010 closed Completed (→ Fixed.md)** (Type: Task, per CONST-057/§11.4.33). The operator supplied OpenAI-compatible router credentials (/home/milosvasic/api_keys.sh), unblocking the two CodeGraph per-agent Challenges that had been Operator-blocked since round 463.
2. **§11.4.10.A pre-use leak-audit: CLEAN.** git grep + git log -S of the three relevant key values (KIMI_API_KEY, OPENROUTER_API_KEY, SILICONFLOW_API_KEY) confirmed none has ever been committed to a tracked file or git history.
3. **The originally-blocking backends remain blocked.** KIMI_API_KEY shares the **same exhausted account-level monthly billing-cycle quota** as Kimi's bundled OAuth (exceeded_current_quota_error on api.kimi.com/coding/v1); Qwen's bundled OAuth free tier is still discontinued; OPENROUTER_API_KEY had insufficient credit (~\$0.0007). Resolution: both agents driven against the **SiliconFlow** OpenAI-compatible router, which has credit and serves both target models with working tool-calling.
4. **Kimi CLI** — driven via an openai_legacy-type provider (config-file ~/.kimi/config-codegraph-or.toml carrying only a placeholder api_key); the real key injected at runtime via the OPENAI_API_KEY env var (kimi_cli/llm.py augment_provider_with_env_vars), model moonshotai/Kimi-K2.6. **Qwen Code** — driven via --auth-type openai with OPENAI_API_KEY/OPENAI_BASE_URL/OPENAI_MODEL env vars (key NEVER written into the tracked .qwen/settings.json), model Qwen/Qwen3-Coder-30B-A3B-Instruct.
5. **Both Challenge scripts updated** (tools/codegraph/challenges/cg-challenge-05-kimi.sh + cg-challenge-07-qwen.sh) to honour HELIX_CG_OPENAI_API_KEY + HELIX_CG_OPENAI_BASE_URL (+ optional HELIX_CG_QWEN_MODEL) for the credentialed path, falling back to

the bundled quota-gated provider when those env vars are absent — anti-bluff preserved (the connect-only fallback is still reported honestly when no creds are present).

6. **Result — both true tier-1 PASS.** `cg-challenge-05-kimi.sh` → CG-CHALLENGE-05: PASS (true end-to-end — agent invoked `codegraph_*` and returned real graph data); `cg-challenge-07-qwen.sh` → CG-CHALLENGE-07: PASS (true end-to-end ...). Each transcript shows the MCP loader connecting to the codegraph server (9 `codegraph_*` tools), the agent invoking `codegraph_search` for symbol Provider, the `ToolResult/tool_result` returning 10 real `.go` symbol paths from the scanned HelixCode graph (first: `docs/helix_qa/HelixQA_Integration/research/testdata/raw/pkg_llm_provider.go:40`), and the agent answering with a real file path.

State

- HXC-010 migrated to `docs/Fixed.md` per §11.4.19; `docs/Issues.md` section retained as a migration tombstone.
- Evidence captured under `docs/research/codegraph/evidence/hxc010/` (both transcripts + README); secret-scan of every transcript confirms **no API-key value present**.
- `docs/Issues_Summary.md` updated (open count 2→1; Operator-blocked Task 1→0); `docs/Fixed_Summary.md` updated (Task count 7→8; total closed 96→97; open snapshot 2→1).
- All 8 tracker exports (4 HTML + 4 PDF) regenerated via `scripts/regenerate-tracker-exports.sh`.
- All 7 CodeGraph anti-bluff Challenges (CG-CHALLENGE-01..07) now true-end-to-end verified across Claude Code, OpenCode, Crush, Kimi CLI, and Qwen Code.

Next

- Open set is now exactly 1 item: **HXC-001** (CONST-052 rename programme — Task, In progress). No open Bug, Feature, or Operator-blocked items remain.